



Southwest Florida Regional Common Treatment Guidelines

Revised 8/2026

These guidelines are of little value in the absence of great training processes in each Department that will utilize them, and the Quality Improvement (QI) processes that ensure their effective and safe use.

We acknowledge that the dedicated, devoted, and consistent efforts of the protocol sub-committee members, who shepherded these guidelines and their updates through the Medical Care Council. The committee sought the contributions of the emergency, pediatric, trauma, obstetric and critical care physicians in the Lee Health system, and those in Charlotte and Hendry and Collier counties, who collectively sought the input of their partners. It is an ongoing effort for these contributors to bring new developments to confirm the best practices from around the country.

We acknowledge the protocols of EMS systems like MedStar in Fort Worth, TX and Austin Travis County, TX for their guidance and material for our consideration and application.

We as an emergency care community move forward with gratitude to those who will make devoted efforts to breathe life into these guidelines perhaps, sometimes literally for the well-being of our patients and for the benefit of our entire community.

Guideline Referenced	Annotated Update Description	Date
Whole Blood	Removed PEARL to allow for treatment of medical etiologies	Aug-26
Freestanding ED	Added D38, to be open 10/2026	Aug-26
Main Campus Hospitals	Added D3 as a Primary Stroke Center Added Helipad for D9 Updated D5 phone number	Aug-26
Pharmacology Reference	Removed IV/IO routes for Epinephrine 1:1000 Added Oral Glucose	Aug-26
Reactive Airway	Added Racemic Epinephrine	Aug-26
BLS Procedures	Updated the Neo-Tee to reflect current procedure	Aug-26
Theory & Operational	Added Special Medical Guidance	Aug-26
Transport Destinations	Added STARS language Updated Trauma Arrest guidance based on Trauma Services recommendations	Aug-26
Medical Cardiac Arrest	Added refractory VT as indication for DSD	Aug-26
ALS Procedures	Added refractory VT as indication for DSD Added using same finger to DSD Updated Synchronized Cardioversion with new joules settings	Aug-26
Tachycardia	Updated the recommended joules settings for cardioversion	Aug-26
Allergic Reaction & Anaphylaxis:	Changed from set epi dose to the same dose range as other guidelines	Aug-26
General Changes	Formatting updates. Removed “Advanced Procedures” language.	Aug-26
Main Campus Hospitals	Added D17 as STEMI Facility Added D7 as a Primary Stroke Center Removed D11 for Pediatric Admits	Jun-26
Pharmacology Reference	Added Hydrocortisone	Jun-26

Guideline Referenced	Annotated Update Description	Date
ALS Procedures	Removed pericardiocentesis procedure from the guideline.	Mar-26
Reactive Airway Disease	Adjusted dose of nebulized epinephrine for croup to 5mg.	Mar-26
TXA Dosage	Made the wording the same for TXA dosing across the different guidelines.	Nov-25
Pediatric Trauma Scorecard	Created a new pediatric scorecard with collaboration of Golisano leadership.	Nov-25
Pharmacology Reference	Removed indications that are not related to clinical guidelines.	Nov-25
Hypo/Hyperthermia	More explicitly explained that cold water immersion is not indicated for medical fever.	Nov-25
Dive Emergencies	Added the new Dive Emergencies guideline	Jun-25
General Changes	Adjusted from Lee County Language to SWFL Regional language. Added MCC logo to cover page. Date updated (footer)	Jun-25
Hypoperfusion	Removed “use for symptomatic bradycardia” from hypoperfusion guideline. Changed name of Perfusion guideline to “Hypoperfusion”	Jun-25
Peripartum Hypertension	Changed name of Eclampsia guideline to “Peripartum Hypertension” Updated Pre-Eclampsia/Eclampsia guideline to include additional treatments	Jun-25
BLS Procedures	Added Mechanical CPR Added KangooFix Added Pedi-Mate Slushman Traction - added the device the Cape is using.	Jun-25
Death in the Field	Simplified language. Removed language about when to call for medical orders since this was covered by when you do need to call. Added a Detailed DNRO (Do Not Resuscitate Order) Section - o Emphasis on Avoiding Discontinuation in the Back of the Unit & Maternal Exception - o Changed EtCO₂ Threshold from 10 mmHg to 20 mmHg for Certain TOR Decisions - o Clarified Medical Control Requirements for Final Rhythm - o Updated Point-of-Care Ultrasound (POCUS) Documentation - o Improved DNRO Delay Instruction	Jun-25

Guideline Referenced	Annotated Update Description	Date
Care Disposition	Removed definitions	Jun-25
Drug Reference: general	Removed dosages from pharmaceutical references	Jun-25
Drug Reference: Ketorolac	Added contraindication for age over 75 y/o.	Jun-25
Drug Reference: Epinephrine 1:10,000	Removed Periarrest information from 1:10,000 Epi since Push-Dose includes this.	Jun-25
Drug Reference: Epinephrine 1mg/mL	Removed infusion indications from 1mg/mL reference and created Infusion reference.	Jun-25
Drug Reference: NEW	Added Acetaminophen, Ancef, and Labetalol to pharmaceutical reference	Jun-25
Traumatic Injury	Added Cefazolin for Open Fractures for adults and Pediatrics. Added TXA for upper respiratory tract bleeding.	Jun-25
Mass Casualty Incidents	Changed the Level 1 & Level 2 verbiage from “Victims” to “Patients”	Jun-25
Reactive Airway Disease	Added TXA for treatment of hemoptysis and respiratory tract bleeding	Jun-25
Transport Destination	Added Pediatric transport PEARL for traumatic injuries under the transportation destination matrix; Added freestanding language; Adjusted stroke language.	Jun-25
Main Campus Hospitals	Added Physicians Regional CB as STEMI capable; Lee and Cape added as primary stroke; Removed ShorePoint Punta Gorda; Renamed Lehigh Regional to new name with HCA. Updated Destination Matrix to include all current stroke center designations	Jun-25
Freestanding EDs	Added HCA Freestanding West Fort Myers (D37).	Jun-25
Emergency Personnel Rehab	Updated guideline with new patient criteria.	Jun-25
Drug Assisted Airway Management	Removed extra sedation dose; added PEARL; increased Ketamine sedation to 150mg.	Jun-25

Guideline Referenced	Annotated Update Description	Date
Pain and Anxiety Management	Added IV Acetaminophen for Adults and Pediatrics.	Jun-25
Overdose & Poisoning	Corrected atropine dose in Cholinergic section to match pharm	Jun-25
Stroke	Added criteria to allow for transport to Primary and Thrombectomy capable stroke centers.	Jun-25
Seizure	Updated formatting and added routes for Lorazepam and Diazepam Increased Adult Dose of Levetiracetam (Keppra) to 2 g. Removed Pediatric dose of Levetiracetam	Jun-25
Tachycardia	Added range for cardioversion of monomorphic WCT Added pediatric dosing options, lidocaine option, and matched timing and dosages with references and Handtevy. Added Medical Control orders for refractory stable tachycardia	Jun-25
Hypo & Hyperthermia	Added pediatric acetaminophen for fever with a PEARL	Jun-25
Allergic Reaction & Anaphylaxis	Added repeat option for Albuterol. Added epinephrine infusion for refractory anaphylaxis	Jun-25
Medical Cardiac Arrest	Adjusted language for pediatric amio to include max dose; added pediatric mag sulfate dose.	Jun-25
Pseudo-PEA	Updated Pseudo-PEA to mirror medication dosages in perfusion guideline	Jun-25
Whole Blood	Added contraindication; added TXA pediatric max dose.	Jun-25
Bradycardia	Matched pediatric atropine language to pharmacy references. Added Push-Dose Epinephrine as a PEARL	Jun-25
Trauma Circulatory Arrest	Added max TXA and calcium for pediatrics.	Jun-25
Spinal Motion Restriction	Updated to reduce use of backboards and collars per trauma service recommendations.	Jun-25
Burns, Electrocution, Smoke Inhalation	Added max dose to pediatric cyanokit.	Jun-25

Guideline Referenced	Annotated Update Description	Date
Whole Blood	Publication of whole blood guideline; grammatical and abbreviation corrections.	Oct-24
Perfusion Guideline	Updated push dose epinephrine dosing for both pediatrics and adults.	Jul-24
Congestive Heart Failure	Removed pressor due to conflict with perfusion guideline.	Jul-24
Traumatic Circulatory Arrest	Renamed guideline, added PEARL to CPR, prioritized bilateral decompression/finger thoracostomy, clarified indications.	Jul-24
Norepinephrine Pharmacology	Corrected dosing to match perfusion guideline.	Jul-24
Push Dose Epinephrine Pharmacology	Updated dosing.	Jul-24
General System-Wide Update	Updated all document dates to 2024; removed crystalloid from most guidelines; aligned all treatments to Paramedic level; restructured 'Universal Care' into new guideline sections.	Mar-24
ALS - Escharotomy Procedure	Added as a new procedure.	Mar-24
NCH North Naples	Added as STEMI receiving center.	Mar-24
Drug Reference: Levetiracetam	Corrected dosing reference.	Mar-24
Spinal Motion Restriction	Replaced 'lacks decisional capacity' with 'altered mental status' per NEXUS.	Mar-24
Pain and Anxiety	Added Ketamine as anxiolytic for procedural compliance; added procedural anxiety indication.	Mar-24
Chest Pain / ACS	Added BLS max dose for Nitro.	Mar-24
Procedural Sedation	Removed non-invasive oxygenation/ventilation from indications; added pediatric IM/IN Ketamine dosing.	Mar-24
Hyperactive Delirium	Added RASS language to guideline.	Mar-24

Annotated Updates

2026

Guideline Referenced	Annotated Update Description	Date
Death in the Field	Updated TOR language for clarity.	Mar-24
Pit Crew	Updated to reflect new training standards.	Mar-24
Abdominal Pain / Pain Management	Added PEARL encouraging analgesia use even if masking concern exists; included nausea/vomiting management.	Mar-24
Universal Care	Specified RASS as sedation scale.	Mar-24
Stroke Alert Destination Marking	Removed 'C' and 'T'; now uses check marks only.	Mar-24
Traumatic Injury	Added PEARL to confirm POCUS prior to decompression, if available.	Mar-24
Vaginal Bleeding	Added postpartum hemorrhage to differentials.	Mar-24
Pre-Eclampsia	Updated PEARL to SBP > 140 or DBP > 90.	Mar-24
DAI Guideline	Added standardized adult dosing.	Mar-24
Medical Cardiac Arrest	Added timing considerations for DSD.	Mar-24

Theory and Operational Guidelines

- Intent and Usage
- Universal Care
- Patient Safety
- Airway, Ventilation, and Oxygenation
- Perfusion Theory
- Patient Autonomy
- Special Medical Guidance
- Care Disposition
- Death in the Field
- Transportation Destination
- Main Campus Hospitals
- Freestanding Emergency Departments
- Air Medical Transport
- Mass Casualty Incidents
- Emergency Services Rehab
- Lee County School Board Accident Waiver
- Patient Restraint

General Guidelines

- Pain and Anxiety
- Procedural Sedation
- Nausea and Vomiting
- Drug-Assisted Airway Management
- Hypo and Hyperthermia
- Hypoperfusion
- Whole Blood

Medical Guidelines

- Abdominal Pain
- Diabetic Emergencies
- Behavioral Emergencies
- Hyperactive Delirium
- Allergic Reaction and Anaphylaxis
- Reactive Airway Disease
- Seizure
- Stroke
- Sepsis
- Overdose Poisoning

Cardiac Guidelines

- Tachycardia
- Bradycardia
- Chest Pain Acute Coronary Syndrome
- Congestive Heart Failure

Resuscitation

- Medical Cardiac Arrest
- Traumatic Circulatory Arrest
- Pit Crew Resuscitation
- Pseudo PEA
- Post-Arrest

Trauma Guidelines

- Spinal Motion Restriction
- Traumatic Injury
- Isolated Spinal Cord Injury
- Isolated Closed Head Injury
- Burn, Electrocution, Smoke Inhalation
- Isolated Eye Injury
- FL Adult Trauma Score Card
- FL Pediatric Trauma Score Card

Environmental Guidelines

- Bites, Envenomation
- Drowning, Submersion
- Dive Emergencies

OBGYN

- Peripartum Hypertension and Eclampsia
- Childbirth
- Vaginal Bleeding

Pharmacology Reference

ALS Procedures

BLS Procedures

The delivery of Emergency Medical Services (EMS) is, by nature, inherently dynamic. Because of this, the Southwest Florida Regional Common Treatment Guideline (SWFLRCTG) is designed to be a clinical job aid and not intended to be an educational document. The SWFLRCTG is a standardized approach to best practice patient care that encompass evidence-based guidelines (EBG). The focus of the SWFLRCTG is patient-centric and supports the evolution of new EMS research. The SWFLRCTG serves as a resource to clinical medicine while maximizing patient care and ensuring patient safety and outcome regardless of existing resources or capabilities.

It is impractical to write a guideline for every condition or specific case. As such, the SWFLRCTG outlines care for a typical case or condition. As a guideline continues, the assumption can be made that previous steps were ineffective, or the patient condition changed. The order of treatment listed may not be appropriate for all situations. In fact, not all procedure options may be indicated in every situation. The provider's clinical judgment, and ability to consult with medical control as needed, must be relied upon to determine which authorized treatment procedure is appropriate for a given condition or situation.

The care concepts and patient safety guidelines are included in each clinical guideline. This reduces the need for reiteration of basic principles, history and physical exam, and other considerations. One fundamental goal of the SWFLRCTG is to promote critical thinking of all pre-hospital providers: thus, developing technicians into clinicians. This development begins with a framework and the most basic element in medicine – History and Physical Exam (H&P). Without this, the provider cannot reasonably determine which guideline to follow. All providers should work collaboratively with the patient's interest at the forefront of every decision.

The organizations that drive the SWFLRCTG are the American Heart Association (AHA), National Association of EMS Physicians (NAEMSP), American College of Emergency Physicians (ACEP), American College of Osteopathic Emergency Physicians (ACOEP), American College of Surgeons–Committee on Trauma (ACS-COT), Lee Health, and regional county EMS agencies.

Goals:

To facilitate appropriate initial assessment and management of any EMS patient and link to appropriate specific guidelines as dictated by the findings within the universal care guideline

The following represents age classification:

- Adult: Age >13 years or signs of puberty
 - Pediatric: Age 1 – 13 years or signs of puberty
 - Infant: Age 1 month – 1 year
 - Neonate: Age birth – 1 month

PEARL | For purposes of admission criteria, Lee Health considers any patient less than 18 y/o to be "pediatric"

PEARL | For purposes of Trauma Alert criteria, adult is age 16 and above, pediatric is age 15 and below

General Actions:**Response**

1. Review dispatch information
2. Consider need for additional resources

Scene Arrival and Size-Up

1. Use appropriate body substance isolation (BSI)
2. Use appropriate personal protective equipment (PPE)
3. Evaluate and ensure scene safety
4. Determine number and location of persons involved versus patients
5. Consider need for additional resources

Patient Approach

1. Determine mechanism of injury (MOI) and/or nature of illness (NOI)
2. If appropriate, begin triage and initiate mass casualty incident (MCI) procedures – RAMP Triage

Primary Assessment and Life-Saving Interventions

1. General Impression – Sick versus Not Sick:
 - A. Appearance
 - B. Work-of-Breathing
 - C. Circulation to Skin
2. Mental Status:
 - A. Awake/Alert
 - B. Responds to Verbal Stimuli (RVS)
 - C. Responds to Painful Stimuli (RPS)
 - D. Unconscious/Unresponsive



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C-A-B, D-E Assessment

3. **Circulation Status:**
 - Central and Peripheral Pulses – present or absent, regular or irregular, rate & quality
 - Absent, Hypotensive or Hypoperfused? Proceed to appropriate Guideline
 - Major Hemorrhage
 - Hemorrhaging? Proceed to appropriate Trauma Guideline
 - Skin - color, temperature, condition
4. **Airway Status:**
 - Natural
 - Artificially Secured
 - Compromised
 - Proceed to Airway | Ventilation | Oxygenation Management Guideline
 - Obstructed
 - Proceed to Airway | Ventilation | Oxygenation Management Guideline
5. **Breathing Status:**
 - Work-of-Breathing
 - Respirations – present or absent, regular or irregular, rate & quality
 - Auscultate Lung Sounds
 - Adventitious? Proceed to appropriate Guideline
6. **Disability Status:**
 - Determine Baseline or Variation from Baseline Mentation
 - Gross Motor/Sensory Function:
 - Moves all extremities? Focal loss/defect? Eyes? Facial symmetry?
 - Blood Glucose
 - Cervical Motion Restriction
 - Pupil Response
 - Glasgow Coma Score
7. **Exposure:**
 - Evaluate illness or injury, remove clothing as necessary
 - Medic Alert bracelets or identification

PEARL | Critically ill or injured patients shall receive initial stabilization and resuscitative measures prior to movement

SAMPLE History and Physical Examination

1. Obtain a SAMPLE and OPQRST History:

Signs/Symptoms	Onset
Allergies	Provocation and Palliation
Medications	Quality
Pertinent Medical History	Region, Radiation, or Referred

Last Oral Intake	Severity
Events Leading To Present Illness or Injury	Timing

2. Conduct an Adult: Head-to-Toes exam or **Pediatric: Toes-to-Head exam**
3. Conduct a focused, detailed or ongoing systems exam:
 - A. Neurological
 - AVPU
 - Stroke Assessment
 - Pain Scale
 - Richmond Agitation-Sedation Scale (RASS)
 - B. Pulmonary
 - Auscultate Lung Sounds
 - C. Cardiovascular
 - D. Gastrointestinal & Genitourinary
 - E. Integumentary
 - F. Musculoskeletal (Trauma Exam)
4. Assess Vital Signs:
 - A. Pulse
 - B. Blood Pressure
 - C. Respirations
 - D. Skin Color, Temperature and Condition
 - E. Capillary Refill
5. Non-Invasive Monitor Assessment (as applicable or indicated):
 - A. Cardiac
 - Limb lead monitoring
 - 12 Lead ECG
 - B. Blood Pressure
 - C. Capnography
 - D. Pulse Oximetry
 - E. Blood Glucose
 - F. Temperature
6. Collect and transport documentation related to patient's history (e.g., emergency information form, medical records, Medic Alert, DNR form, etc.)

Impression

1. Develop a triple differential impression of the case
 - e.g., Altered Mental Status: *Hypoglycemia vs. Stroke vs. Organic Brain Syndrome*
 - e.g., Acute Coronary Syndrome: *STEMI vs. Unstable Angina vs. Pulmonary Emboli*
 - e.g., Shoulder Injury: *Shoulder Fracture vs. Shoulder Dislocation vs. Shoulder Contusion*

Treatment

- Refer to appropriate clinical guideline(s)
- General control measures and principles:
 - A. Establish an airway as prescribed by the Airway | Ventilation | Oxygenation Management Guideline

(continued)

- B. Ensure adequate ventilation as prescribed by Airway | Ventilation | Oxygenation Management Guideline
 - Ventilation target: etCO₂ 35mmHg – 45mmHg; normal capnograph
 - C. Administer oxygen as prescribed by Airway | Ventilation | Oxygenation Management Guideline
 - Oxygenation target: SpO₂ ≥ 92% in COPD patients and ≥94% for all other patients
 - D. Correct tension pneumothorax with pleural decompression
 - E. Correct open pneumothorax with an appropriate vented occlusive dressing
 - F. Establish vascular access as appropriate and indicated for condition
- PEARL | *No medications will be administered directly via medication port or saline lock***

Assign Clinical Priority

1. Priority 1 (Red) — unstable advanced life support patient; requiring immediate emergent medical attention for a life and/or limb threatening illness or injury
2. Priority 2 (Yellow) — stable advanced life support patient; requiring medical attention but not immediately endangering patient's life
3. Priority 3 (Green) — basic life support patient; requiring non-emergent medical attention

Determine Disposition

1. Mode—Consider mode of transport (air, land, water, etc.)
2. Status—Evaluate need for emergent (lights and sirens) versus non-emergent transportation

Communications

1. Notification to the receiving hospital should be made for all patient transports
2. Medical Control consultation (on a recorded line) is encouraged for any out-of-the-ordinary cases

Reassessment

1. Re-vital sign unstable patients at appropriate intervals or at least every 5 minutes
2. Re-vital sign stable patients at a minimum of every 15 minutes
3. A minimum of 2 assessments are required for every patient transport

Goal(s):

To provide a consistent and standardized foundation for patient, provider, department and system safety.

General Actions:**Providers will:**

- maintain a heightened situational awareness for patient and provider safety
- perform only those procedures for which they are educated (trained to perform), certified (competent to perform), licensed (legal authority to perform), and credentialed (medical director authorized to perform)
- be aware of legal issues and patient rights as they pertain to and impact patient care (e.g., Patients with functional needs, Children with special needs, Baker & Marchman Act patients)
- ensure that stretcher safety restraints, side rails, and cot fastener systems are used in accordance with manufacturer recommendations
- be prepared to adjust management and medications based on patient age, ideal body weight, and co-morbid factors (e.g., adrenal and catecholamine deficiencies, autoimmune disorders, heart failure, end-stage renal disease or renal insufficiency, end-stage liver disease, etc.)
- ensure six (6) medication rights before the administration of any medication:
 - 1) Right patient
 - 2) Right drug
 - 3) Right dose
 - 4) Right route
 - 5) Right time
 - 6) Right documentation
- perform a partner cross-check before the administration of any medication and/or critical procedure
- most medications (unless specified differently) in this reference should be dosed (when applicable) based on ideal body weight. Ideal body weight categories can be found in the Handtevy reference where pediatrics is categorized by age and adults are categorized as small adult, large adult. Maximum dose of medication should not exceed the adult dose except where specifically stated in a patient care guideline
- report any medication errors, clinical misadventures, near miss events, or unanticipated patient outcomes immediately to the receiving physician and respective department supervisor(s)

Goal(s):

To provide evidence-based and reasoned logic core principles for Progressive Airway, Ventilation and Oxygenation management.

General Actions:

AIRWAY

Airway management is a clinical mindset and a constellation of skills, tools and techniques that are deployed to establish and/or manage non-natural airways. Airway management is not one treatment modality; it is a progression of interventions ranging from least invasive (BLS) to the most invasive (ALS) as necessary to achieve sufficient ventilation and adequate oxygenation. Maintenance of the airway throughout the prehospital time, including when the patient is moved, is a critical element of care, and is a priority management item for all personnel, not just the paramedic. Loss of the airway due to mechanical displacement of an airway; change in patient condition; aspiration of blood, gastric content, or other fluids; and swelling along the airway, should be anticipated and monitored.

PEARL | When placing an advanced airway, every effort must be made to avoid iatrogenic hyper/hypocapnea, hypotension, bradycardia and SpO2 desaturation events

PEARL | Waveform capnography (etCO2) and pulse oximetry are required for all advanced airway/ventilation cases – colormetric etCO2 device may be used for initial CO2 detection when continuous waveform capnography is not immediately available

Indications for prehospital endotracheal intubation can be narrowed to the following:

- respiratory failure,
- inability to manage secretions with conventional methods,
- inadequate gag reflex resulting in an inability to protect the airway,
- suspicion for laryngeal edema

PEARL | Airway axis alignment is crucial to endotracheal intubation – the “ear hole, to sternal notch” position substantially increases the likelihood of obtaining a better laryngeal view

PEARL | Video Laryngoscopy (VL) is preferential to Direct Laryngoscopy (DL), however all providers should be familiar with the abilities and limitations of both devices

PEARL | An endotracheal intubation attempt is defined as passing the laryngoscope blade and/or endotracheal tube beyond the teeth with the intent to intubate the trachea

PEARL | Paramedics should confirm placement of advanced airways that are not performed in their presence

VENTILATION AND OXYGENATION – AN IMPORTANT RELATIONSHIP

Ventilation is the mechanical aspect of breathing in which air moves into the lungs and CO2 (normal byproduct of metabolism) moves out of the lungs. Proper ventilation requires both adequate tidal volume and respiratory rate. Oxygenation is defined as, “The addition of oxygen to any system, including the human body. Oxygenation may also refer to the process of treating a patient with oxygen, or of combining a medication or other substance with oxygen.”

With ventilation serving as the mechanical means of adding oxygen to the body, the patient must have sufficient oxygen, and the ability for that oxygen to be utilized (O_2/CO_2 exchange). While ventilatory volume and rate are the key components, other factors can affect whether the patient is being adequately oxygenated. Even if the ventilation volume and rate are adequate, every patient must be evaluated for the need to have supplemental oxygen delivered and the most appropriate mechanism for that to occur. Considerations in determining a patient's need for supplemental oxygen are determined from the patient's presenting condition coupled with History and Physical Exam.

The lack of adequate CO_2 causes a drop in the acid levels resulting in alkalosis. Iatrogenic hyperventilation by prehospital providers is very controversial for the following reason. CO_2 is a potent vasodilator. When CO_2 drops because of iatrogenic hyperventilation (aggressive positive pressure ventilation), blood vessels constrict. When arterial vessels constrict, blood flow to vital organs is minimized. In the case of a brain injured patient, iatrogenic hyperventilation will reduce blood flow to the injury/ischemic zone resulting in an increase in morbidity/mortality and poor patient outcome.

When inadequate oxygenation is recognized, it is essential to supplement the patient's oxygen intake. Primary treatment goals for patients suffering from inadequate oxygenation include:

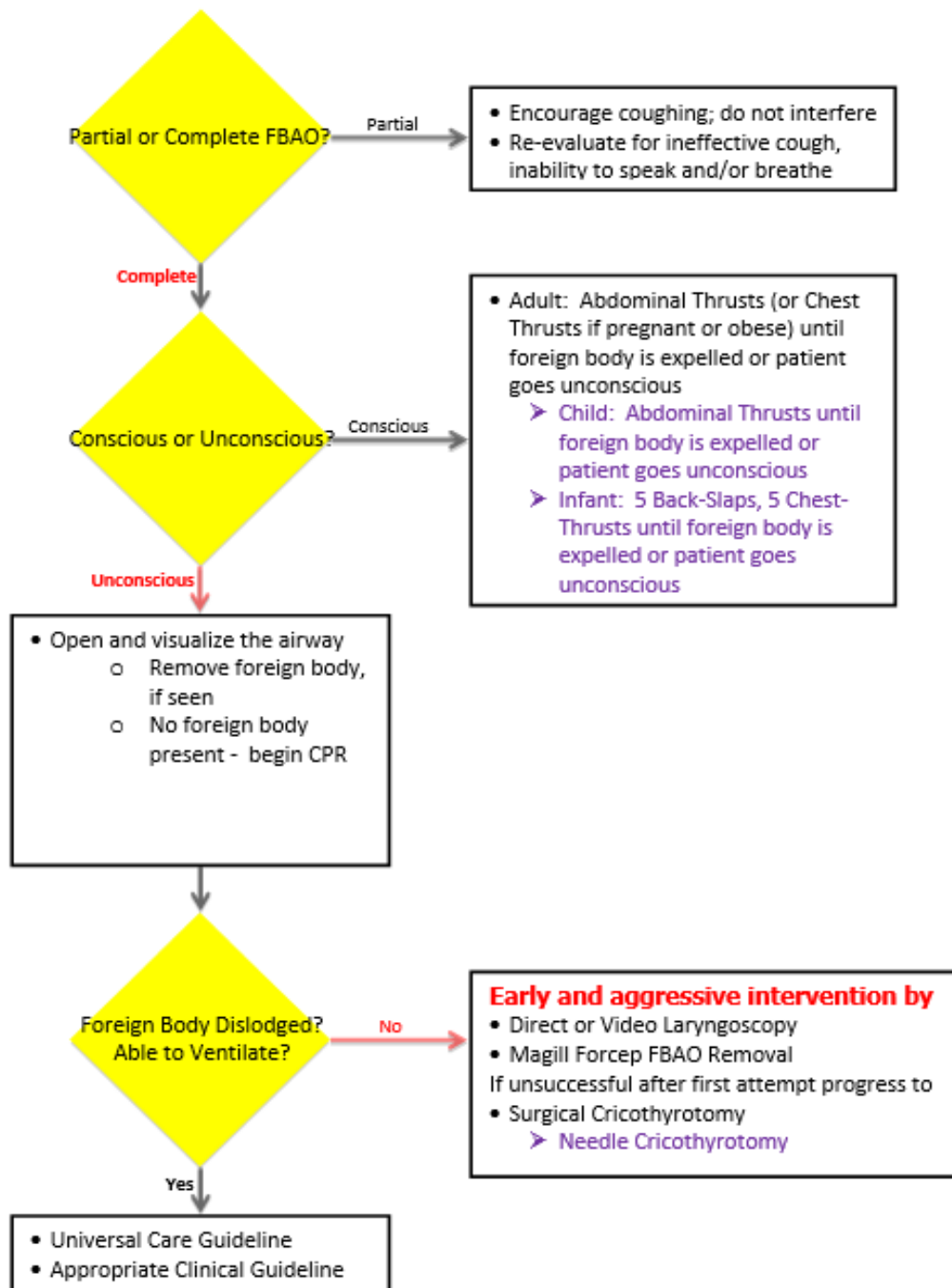
- Preventing or correcting hypoxia
- Optimizing $etCO_2$ and SpO_2
- Minimizing the effects of secondary and/or iatrogenic injury
- Decreasing airway resistance

Positive End-Expiratory Pressure, or PEEP, is an effective way to improve oxygenation in patients that are non-invasively or invasively ventilated. In patients who have respiratory embarrassment and increased work-of-breathing, PEEP stents open closed alveoli and recruits lung thus increasing surface area for gas exchange. PEEP also increases functional residual capacity (FRC) which improves pulmonary reserve between breaths. In prehospital care, the range of PEEP is generally 5 – 15cmH₂O. Providers should routinely start low and titrate as needed. PEEP is not a “if a little is good, more must be better” theory. To that end, tight-lung patients (reactive airway disease) typically do better at 5cmH₂O while wet-lung patients (congestive heart failure/pulmonary edema) may require 7.5 – 15cmH₂O. PEEP greater than 15cmH₂O can result in an increase in intrathoracic pressure thus causing a decrease in venous return and cardiac output.

VENTILATION/PERFUSION – YET ANOTHER CRITICAL RELATIONSHIP

A common pitfall in ventilation is to over-ventilate patients by providing too much tidal volume (V_t) or too fast a minute rate (V_f). The physics that allow mammals to move air in and out of the lungs can also have a major impact on blood circulation. When a normally breathing patient takes a breath, intrathoracic pressure decreases allowing air to be drawn into the lungs because of the pressure gradient. In patients that receive positive pressure ventilation (PPV), intrathoracic pressure is increased as the lungs are inflated. This increase can squeeze the heart and impair filling and forward blood movement. Unregulated PPV will have a dramatic adverse effect on circulation/perfusion. When attention is not paid to PPV volume and rate, the patient can be harmed because of an imbalance between alveolar ventilation and pulmonary capillary blood flow. This imbalance is known as ventilation/perfusion (V/Q) mismatching. Iatrogenic V/Q mismatching can be mitigated using controlled mechanical ventilation (CMV) devices or automated transport ventilators (ATVs). Ventilation volume and rate should be guided using waveform capnography or $etCO_2$ in concert with American Heart Association Guidelines.

Foreign Body Airway Obstruction



Goal(s):

- Correct reversible conditions that compromise perfusion
- Obtain and/or maintain systemic perfusion that sufficiently maintains vital organs

General Actions:

- There are various causes of perfusion deficits. Consider the most appropriate therapeutic intervention for the underlying pathophysiology
 - Distributive shock treatment should focus on correcting the underlying cause and administering pressors
 - Obstructive shock treatment should focus on correction of the obstructive phenomenon when possible (decompression of tension phenomenon) and rapid transport for causes that cannot be corrected (cardiac tamponade, pneumothorax embolism, etc.)
 - Hypovolemic shock treatment should focus on replenishing fluid deficits.
 - Cardiogenic shock treatment should focus on optimizing cardiac output through volume optimization and pressors and correction of arrhythmias
- A mean arterial pressure (MAP) above 65 is effective for most patients to obtain critical organ perfusion. Hemodynamic interventions are generally not necessary under these circumstances.
 - Consider that slow infusions do not have any significant effect on volume expansion, whereas bolus infusions will have a rapid effect on increasing perfusion. Perform reassessment between boluses
 - Vasopressors applied to a hypovolemic system are far less effective than those applied under euvolemic conditions
 - IV fluids may serve to disrupt clot formation and dilution of clotting factors; therefore, permissive hypotension (maintaining SBP greater than 100 and a MAP greater than 65, or to maintain peripheral pulses) should be considered in the setting of bleeding (traumatic and non-traumatic)
 - A single hypotensive episode in the context of trauma carries with it an increased risk of mortality
- Various medications that the patient may already be on or that we may administer cause hemodynamic changes; use caution as appropriate
- Hypoperfusion may be the cause or the effect of an underlying cardiac etiology including inotropic or electrophysiologic deficit
- Consider the potential for positive pressure ventilation to decrease venous return to the heart, thus causing a reduction in cardiac output. Patients undergoing positive pressure ventilation should be watched closely for signs of deterioration from this phenomenon

Goal(s):

- Respect patient autonomy to provide legal and ethical treatment
- Obtain the most appropriate level of consent for treatment given or withheld
- Determine situation-specific patient cognition, competency, and capacity when needed

General Actions:

Definitions

- **Patient:** A patient shall be defined as an individual who meets one or more of the following criteria:
 - 1) Any individual with a medical or traumatic complaint
 - 2) Any individual with an illness or injury
 - 3) Any individual with a new altered mental status
 - 4) Any individual in the same event as a significantly ill and/or injured party (e.g., motor vehicle crash, structural collapse, explosion, toxic fume environment, etc.)
 - 5) Any individual who, at the discretion of the highest medical authority providing direct patient care, demonstrates a high index of suspicion for illness or injury (EMT or Paramedic judgment)
- **Responsible Party:** A designated decision maker (DDM) when a patient is not of decisional capacity or has legally transferred their healthcare decision making to another party (parent, legal guardian, power of attorney, healthcare surrogate, etc.)

Patient Decision Making Ability

Patients have the right to determine if and the details of how they receive healthcare, including pre-hospital care (informed consent). Patients who have appropriate cognition, competency, and capacity to make decisions must be allowed to do so

- **Cognition:** the ability to acquire knowledge and understanding through thought
 - Determined by the ability to answer questions of fact
- **Competency:** the legal right to make a decision
 - Determined legally. Adult (18 y/o or older) or legally emancipated patients are assumed to have legal competency unless it has been removed by a court
 - A parent or legal guardian (responsible party) who refuses care on behalf of their minor child (or children).
 - Legal competency can be revoked for individual patients due to an underlying disease process or revoked for an entire class of individuals (children, prisoners, suicidal individuals, etc.)
- **Capacity:** the ability to weigh the integrated aspects of a decision and express a choice
 - Requires the simultaneous understanding of:
 - The current situation
 - The available choices, including the choice to do nothing (alternatives)
 - The predictable outcomes of each available choice (risks and benefits)

Factors Used to Assess Capacity

Capacity is best assessed by non-leading open-ended questions.

- **Orientation:** All patients undergoing the refusal process must be awake, alert and oriented x4 (person, place, time and situation) with the ability to understand the nature and consequences of their actions by refusing evaluation, treatment, and/or transportation
- **Gait and/or Coordination:** Staggering gait, or inability to stand and ambulate may indicate an impairment that alters decision making capacity
- **Speech Pattern:** Slurred, incoherent or otherwise inappropriate speech patterns may indicate an impairment that alters decision making capacity
- **Insight and Judgment:** Patients must express good insight into the nature of their condition, and convey a reasonable plan to deal with their condition
- **Psychiatric Decompensation:** Patients experiencing suicidal or homicidal ideations or gestures, audio or visual hallucinations, and/or other forms of delusional behavior that alters decision making capacity

Implied Consent

- If a patient is determined to be incompetent and/or lacks decisional capacity, they may be evaluated, treated, and transported under "implied consent" (what the reasonable individual would consent to under the same circumstances)
- If the patient is evaluated, treated and transported on the basis of implied consent, providers should use reasonable measures to ensure safe transport to the closest appropriate facility
- Patients who lack one or more aspect of autonomous decision making must have a responsible party identified to make choices on their behalf. If no one else is present (or if those present are unable or unwilling), the EMS provider shall fill the role of decision maker. This is implied consent
- Patients who are unable (or unwilling) to engage in the necessary interaction to determine cognition, competency, and capacity will not be allowed to make autonomous decisions
- Any time actions are taken against a patient's will, the absence of cognition and/or competency and/or capacity must be clearly documented
- Healthcare systems, services, and individual providers are not obligated to offer or provide care that they believe is harmful to the patient (the risk outweighs the benefit to the patient). However, every effort should be made to respect the patient's wishes
- Special situations
 - In the case of patients who do not have legal competency because their rights have been revoked by a specific court, the individual designated by the court shall be the designated decision maker, and that person must be able to demonstrate situational cognition and capacity (give informed consent).
 - In the case of patients who do not have legal competency because they are children, the parent/guardian/responsible party is the designated decision maker, and that person must be able to demonstrate situational cognition and capacity.
 - In the case of patients who do not have legal competency because they are prisoners (or are under arrest), the responsible law enforcement officer is the designated decision maker, and the law enforcement officer must be able to demonstrate situational cognition and capacity.

- A person with Designated Healthcare Power of Attorney (DPOA) rights may make decisions for a patient but may only do so if the patient is unable to meet the requirements of autonomous decision making on their own. A patient able to make their own decisions is not subject to the decision making of a DPOA.
 - When bystanders report that legal paperwork exists but is not readily available, EMS Providers should trust that this information is correct, unless there are situational cues to suggest that they are being misled.
- Occasionally a patient will put their decision making into writing (especially near the end of life) in the form of legal documents or other paperwork/forms. They may also have conversations with family/friends about how they would like to be treated in various situations. When such wishes are expressed through formal paperwork (DNRO form), these wishes should be immediately respected. When patient wishes are expressed through other forms of communication (other written advanced directives – POLST, 5 Wishes, etc., or through family/friends), it is the responsibility for the EMS providers to contact medical direction for consult.
- Substance use (alcohol, drugs, etc.) does not independently eliminate the possibility that a patient can have intact decision-making abilities. However, all necessary components of autonomous decision making must be clearly demonstrated if that patient is to be permitted to make decisions
- All levels of EMS providers are empowered to make the determinations outlined in this care concept but should exercise caution as the complexity and/or severity of the situation escalates in regards to risks, benefits, and alternatives. Consult with a higher-level provider as needed

Emancipation

- Medical: A female less than 18 years of age who is unmarried, pregnant and/or has a minor child may consent to medical care relating to her pregnancy and can make medical decision on behalf of the unborn or born child
- Legal: A person less than 18 years of age but at least 16 years of age who is married, enlisted in military service or has been declared emancipated by court order

PEARL | No minor less than 16 years of age can be emancipated in Florida; pregnant minors still have consent and autonomy related to her pregnancy or her child

Goal(s):

To provide operational clarity that providers may continue, honor, or facilitate previously prescribed therapies and established medical plans for patients with chronic or specialized medical conditions, even when those therapies are not explicitly listed within an individual treatment guideline.

This guideline ensures continuity of care, respect for patient autonomy, and alignment with medical direction.

SCOPE AND SAFETY CONSIDERATIONS

This guideline does not authorize:

- Initiation of novel therapies outside scope.
- Administration of medications not prescribed to the patient.
- Deviation from statutory DNR or maternal exception requirements.

This guideline supports:

- Established patient:physician relationship
- Continuity of care.
- Respect for patient autonomy.
- Individualized medicine within structured medical oversight.
- Specialized care for medically complex populations.

General Actions:

1. EMS providers may administer or assist with previously prescribed therapies for chronic medical conditions when:
 - The therapy is consistent with the patient's established medical plan.
 - The therapy is available to the patient at the time of EMS encounter.
 - There are no contraindications identified during the current assessment.
 - The therapy falls within Florida scope of practice and agency authorization.
2. When a chronic therapy or specialized intervention is not specifically listed in an individual SWFL common treatment guideline, the absence of listing does not automatically prohibit its continuation if it is:
 - Previously prescribed.
 - Medically indicated.
 - Consistent with patient-specific documentation or established plan of care.
3. When uncertainty exists, providers shall contact Medical Direction.

STARS (Special Needs / Individualized Care Plans).

There may be other specialty conditions and patient registries that provide medical guidance by physicians that have primary responsibility for a patient with an ongoing medical condition. Guidance is contained below. If there are any questions about the validity of the medical guidance, contact Medical Direction.

1. Patients enrolled in STARS or similar individualized care programs may have condition-specific treatment plans, written by physicians who have primary responsibility for the patient, and confirmed by EMS Medical Direction
2. If a STARS plan differs from standard dosing within the Common Treatment Guidelines, the individualized plan should be followed, including recommended doses
3. If documentation is unavailable or unclear, contact Medical Control.

Documentation shall include:

- Confirmation of STARS enrollment.
- Specific elements of the plan followed.
- Any deviations and rationale.

VENTRICULAR ASSIST DEVICE (VAD) PATIENTS

Patients with implanted ventricular assist devices require specialized management.

1. Providers shall:
 - Follow manufacturer best practices.
 - Utilize the patient's VAD emergency guidebook when available.
 - Contact the patient's VAD coordinator as early as possible.
2. Traditional assessment parameters (e.g., palpable pulse, standard blood pressure measurement, pulse oximetry) may not apply.
3. Clinical decisions should prioritize:
 - Mental status.
 - Perfusion assessment.
 - Device function indicators and alarms.

If uncertainty exists regarding device management, immediately contact Medical Control and the VAD center.

Goal(s):

- To provide guidelines and definitions for treatment and transport.
- To establish guidelines for the management and documentation of situations in where patients or potential patients refuse evaluation, treatment, and/or transportation to a hospital in accordance with state and local statute

General Actions:

Disposition Determination

There are three components to a valid refusal of care and/or transport. In the absence of any of these components, the refusal can be deemed legally invalid; thus, resulting in high liability for the providers, their respective Department, and their respective medical director. The three components are competence, capacity, and informed decision.

- Informed Decision as defined by:
 - Patients or Responsible Party must be fully informed about their medical condition; the risks and benefits associated with the proposed treatment or transportation; and the risks associated with refusing evaluation, treatment, and/or transportation.

Patients Able to Refuse Care

- Must be competent
- Must have decisional capacity
- Must be informed of the risks associated with their desired disposition

Patients NOT Able to Refuse Care

- Incompetent – less than eighteen (18) years of age or not legally emancipated
- Lacks Decisional Capacity as defined

Refusal of Care Procedure

- Perform a Primary Assessment, History and Physical Examination, including Vital Signs
- Fully inform the patient (or Responsible Party) about his/her medical condition, the risks and benefits associated with the proposed treatment and the risks associated with refusing evaluation, treatment, and/or transportation

PEARL | Providers should use a recorded phone-patch when a Responsible Party is not on-scene for an incompetent (minor) refusal event if resource is available.

- Ensure the patient or responsible party fully understands the potential consequences of their decision.

PEARL | This is best assessed via non-leading open-ended questions:

“Can you explain back to me the potential risks of refusing transport to the emergency room?”

- Attempt to convince the patient or responsible party to consent, including enlisting the help of family, friends, or supervisor.

- If the patient is persistent in refusing EMS transport, assist them with an alternative plan for care
- Where it is possible, patients will be left in the care of family, friends, or responsible parties
- Document the patient care/transport refusal completely and accurately

Refusal of Care Documentation

- Document the patient's condition
- Document a full assessment and/or specific assessments that were refused
- Document facts demonstrating legal competency and decisional-making capacity
 - Include steps used to determine decision-making and medical capacity
 - Include the understanding of the consequences of refusing care/transport
- Document the strategies used by the provider(s) to convince the patient to make the most appropriate treatment/transport decision
- Document how the patient was informed to seek immediate medical attention or call 9-1-1 if they need EMS or if their condition changes or worsens. Document their acknowledgement.
- The patient's alternative to EMS transport shall be documented (POV, primary physician, etc)
- Patient (or DDM) must sign the patient refusal form
 - If the patient (or DDM) refuses to sign, document the circumstances
- The primary provider must sign the patient refusal form
- A witness must sign the patient refusal form

Goal(s):

This guideline is divided into three separate sections that cover the different situations involving death in the field. All patients found in cardiac arrest will receive cardiopulmonary resuscitation unless an exception is met as outlined in the following sections:

- Do Not Resuscitate Orders (DNRO)
- Determination of Death
- Termination of Resuscitation

General Actions:

DO NOT RESUSCITATE ORDERS (DNRO)

Legislative authority; under Florida Administrative Code (FAC)

64J-2.018 Do Not Resuscitate Order (DNRO) Form and Patient Identification Device.

- A valid DNRO indicates the patient does not wish to be resuscitated in the event of cardiopulmonary arrest.

PEARL | *Every effort should be made to follow the patient and/or family's requests.*

Patients with impending cardiopulmonary arrest shall not be withheld care without medical control order. Providers shall consult medical control to discuss the best course of care for the patient up to and including the following below:

Comfort Care Measures

- Positioning & Basic Airway Maintenance (suctioning secretions, ensuring patient comfort).
- Active Airway Management (e.g., bag-valve-mask ventilation).
- Cardiac medications intended to support circulation (e.g., epinephrine, vasopressors).
- Pain Management (e.g., opioids, NSAIDs).
- Oxygen Therapy (low-flow or high-flow as needed).
- IV fluids for hydration or symptom relief (not for resuscitation).
- Hemorrhage Control (direct pressure, wound dressings).
- Emotional Support & Comfort Measures (support for patient and family).

In cases of uncertainty about DNRO validity (e.g., from another state) or verbal reports of an existing DNRO without physical documentation, providers should initiate resuscitation and contact medical control for guidance.

DETERMINATION OF DEATH

The EMT or paramedic may determine that the patient is dead/non-salvageable and decide not to resuscitate the patient under the following guidelines.

- The patient may be determined to be dead/non-salvageable and will not be resuscitated or transported if all five (5) presumptive signs of death and at least one (1) conclusive sign of death are identified.
 - The five presumptive signs of death that MUST be present are:
 - Unresponsiveness
 - Apnea
 - Pulseless

- Absent cardiac movement confirmed by ultrasound or absent heart tones auscultated by stethoscope, if ultrasound is not available (*60 seconds at Point of Maximal Impulse*)
- Absent pupillary reflexes
- In addition to the five presumptive signs of death, at least one (1) of the following conclusive signs of death **MUST** be present:
 - Injuries incompatible with life (e.g., decapitation, massive crush injury, incineration, multiple penetrating injuries to head/torso, or penetrating or blunt injury with evisceration of brain, heart, or lung)
 - Tissue decomposition
 - Rigor mortis (post mortem rigidity)
 - Livor mortis (post mortem lividity)
 - Algor mortis (post mortem coldness)

PEARL | Patients with suspected hypothermia, barbiturate overdose, or electrocution require resuscitation, unless they have injuries incompatible with life or tissue decomposition

PEARL | The local law enforcement agency that has jurisdiction will be responsible for the body once death has been determined. The body is to be left at the scene until a disposition has been made by the Medical Examiner's Office or the local jurisdiction.

TERMINATION OF RESUSCITATION (TOR)

- Resuscitation that is started in the field by EMS personnel cannot be discontinued without an order from medical direction except where specified. EMS personnel are not obligated to continue resuscitation efforts that were started inappropriately by others at the scene.

Criteria For Termination of Resuscitation

1. Duration of Attempts
 - Appropriate BLS and ALS interventions must have been provided for at least 20 minutes on scene unless the scene conditions warrant (some patients require a longer attempt).
2. Reversible Causes
 - Consider and address any H's and T's
3. Assessment of Cardiac Activity
 - Absent cardiac movement confirmed by POCUS
 - If POCUS is unavailable, absent heart tones on auscultation (60 seconds at the Point of Maximal Impulse)
4. Final Rhythm & EtCO₂
 - Medical Control Orders Required:
 - Final rhythm is shockable (VF/VT)
 - Final rhythm is PEA > 20 bpm
 - Final EtCO₂ is ≥ 20 mmHg.

PEARL | If a recording feature is available, any POCUS used to confirm lack of cardiac movement must be recorded if feasible, and the procedure documented on the patient chart.

- **Place of Business or Public Place:**
 - Resuscitations should not be terminated in a business (e.g., retail store, restaurant) or public space (e.g., beach, park, place of worship, stadium) unless the scene is deemed a crime scene, or the patient cannot be moved to a location where the continued presence doesn't severely disrupt business/public operations.
- **Transport Considerations:**
 - Providers should avoid discontinuing resuscitation in the back of the unit if possible.
 - Maternal resuscitations (pregnancy >20 weeks) shall not be terminated in the field.

FIELD PROCEDURE WHEN TERMINATING

- Provide compassionate care and appropriate grief support to the patient's family, bystanders, or others at the scene. Offer referral resources as available.
- Once death is determined:
 - Cover the body with a sheet or other suitable item, unless the death is a suspected crime scene or otherwise directed by law enforcement not to cover.
 - Do not remove property from the body or the scene.
 - Immediately notify law enforcement if not already present and remain on scene until they arrive.

PEARL | The local law enforcement agency with jurisdiction is responsible for the body once death is determined. The body is generally left at the scene until a disposition is made by the Medical Examiner's Office or local jurisdiction.

Medical Examiner Notification

- If law enforcement has not committed to contacting the ME, make contact with the Medical Examiner's office.

Documentation

- Complete an electronic patient care report (ePCR) as soon as possible, documenting the criteria for TOR, times and interventions, final rhythm, EtCO₂ measurements, presence/absence of cardiac ultrasound activity, and any instructions from Medical Control.

If the Medical Examiner's Office Accepts Jurisdiction

- All disposable medical equipment (e.g., artificial airways, ECG/defibrillator pads, vascular lines) must remain in place.

Goal(s):

- First consideration: patients shall be transported to a local facility of their choice
- Second consideration: patients should be transported to the closest appropriate facility for treatment of their primary illness and/or injury

General Actions:

- Mode of transport is determined by the highest medical authority providing direct patient care; it should not be determined by any other emergency responder(s), bystander(s), or family member(s)
- Status of transport (lights and siren use) is determined by the EMS Transport Provider, with the highest medical authority providing direct patient care. The decision to run lights and sirens should be justified by the need for time sensitive medical intervention that is beyond the capabilities of the transport unit.
- **Freestanding Emergency Departments:**
 - Freestanding emergency departments are not properly equipped to handle patients that require active sitter assistance (Baker Acts, potential Baker Acts, or Marchman Acts). If in doubt, providers are encouraged to contact the receiving facility for consultation regarding patient acceptance.
- **Cardiopulmonary Arrest:**
 - Patients, regardless of age, who are transported in cardiopulmonary arrest or deteriorate to cardiopulmonary arrest in transit, shall be transported to the closest facility.
- **Cardiac:**
 - *STEMI Alert*, Acute Coronary Syndrome (ACS), and Return of Spontaneous Circulation (ROSC) patients shall be transported to the closest STEMI/Percutaneous Coronary Intervention (PCI) facility.
- **Stroke: VAN**
 - Positive Stroke Alert patients and suspected hemorrhagic strokes shall be transported to the closest Comprehensive Stroke Center (CSC) or Thrombectomy Capable Stroke Center (TSC). All other stroke alerts shall be transported to the closest stroke center (primary or comprehensive).
- **Trauma:**
 - *Trauma Alert* patients, regardless of age or pregnancy status, shall be transported to the closest trauma center.
- **Trauma Arrest:**
 - Asystolic trauma arrests should not be transported.
 - If transport is indicated, initiate immediate transport after critical life-saving interventions. Do not remain on scene to continue resuscitation.
 - Destination:
 - **Trauma center ≤ 15 minutes:** Transport to the trauma center.
 - **Trauma center > 15 minutes:** Transport to the closest appropriate main campus hospital.

- **Sepsis:**
 - *Sepsis Alert* and severe sepsis patients shall be transported to any admit facility.
- **Obstetrical/Gynecology:**
 - Obstetrical patients ≥20 weeks gestation shall be transported to an OB/GYN receiving facility; high-risk obstetrical patients shall be transported to a neonatal intensive care receiving facility.
- **Pediatric:**
 - Children that may require admission should be taken to the children's hospital. **Pediatric:** Children that may require admission should be taken to the children's hospital.
- **STARS Patients:**
 - STARS patients should be taken to Golisano Children's Hospital
- **Hazmat:**
 - Patients exposed to hazardous materials can be transported to any facility following appropriate prehospital decontamination. If there are multiple contaminated patients, they should be removed to the least number of hospitals that can manage them, to avoid contaminating multiple facilities.
- **Hyperbaric:**
 - Patients with a high probability for hyperbaric oxygen therapy can be transported to any facility.
- **Envenomation:**
 - Venomous snake and spider bites/stings can be transported to any facility. Mammal and marine bites/stings can be transported to any facility.

Main Campus Hospitals

Lee County	STEMI Alert/PCI	Stroke Alert	Trauma Alert	Pediatric Admit	OB/GYN	Neonate	Oncology Admit	Helipad
Lee Memorial Hospital D1 ED Phone: 239.343.2329		✓P						✓
Golisano Children's Hospital of SWFL D2 ED Phone: 239.343.6258				✓		✓		✓
HCA Lehigh Hospital D3 ED Phone: 239.368.4410		✓P						✓
Cape Coral Hospital D4 ED Phone: 239.424.2222		✓P			✓			✓
Gulf Coast Medical Center D5 ED Phone: 239.343.0485	✓	✓C	✓L2				✓	✓
HealthPark Medical Center D7 ED Phone: 239.343.6279	✓	✓P			✓			✓
Collier County	STEMI Alert/PCI	Stroke Alert	Trauma Alert	Pediatric Admit	OB/GYN	Neonate	Oncology Admit	Helipad
Naples Community Hospital - Baker DT D6 ED Phone: 239.624.2611	✓	✓C					✓	✓
Physicians Regional Medical Center - PR D13 ED Phone: 239.304.4737	✓	✓C						✓
Naples Community Hospital - North D14 ED Phone: 239.552.7709	✓	✓P		✓	✓	✓		✓
Physicians Regional Medical Center - CB D15 ED Phone: 239.354.6191	✓	✓P						✓
Charlotte County	STEMI Alert/PCI	Stroke Alert	Trauma Alert	Pediatric Admit	OB/GYN	Neonate	Oncology Admit	Helipad
HCA Florida Fawcett Hospital D9 ED Phone: 941.627.6131	✓	✓C					✓	✓
AdventHealth Port Charlotte D11 ED Phone: 941.766.4255	✓				✓	✓		✓
Sarasota County	STEMI Alert/PCI	Stroke Alert	Trauma Alert	Pediatric Admit	OB/GYN	Neonate	Oncology Admit	Helipad
Englewood Community Hospital D8 ED Phone: 941.473.5810	✓	✓P						✓
Sarasota Memorial Hospital D16 ED Phone: 941.364.5591	✓	✓C	✓L2	✓	✓	✓	✓	✓
Sarasota Memorial Hospital - Venice D17 ED Phone: 941.303-4390	✓	✓P			✓		✓	✓

Freestanding Emergency Departments

Lee County	STEMI Alert/PCI	Stroke Alert	Trauma Alert	Pediatric Admit	OB/GYN	Neonate	Oncology Admit	Helipad
Coconut Point (Lee Health) 23450 Via Coconut Point, Estero, FL D31 ED Phone: 239.468.0100								✓
NCH Bonita (NCH Healthcare System) 24040 S. Tamiami Trail, Bonita Springs, FL D32 ED Phone: 239.624.6924								No
AdventHealth Cape Coral ER 2521 Del Prado Blvd N., Cape Coral, FL D35 ED Phone: 239.242.1357								No
HCA Freestanding Cape Coral 322 SW Pine Island Rd, Cape Coral, FL D36 ED Phone: 941.249.6858								No
HCA Freestanding West Fort Myers 8919 College Pointe Ct, Fort Myers, FL D37 ED Phone: 239.416.1150								No
HCA Freestanding South Fort Myers 16944 Vintage Commerce Blvd, Fort Myers, FL D38 ED Phone: 239.369.2101								No
Collier County	STEMI Alert/PCI	Stroke Alert	Trauma Alert	Pediatric Admit	OB/GYN	Neonate	Oncology Admit	Helipad
NCH Northeast (NCH Healthcare System) 15420 Collier Blvd, Naples, FL D34 ED Phone: 239.624.8728								No
Sarasota County	STEMI Alert/PCI	Stroke Alert	Trauma Alert	Pediatric Admit	OB/GYN	Neonate	Oncology Admit	Helipad
North Port ER (Sarasota Memorial Hospital) 2345 Bobcat Village Center Rd, North Port, FL D33 ED Phone: 941.257.2800				✓	✓		✓	✓

Goal(s):

- To provide a guideline for the use of air medical transport

General Actions:

Procedure & Criteria

- Place “air medical transport” on standby when:
 - Call information obtained by dispatch suggests the need for air medical transport
- Request “air medical transport” within the first 2 minutes of patient contact for:
 - Priority 1 patients that exceed a ground transport time of 30 minutes unencumbered (not entrapped or otherwise in need of extrication) or,
 - Priority 2 patients that are inaccessible by roads

Notes

1. Any on-scene first responder may request air medical transport
2. Any supervisor, on-scene or not, may request air medical transport based on available information at that time
3. Lee Control may provide information to air medical transport and request an auto-launch
4. Air medical crews may request information from Lee Control and decide to auto-launch
5. After initial assessment, the highest medical authority providing direct patient care should cancel air medical transport if the patient’s condition does not warrant the service or meet the criteria

PEARL | The following patients are not appropriate for air medical transport:

- Cardiopulmonary Arrest patients (CPR in-progress)
- HAZMAT patients (regardless of decontamination status)
- Priority 3 patients
- Imminent OB delivery

Landing Zones (LZ)

1. Fire department personnel are responsible for preparing/securing LZs and assuming the LZ Controller role
 - A. All LZs should be a minimum of 100’ x 100’ (day or night)
 - LZs must be illuminated at the corners with strobe and/or a steady-burn light source
 - Hard surface LZs (highway, parking lots, etc.) are preferential to soft surface LZs
 - B. Once established, the LZ Controller will ensure LZ security the duration of the event
 - C. When requested by the pilot, the LZ Controller will provide a LZ report; this report should include the type of LZ (hard versus soft surface), wind direction and speed and any potential hazards that may be identified from the ground (wires, fences, signs, etc.)
 - D. After the patient has been loaded in the aircraft, the pilot will advise the LZ Controller that the aircraft is ready to depart; the LZ Controller should clear the aircraft for take-off by looking around the LZ and to the sky for any other aircraft traffic in the vicinity

PEARL | If at any time the LZ becomes unsafe for takeoff or landing, the LZ Controller will transmit “ABORT, ABORT, ABORT” over the radio and halt the operation until the unsafe condition is corrected

Goal(s):

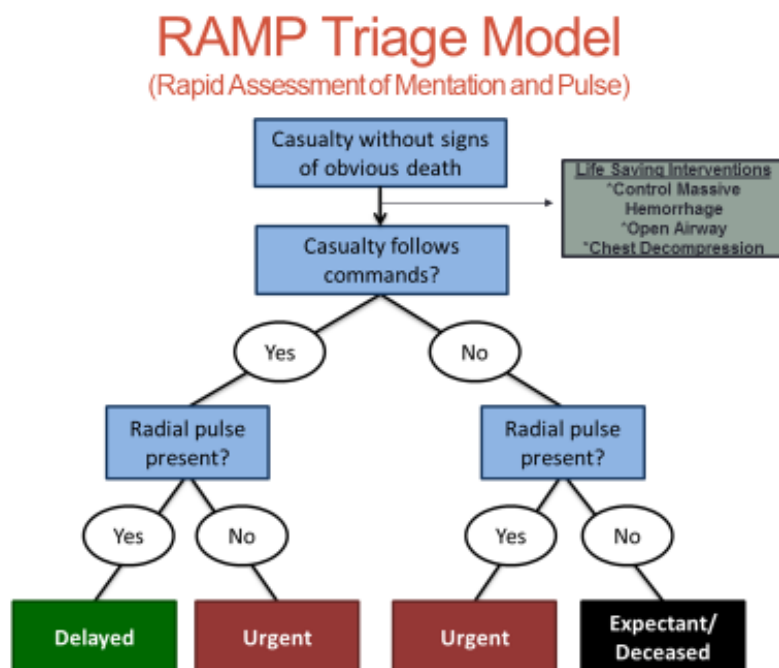
To provide a consistent and standardized foundation for triage, treatment, and transportation of victims of multiple/mass casualty incidents (MCI) that reasonably aligns with the Florida Operations Guide (FOG).

General Actions:

- If first on-scene, ensure radio transmitted scene size-up prior to exiting vehicle
- Establish Incident/Unified Command
- Establish TAC Communications
- Perform a Needs Assessment based upon:
 - Level 1 MCI: 5 – 10 Patients
 - Level 2 MCI: 11 – 20 Patients
 - Level 3 MCI: 21 – 100 Victims
 - Level 4 MCI: 101 – 1000 Victims
 - Level 5 MCI: 1000+ Victims
- Perform RAMP Triage: Rapid Assessment of Mentation and Pulse
 - R – Rapid
 - A – Assessment of
 - M – Mental Status and
 - P – Pulse (radial)

PEARL | Due to the nature and inherent complexity of MCIs, it may become necessary to suspend or modify additional guidelines to address the needs of the incident

PEARL | For ASHER events, patient transports should be prioritized using the ACE acronym (abdominal, chest, extremity)



Goal(s):

- To establish procedures for Emergency Services Personnel Rehabilitation
- This guideline is intended to provide proper rehab for all emergency services personnel at any given time necessary that meets or exceeds NFPA 1584

General Actions:

Responsibilities

- Emergency Services Personnel (ESP):
 - Are responsible for reporting to Rehab (AFTER DECON) when ordered by a commanding officer
 - Are to advise the commanding officer when any member of his/her crew is in need of rehab
- Incident Commander (IC)/Unified Command (UC):
 - Must ensure all personnel receive the proper rest, refreshments, medical evaluation, monitoring, and clearance
- Rehab Supervisor (RS):
 - Should be led by a paramedic who reports directly to the IC and/or the Incident Safety Officer (ISO). Function includes:
 - Locate/establish a rehab location
 - Report to the IC/UC/ISO and obtain rehabilitation requirements
 - Document all member names that report to rehab on SWFL Regional Incident Rehab Worksheet
 - Check vital signs, monitor for heat stress, and evaluate for any medical issues/injury
 - Provide medical care and transportation to medical facilities as required
 - Inform the IC/UC and ISO when personnel require any treatment and transportation to a medical facility
 - Any member with any medical complaints/injuries will be deemed a patient and will require a Patient Care Report to be completed

Establishment of the Rehab Group

- Location:
 - If a specific location has not been designated, the RS shall select an appropriate location based on site characteristics and designations such as fire apparatus, ambulance, nearby garage, or make-shift rehab structure.
- The RS shall notify the IC/UC where the rehab area has been established
- Site Characteristics:
 - Preferably upwind
 - Far enough away from hot zone/tactical area that members may safely remove their Personal Protective Equipment (PPE)
 - Large enough to accommodate the number of personnel expected with a separate area for members to remove PPE
 - Preferably shaded; protected from elements

- Away from exhaust fumes
 - Provide access to SCBA/SCUBA replenishment/refill equipment
 - Easy ingress and egress for ambulance traffic
 - Able to accommodate prompt re-entry back into the operation upon complete rehabilitation
 - Away from spectators and the media
- Resources:
 - Fluids/food – potable drinking water, sports beverages, ice, food, and snacks
 - Medical monitoring equipment
 - Tarps
 - Water supply for active cooling (wet towels, misting fans, ice vests, forearm immersion chairs)
 - Blankets and warm, dry clothing for winter months
 - Chairs (if available)

Rehab Procedure

- Decon prior to rehab
 - No PPE that has been exposed to an immediate dangerous to life or health (IDLH) atmosphere is allowed in the rehab area. Full decon including removing all PPE and wiping face, neck, and hands with wipes
 - If rehab is done outside of an apparatus, personnel may leave trousers on and report to rehab after full decon and must stay outside of apparatus unless immediate medical care is required
- Initiate Medical Monitoring:
- Normal Parameters as noted on the SWFL Regional Rehab Worksheet
- Assess all members and ask about any symptoms of illness or injury
- Any member who is unconscious, displaying altered level of consciousness, reported by another member as acting abnormally, having signs/symptoms of heat stroke or chest pain, will be transported to a medical facility after treatment is initiated
- Initiate Cooling:

PEARL | *Any concern for heat stroke requires immediate water immersion*

- Passive
 - Remove to a cooler environment
- Active
 - Cool, wet towels
 - Forearm immersion buckets/tubs
 - Misting fans
 - Ice vests

- Immersion cooler
- Begin Hydration:
 - Water/fluids (if fluid is too cold it could cause stomach cramp and vomiting)
- Rest time:
 - After 10 minutes of rest, initial vitals will be taken
 - Normal Vital Signs, may be released
 - Abnormal Vital Signs, 10 additional minutes in rehab
 - Abnormal Vital Signs, move to Medical Treatment Area (now a patient)
- Release:
- All members that cannot be cleared shall be reported to the IC/UC and ISO

PEARL | The RS and ISO retain final authority to ground any member

- Completed SWFL Regional Incident Rehab Worksheets shall be signed by the IC/UC or ISO

SWFL Regional Incident Rehab Worksheet

INCIDENT NUMBER:

DATE:

Name						
Assigned Unit						
Decon Complete	Y N	Y N	Y N	Y N	Y N	Y N
Initial Evaluation Time						
Pulse Rate						
SpO2 Level						
SpCO Level						
Injuries	Y N	Y N	Y N	Y N	Y N	Y N
C/O Illness	Y N	Y N	Y N	Y N	Y N	Y N
2nd Eval Time						
Pulse Rate						
SpO2 Level						
SpCO Level						
Injuries	Y N	Y N	Y N	Y N	Y N	Y N
C/O Illness	Y N	Y N	Y N	Y N	Y N	Y N
3rd Eval Time						
Pulse Rate						
SpO2 Level						
SpCO Level						
Injuries	Y N	Y N	Y N	Y N	Y N	Y N
C/O Illness	Y N	Y N	Y N	Y N	Y N	Y N
Initials of Rehab Officer						

VITALS TAKEN AFTER 10 MINUTES OF REHAB AND REPEATED AS NEEDED EVERY 10 MINUTES

SIGNS/ SYMPTOMS THAT REQUIRE TRANSPORT AFTER TREATMENT

Unconscious, Altered Mental Status, Chest Pain, Heat Stroke Symptoms

SIGNS/ SYMPTOMS TO CONSIDER TRANSPORT TO ED

SOB

Dizziness

Weakness

Significant Cramps

Nausea

Vomiting

Sustained heart rate over 120 BPM or irregular

SpO2 below 91% on room air

SpCO over 5%

****NO PERSON SHALL BE RELEASED FROM REHAB UNTIL CLEARED BY THE REHAB OFFICER****

The IC/ISO signing below overrides the recommendation made by the Rehab officer on member(s) _____, and takes full responsibility for that decision.

IC/ISO (Print): _____ (Sign): _____

Rehab Officer (Print): _____ (Sign): _____

Goal(s):

- To establish a guideline for the management and documentation of accidents/crashes involving a Lee County School Board vehicle with Lee County School Board students and/or employees on-board

General Actions:**Definitions**

- **Lee County School Board Administrator/Representative:** An administrator/representative of the Lee County School Board who is dispatched to the scene of all accident/crashes involving a Lee County School Board vehicle and is responsible for and assumes custody of the students on the bus
- **Lee County School Transportation Accident-Student Responsibility Affidavit:** The authorized 2-part form used for non-patient deemed students who will be remaining in the custody of the Lee County School Board
- **Legal Custodian:** 1) School Administrator/Representative, 2) Parent or legal guardian (responsible party)
- **Patient:** a patient shall be defined as an individual who meets one of more of the following criteria:
 - ❖ Any individual with a medical or traumatic complaint
 - ❖ Any individual with an illness or injury
 - ❖ Any individual with a new altered mental status
 - ❖ Any individual in the same event as a significantly ill and/or injured party (e.g., motor vehicle crash, structural collapse, explosion, toxic fume environment, etc.)
 - ❖ Any individual who, at the discretion of the highest medical authority providing direct patient care, demonstrates a high index of suspicion for illness or injury (EMT or Paramedic judgement)
- **No Care Required:** Unit arrives on-scene and the Person Involved (PI) does not meet “patient” criteria as prescribed above

Procedure

- 1) All Lee County School Board students and employees involved in an accident/crash while on a Lee County School Board vehicle shall be evaluated as prescribed by the Lee County Common Treatment Guidelines
- 2) Students that do not meet patient criteria, in accordance with the definition, may be left in the custody of a legal custodian
- 3) Students that are not patients or transported shall have their names printed on the Lee County School Transportation Accident-Student Responsibility Affidavit form
- 4) The Lee County School Transportation Accident-Student Responsibility Affidavit form will be filled out completely; including the bus number and school name
- 5) A legal custodian on-scene shall print their name and sign the form at the bottom acknowledging custody of the students
- 6) If multiple Lee County School District vehicles are involved, a separate Lee County School Transportation Accident-Student Responsibility Affidavit form shall be completed for each vehicle
- 7) All Lee County School Transportation Accident-Student Responsibility Affidavit forms shall be scanned and attached to an electronic patient care report (ePCR) for the department/service handling the event – Disposition: No Care Required
- 8) Any Lee County School Board student and employee that is deemed to be a patient, shall have a separate ePCR completed – regardless of the disposition (Transport, No-Transport or Refusal of Care)

Lee County School Board Accident Waiver

2026

Agency _____ PCR/RUN # _____ Date _____	
School _____ Bus # _____	
The students listed below have been evaluated by Emergency Responders and it has been determined that no complaints or injuries were found present at the time of exam, thus the need for transport to an Emergency Department by ambulance was deemed unnecessary. The below signed takes legal custody of students listed below and hereby releases and holds harmless Emergency Medical Service (EMS), The EMS Care Providers, The EMS Medical Director(s), the responding Lee County Fire/Rescue Districts(s), the Lee County Board of County Commissioners, the City of Cape Coral, the City of Ft. Myers, and the Medical Control Physician(s) from any liability for any medical consequences, which may result in any way related to the non-transport of listed students.	
1.	21.
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13.	33.
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20.	40.
SCHOOL BOARD REPRESENTATIVE	RESCUE SERVICE REPRESENTATIVE
Printed Name _____	Witness _____
Signature _____	Signature _____

Goal(s):

- To establish a guideline for the management and documentation of restraining patients
- Primary consideration: The use of patient restraints is authorized in all instances where a patient's behavior may jeopardize the safety of the patient or crew
- Secondary consideration: Restraints may be used when a patient lacks decisional capacity to make rational decisions and exhibits behavior that may disallow necessary medical treatment

General Actions:

- Crew safety
- Request law enforcement
 - PEARL | *Carefully evaluate the risk-benefit of mechanical patient restraint versus chemical/pharmacological restraint*
- When appropriate, attempt less restrictive means of management including, verbal de-escalation

Patient Positioning

- Patients will be restrained in the supine, head-up position
- Patients may be restrained in a lateral recovery, head-up position as an alternative
- Patients will be mechanically restrained using a commercial soft restraint system or, if in custody, hand-cuffs or shackles as deemed appropriate by law enforcement (handcuffs on patients require law enforcement presence)
- Patients will never be restrained in the prone position

Assessment and Documentation

- When a patient is restrained, the restraints shall be placed only tight enough to secure the extremity without compromising neurovascular function. Providers will provide ongoing monitoring of the patient, so that breathing status is not compromised. Distal neurovascular function shall be checked and documented after application and every 10 minutes thereafter using the following test procedures:
 - Pulse – upper and lower extremities must result in peripheral perfusion: distal pulses and capillary refill time of less than 2 seconds
 - Motor – Grip strength and the ability to move distally should be equal and strong on most patients
 - Sensation – upper and lower extremities must have good sensation and absence of numbness
- The reason for restraining a patient and the results of all the above tests shall be documented in the patient care report
- Grip strength, sensation and capillary refill tests are to be performed and the results documented every 10 minutes
- In the event of a short transport time, the results of a minimum of 2 sets are to be documented with one set to be completed upon arrival at the receiving facility

Hospital Notification

- The receiving facility shall be notified prior to arrival that a patient is in restraints

Differential Impressions:

- Musculoskeletal Pain
- Skin/Integumentary Pain
- Abdominal Pain
- Procedural anxiety
- Neurogenic Pain
- Sickle Cell Crisis
- Intraosseous Device Pain
- Severe Anxiety

Clinical Management Options:

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- Theory and Operational Guidelines
- Ice pack therapy (for simple Musculoskeletal & Skin/Integumentary Trauma Pain)

PEARL | *Pain medication should not be held in fear of masking an injury/illness*

- Acute opiate pain analgesia

Primary option	Substitute 1	Substitute 2
Fentanyl 1mcg/kg IV, IO, IM, IN every 10min PRN Pediatric: 1mcg/kg IV, IO, IM, IN; every 10min PRN	Hydromorphone 1mg IV, IO, IM, IN every 20min PRN Pediatric: 5mcg/kg IV, IO, IM, IN every 20min PRN	Morphine 0.2mg/kg IV, IO, IM, IN every 20min PRN Pediatric: 0.1mg/kg IV, IO, IM, IN every 20min PRN

- Acute non-opiate pain analgesia

PEARL | *Ketorolac is contraindicated in patients > 75 years old.*

Option 1	Option 2	Option 3
Acetaminophen 1g IV, IO over 15min Pediatric: 15mg/kg IV, IO, PO max 500mg over 15min if IV, IO	Ketorolac 15mg IV, IO, IM Pediatric: 0.5mg/kg IV, IO, IM (Max 15mg)	Ketamine 0.2mg/kg IV, IO, IM, IN every 20min Pediatric: 0.2mg/kg IV, IO, IM, IN every 20 min

PEARL | *Acetaminophen PO may be administered to any pediatric that maintains the ability to swallow.*

- Intraosseous Device Comfort:
Lidocaine 40mg IO over 2min; allow to dwell in IO space for 60 seconds
Pediatric: 0.5mg/kg (40mg maximum) IO over 2min; allow to dwell in IO space for 60 seconds
- Procedural anxiolytic
Ketamine 0.2mg/kg IV, IO
- Severe Anxiety:
Midazolam 2mg IV, IO, IM, IN; repeat every 10 minutes PRN

Medical Control Actions/Orders/Requests:

- Consult as needed/indicated
- Pediatric severe anxiety**

Clinical Indications:

- Synchronized Cardioversion
- Complex Extremity Entrapment/Extraction
- Combative Closed Head Injury/Traumatic Brain Injury
- Transcutaneous Pacing

Clinical Management Options:

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- Theory and Operational Guidelines

- Sedation

Option 1	OR	Option 2
Ketamine 1mg/kg IV, IO, every 10min PRN or Ketamine 2mg/kg IM, IN Pediatric: 1mg/kg IV, IO, every 10min PRN Or 2mg/kg IM, IN		Midazolam 5mg IV, IO, Or Midazolam 10mg IM, IN every 5min PRN. Pediatric: 0.2mg/kg IV, IO, IM, IN (Max of 2mg) every 5min PRN

PEARL | Just like another vital sign, RASS should be considered as well as documented multiple times whenever someone is sedated (from medication or medical condition) or aggressive. This should also be considered assessment of your intervention.

 Richmond Agitation Sedation Scale (RASS)		
Scale	Label	Description
+4	Combative	Violent, immediate danger to staff
+3	Very agitated	Pulls or removes tube(s) or catheter(s); aggressive
+2	Agitated	Frequent non-purposeful movement, fights ventilator
+1	Restless	Anxious but movements not aggressive, vigorous
0	Alert and calm	Spontaneously pays attention to care giver
-1	Drowsy	Not fully alert, but has sustained awakening (eye-opening/eye contact) to voice (>10 seconds)
-2	Light sedation	Briefly awakens with eye contact to voice (<10 seconds)
-3	Moderate sedation	Movement or eye opening to voice (but no eye contact)
-4	Deep sedation	No response to voice, but movement or eye opening to physical stimulation
-5	Unarousable	No response to voice or physical stimulation

OBSERVATION
VOICE
TOUCH

Medical Control Actions/Orders/Requests:

- For Ketamine emergence phenomenon
Midazolam 5mg IV, IO, IM, IN
Pediatric: 0.2mg/kg IV, IO, IM, IN

Differential Impressions:

- Central Nervous System origins
- Digestive Tract disorder
- Food poisoning/Alcohol use
- Gastrointestinal distress
- Genitourinary origins
- Infectious origins
- Metabolic origins
- Medication/Toxin induced
- Neurological origins
- Oncology origins
- Pregnancy
- Psychological disorders
- Sepsis
- Stroke
- Traumatic Brain Injury
- Viral origins

Clinical Management Options:

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- Theory and Operational Guidelines
- Alcohol pad sniff to reduce nausea, if tolerated
- **Ondansetron 4mg ODT**
Pediatric: 2mg ODT
- Antiemetic

Primary option	Substitute 1	Substitute 2
Ondansetron 4mg IV, IO, IM repeat at 10 minutes if no relief Pediatric: 0.1mg/kg IV, IO, IM	OR	OR
	Promethazine 25mg IV	Droperidol 1.25mg IV, IO, IM

Medical Control Actions/Orders/Requests:

- **Pediatric Promethazine and Droperidol**

Goals:

- Safely and effectively facilitate airway management.
- Optimize first pass intubation success rates by implementing best practices of airway management including patient selection, assessment, pre and post intubation oxygenation, ventilating, and monitoring.

Clinical Management Options:

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- Theory and Operational Guidelines
 - Airway, Ventilation, Oxygenation
 - Patient positioning (airway axis alignment – “ear hole to sternal notch”)
 - Preoxygenation goal: SpO₂ >94% for 3 minutes prior to ETI attempt if possible.
 - NRB at 25L/min with NC under mask using separate O₂ source at 6 L/min
 - Non-invasive mechanical ventilation
 - For apneic patients, BVM at 15L/min with NC at 15L/min under mask with NPA/OPA

PEARL | Continuous patient monitoring (HR, BP, SpO₂, EtCO₂, ECG) pre and post procedure required. Avoid critical values and be prepared to abort procedure and provide BVM ventilations if any critical values occur.

PEARL | A minimum of two paramedics should be on scene for procedure

- Pre-Intubation Sedation

Option 1		Option 2
Ketamine 150mg IV, IO	OR	Etomidate 20mg IV, IO

- Procedural Timeout
 - The preoxygenation goal is SpO₂ >94% for 3 minutes prior to ETI attempt if possible.

PEARL | Pulmonary reserve will be significantly reduced if SpO₂ goal isn't achieved.

- Intubation Paralysis

Option 1		Option 2
Rocuronium 100mg IV, IO	OR	Succinylcholine 150mg IV, IO

- Post-Intubation Sedation

Option 1		Option 2
Re-bolus Ketamine 150mg IV, IO repeat dose at 15 minutes PRN	OR	Fentanyl 1mcg/kg with Midazolam 5mg IV, IO repeat dose at 15 minutes PRN

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
- [Pediatric Drug Assisted Airway Management](#)

Differential Impressions:

- Environmental Hypothermia
- Infection
- Serotonin Syndrome
- Environmental Hyperthermia
- Anticholinergic Syndrome
- Salicylate Overdose

Clinical Management Options:

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C | <ul style="list-style-type: none"> • Theory and Operational Guidelines |
| | | <ul style="list-style-type: none"> • Hypothermia (adults & pediatrics) <ul style="list-style-type: none"> Passive External Rewarming Remove from cold environment, remove wet clothing Hibler's Method of Thermopreservation PEARL Handle gently to reduce lethal arrhythmias • Hyperthermia (adults & pediatrics) <ul style="list-style-type: none"> Cold water immersion PEARL Cold water immersion should not be delayed and is the primary treatment for environmental hyperthermia and should be titrated to an improvement in mental status PEARL Cold water immersion should not be used for febrile patients Passive External Cooling – fans, misting, and/or ice packs to groin, axilla and neck PEARL Withdrawal cooling as mentation improves |
| | | <ul style="list-style-type: none"> • Hypothermia <ul style="list-style-type: none"> Warm Crystalloid Resuscitation 20mL/kg IV, IO as necessary/indicated Pediatric: Warm Crystalloid Resuscitation 20mL/kg IV, IO as necessary/indicated PEARL Rough patient handling may cause ventricular fibrillation PEARL Hypothermia is susceptible to progressive bradycardias |
| | | <ul style="list-style-type: none"> • Fever <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> Option 1
 Pediatric: Acetaminophen 15mg/kg
 PO max of 500mg </div> <ul style="list-style-type: none"> PEARL IV Acetaminophen is a pain medication and is not indicated for adults with a fever. PEARL PO Acetaminophen is indicated for pediatric patients with a fever (> 100.4°F / 38°C) from a potentially infectious source. This is not indicated for environmental hyperthermia PEARL PO Acetaminophen should not be administered if patient has received a full Tylenol dose within the last 4 hours. PEARL PO Acetaminophen may be administered to any pediatric that can swallow. |
- Drug assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Indications:

- Hypoperfusion

Clinical Management Options:

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- Theory and Operational Guideline
- Patient positioning (supine or Trendelenburg)

- Crystalloid Resuscitation 20mL/kg IV, IO** as necessary/indicated

Pediatric: 20mL/kg IV, IO as necessary/indicated

- Pressor to maintain MAP > 65 mmHg:

Option 1

Norepinephrine Infusion

- 0.1 – 0.5**
mcg/kg/min IV, IO
- 0.03 – 0.05**
mcg/kg/min IV, IO

OR

Option 2

Epinephrine Infusion

- 0.1 - 0.5**
mcg/kg/min IV, IO
- 0.03 – 0.2**
mcg/kg/min IV, IO

OR

Option 3

Push dose Epinephrine (0.01mg/mL)

- 2 mL every 2 min**
PRN
- 0.1mL/kg up to 20**
Kg every 2 min PRN

PEARL | Push dose epinephrine may be administered while infusion is prepared

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Indications:

- Hemorrhagic Shock as evident by known bleeding and:
 - Systolic blood pressure (SBP) < 70 mmHg **or**,
 - **Patients ≥ 10 years of age:** < 90 mmHg with heart rate > 110 **or**,
 - **Pediatric patients between the ages of 5 – 9:** SBP < 80mmHg with heart rate > 110 **or**,
 - ETCO₂ < 25mmHg **or**,
 - **Adults:** Shock index > 1 (HR > SBP)
- Traumatic Circulatory Arrest (for adult and **pediatrics** ≥ 5 yo)
 - Witnessed traumatic circulatory arrest by providers

Contraindications:

- Isolated penetrating injury to the cranium, with a GCS of 3

Clinical Management Options:

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C | <ul style="list-style-type: none"> • Hemorrhagic Shock & Traumatic Circulatory Arrest <ul style="list-style-type: none"> ○ Whole Blood 1 unit IV/IO (repeat as needed)
 Pediatric ≥ 5 years old: 10mL/kg IV/IO (may repeat 1x PRN) max total dose 1 unit
 <i>PEARL Medical control orders are required for blood administration in pediatric patients less than 5 years of age</i>
 <i>PEARL Avoid the administration of blood to patients with an isolated head injury,</i> <ul style="list-style-type: none"> ○ Tranexamic Acid 2g IV, IO over 2 min
 Pediatric: 15mg/kg IV/IO max dose of 1g ○ Calcium Chloride 1g IV/IO
 Pediatric: 20mg/kg max dose of 1g |
|-----------------------|--|
- Clinical PEARLs**
- Blood administration requires a dedicated IV/IO line.*
- Monitor for transfusion reactions. If any symptoms occur, stop transfusion immediately and move to Allergic Reaction/Anaphylaxis guideline if necessary.*
- *Fever, chills, hives, or pain at the transfusion site*
 - *Chest pain*
 - *Shortness of breath*
 - *Nausea or vomiting*

Medical Control Actions/Orders/Requests:

- **Pediatric < 5 years old: 10mL/kg IV/IO (may repeat 1x PRN) max total dose 1 unit**
- Consult as necessary/indicated

Differential Impressions:

- Acute Abdominal Syndrome
- Cholecystitis
- Colitis
- Crohn's Disease
- Diverticulitis
- Pancreatitis
- Peptic Ulcer Disease
- Pelvic Inflammatory Disease
- Renal Colic
- Abdominal Aortic Aneurysm
- Appendicitis
- Bowel Obstruction
- Ectopic Pregnancy
- Incarcerated Hernia
- Rupture Ovarian Cyst
- Traumatic injuries
- Urinary Tract Infection

Clinical Management Options:

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C | • Theory and Operational Guidelines |
| | | • General Guidelines |
| | | PEARL <i>Pain medication should not be held in fear of masking an injury/illness</i> |
| | | • Nausea and Vomiting Guidelines |

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Hypoglycemia (blood glucose <60mg/dL)
- Hyperglycemic Insult
- Iatrogenic Hypoglycemia
- Hyperglycemia (blood glucose >300mg/dL)
- Diabetic Ketoacidosis (DKA)
- Hyperosmolar Hyperglycemia State (HHS)

Clinical Management Options:

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- Theory and Operational Guidelines
- General Guidelines
- bG <60mg/dL:
Oral Glucose 30g PO
Pediatric: 15g PO

- bG <60mg/dL

Primary option	Substitute 1	Substitute 2
Dextrose 10% 125 – 250mL (12.5 – 25g) IV titrated to return of normal mental status Pediatric: 5mL/kg IV titrated to return of normal mental status	Dextrose 50% 12.5 – 25g IV Pediatric: 0.5g/kg IV	D5W 250 – 500mL IV titrated to return of normal mental status Pediatric: 5mL/kg IV titrated to return of normal mental status

OR

OR

PEARL | Consider complex carbohydrates after correcting hypoglycemia and patient returns to normal mentation

PEARL | If hypoglycemia resolves but repeat BGL drops within 10 minutes, consider sulfonylurea toxicity/OD which can be lethal for pediatrics

- bG >300mg/dL with vascular access:
Crystalloid Resuscitation 20mL/kg IV, IO; repeat PRN
Pediatric: 20mL/kg IV, IO; repeat PRN

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Mental Illness
- Psychiatric Emergencies
- Substance Abuse
- Baker Act
- Marchman Act

General Actions:

- Theory and Operational Guidelines
- General Guidelines
- Hyperactive Delirium Guideline as necessary/indicated
- Baker Act consideration:

Florida Statute Chapter 394, Part I, is also known as the Florida Mental Health Act. The Baker Act provides legal procedures for patients with known or suspected mental illness. This includes mental health examinations and treatment and provides authorization to police, physicians, mental health professionals and the courts to dictate certain medical care for persons who pose a threat of harm to themselves or to others.

PEARL | Baker Act is not intended for patients who are competent, are without mental illness, have decisional capacity, and have been informed yet still desire to refuse care against medical advice

PEARL | Law enforcement will provide EMS with a Baker Act Form (3052a), and as required for EMS safety, will accompany or follow the ambulance to the hospital

- Marchman Act consideration:
Florida Statute Chapter 397, Part V, provides legal procedures for patients with known or suspected conditions involving substance abuse. This includes mental health examinations and treatment and provides authorization to police, physicians, mental health professionals and the courts to dictate certain medical care for persons who are impaired and pose a threat of harm to themselves or to others or is so impaired that he is incapable of appreciating his/her need for substance abuse services.

PEARL | Marchman Act is not intended for patients who are competent, have decisional capacity, and have been informed yet still desire to refuse care against medical advice

- Florida Statute 401.455, Emergency examination and treatment of incapacitated persons protects providers for caring for patients if:
 - The patient at the time of examination or treatment is intoxicated, under the influence of drugs, or otherwise incapable of providing informed consent as provided in s. 766.103;
 - The patient at the time of examination or treatment is experiencing an emergency medical condition; and
 - The patient would reasonably, under all the surrounding circumstances, undergo such examination, treatment, or procedure

PEARL | Neither a Baker Act or Marchman Act removes the need for medical treatment and assessment and should be completed as soon as possible

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Hyperactive delirium
- Conditions that result in agitated behavior that pose imminent threat or danger to self or others

Clinical Management Options:E
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- Theory and Operational Guideline
- General Guideline

- Dissociation

Primary
Ketamine 4mg/kg IM max dose of 500mg

OR

Option 1
Droperidol 5mg IM, IV, IO

OR

Option 2
Midazolam 5mg IV or 10mg IM

PEARL | *First-line therapy; no absolute contraindications*

PEARL | *Patients shall never be placed prone*

PEARL | *There is a gradient between sedated and aggressive, and this scale is an extremely helpful tool. Just like another vital sign, RASS should be considered as well as documented multiple times whenever someone is sedated (from medication or medical condition) or aggressive. This should also be considered assessment of your intervention.*

 Richmond Agitation Sedation Scale (RASS)		
Scale	Label	Description
+4	Combative	Violent, immediate danger to staff
+3	Very agitated	Pulls or removes tube(s) or catheter(s); aggressive
+2	Agitated	Frequent non-purposeful movement, fights ventilator
+1	Restless	Anxious but movements not aggressive, vigorous
0	Alert and calm	Spontaneously pays attention to care giver
-1	Drowsy	Not fully alert, but has sustained awakening (eye-opening/eye contact) to voice (>10 seconds)
-2	Light sedation	Briefly awakens with eye contact to voice (<10 seconds)
-3	Moderate sedation	Movement or eye opening to voice (but no eye contact)
-4	Deep sedation	No response to voice, but movement or eye opening to physical stimulation
-5	Unarousable	No response to voice or physical stimulation

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- For Dissociation Emergence – unlikely and rare event:

Primary
Ketamine 1mg/kg IV, IO max dose of 500mg

OR

Option 1
Droperidol 5mg IM, IV, IO

OR

Option 2
Midazolam 5mg IV or 10mg IM

PEARL | *Administered 5 – 10 minutes after initial dose for inadequate dissociation*

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
- Pediatric: Contact Medical Control

Differential Impressions:

- Localized Allergic Reaction
- Systemic Anaphylaxis Reaction
- Anaphylactic Shock
- Angioedema
- Systemic Anaphylactoid Reaction
- Anaphylactoid Shock
- Transfusion Reaction

Clinical Management Options:

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- Theory and Operational Guidelines
- General Guidelines
- **EpiPen IM**
Pediatric: EpiPen, Jr. IM
or
- **Epinephrine (1mg/mL | 1:1,000) 0.5mg (0.5mL) IM lateral thigh; repeat as needed**
Pediatric: 0.01mg/kg max of 0.3mg IM lateral thigh; repeat as needed
- Bronchodilator (if indicated)

Primary option

Albuterol 2.5mg AT
Pediatric: 2.5mg AT
May repeat PRN

OR

Substitute

Levalbuterol 0.63mg AT
Pediatric: 0.31mg – 0.63mg AT

- **CPAP 5 – 15cm/H2O PEEP or CPAP with Pressure Support**
- **Epinephrine 0.1 – 0.5mcg/kg/min** (if refractory to second IM dose)
- **Diphenhydramine 50mg IV, IO, IM**
Pediatric: 1mg/kg IV, IO, IM
- **DuoNeb: Albuterol 2.5mg & Ipratropium 0.5mg AT** (if indicated)
Albuterol 2.5mg & Ipratropium 0.5mg AT (as indicated)
- Steroid

Option 1

Methylprednisolone 125mg SIVP, IM
Pediatric: 2mg/kg SIVP, IM
(max 125mg)

OR

Option 2

Dexamethasone 10mg IV, IO
Pediatric: 0.6mg/kg IV, IO, PO
(max 16mg)

Advanced Procedures

- Drug-assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Asthma and Asthma-Like Syndrome
- Chronic Obstructive Pulmonary Disease
- Toxic Inhalation (vapor, fume, or smoke)
- Upper Respiratory or Pulmonary Viral Infection

Clinical Management Options:

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- Theory and Operational Guidelines
- General Guidelines
- Bronchodilator (as indicated)

Primary option
Albuterol 2.5mg AT (may repeat PRN) Pediatric: 2.5mg AT (may repeat PRN)

OR

Substitute
Levalbuterol 0.63mg AT Pediatric: 0.31mg – 0.63mg AT

- Severe respiratory distress pending respiratory failure
Epinephrine (1:1000) 0.5mg IM; may repeat once
Pediatric: 0.01mg/kg max of 0.3mg IM may repeat once
- CPAP 5 – 7.5cm/H2O PEEP or CPAP with Pressure Support

- DuoNeb: Albuterol 2.5mg & Ipratropium 0.5mg AT (as indicated)
Albuterol 2.5mg & Ipratropium 0.5mg AT (as indicated)

- Steroid

Option 1
Methylprednisolone 125mg SIVP, IM Pediatric: 2mg/kg SIVP, IM (max 125mg)

OR

Option 2
Dexamethasone 10mg IV, IO Pediatric: 0.6mg/kg IV, IO, PO (max 16mg) <i>PO preferred if an IV, IO isn't needed</i>

PEARL | Oral steroids in pediatrics is a highly effective treatment option with a similar onset of action as intravenous and is the preferred route.

- Magnesium Sulfate 2g in 100mL D5W IV, IO Infusion over 10 minutes
Pediatric: 50mg/kg in 100mL D5W IV, IO Infusion over 10 minutes

Hemoptysis/Respiratory Tract Bleeding:

- Tranexamic Acid 1g AT

Croup

Option 1
Pediatric: Nebulized Epinephrine (1mg/mL) ≥ 1 year: 5mg (5mL) AT < 1 year: 2.5mg (2.5mL) AT

OR

Option 2
Pediatric: Racemic Epinephrine (11.25mg/0.5mL) ≥ 1 year: 0.5mL diluted in 3mL of NS AT < 1 year: 0.25mL diluted in 3mL of NS AT

- Drug-Assisted Airway Management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Epilepsy
- Cardiac arrest or dysrhythmias
- Closed Head/Traumatic Brain Injury
- Infectious origins (i.e., Febrile)
- Metabolic origins
- Medication/Toxin induced
- Neurological origins
- Oncology origins
- Pregnancy (i.e., Eclampsia)
- Psychological disorders
- Stroke
- Viral origins

Clinical Management Options:

- E M T** **M E D I C**
- Theory and Operational Guidelines
 - General Guidelines
 - Diabetic Emergencies Guideline

- Anti-convulsant

Primary option	Substitute 1	Substitute 2
• Midazolam 5mg IV, IO; may repeat every 5 minutes PRN Pediatric: 0.2mg/kg IV, IO (max of 5mg); repeat every 5 minutes PRN • Midazolam 10mg IM, IN; may repeat every 5 minutes PRN Pediatric: 0.2mg/kg IM, IN (max of 10mg); repeat every 5 minutes PRN	OR • Lorazepam 2mg IV, IO, IM; may repeat once after 10–15 minutes if seizures persist (Max total dose of 8 mg) Pediatric: 0.1mg/kg IV, IO, IM (max of 2mg per dose) repeat every 5 minutes PRN	OR • Diazepam 5mg IV, IO; may repeat every 5 minutes PRN Pediatric: 0.2mg/kg IV (max of 5mg per dose) repeat every 5 minutes PRN

PEARL: For febrile seizures, refer to hyperthermia guideline only after the seizure is managed.

- Drug-assisted airway management

- **Levetiracetam 2g IV, IO infusion**

PEARL: *Levetiracetam should be second line use for prolonged seizures or seizures refractory to second dose of anticonvulsant administration.*

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
- **Levetiracetam for refractory pediatric seizures requires medical control order.**

Differential Impressions:

- Ischemic Stroke
 - Thrombotic
 - Large Vessel Occlusion
 - Transient Ischemic
- Diabetic Emergency
- Hemorrhagic Stroke
 - Subarachnoid
 - Epidural
 - Subdural
- Cardiac event

Clinical Management Options:

- | | | |
|-------------|--------|--|
| E
M
T | M | • Theory and Operational Guidelines |
| | E | • General Guidelines |
| | D | • bG <60 mg/dL:
Diabetic Emergencies |
| | I
C | • Perform Lee County Stroke Assessment/Checklist |

-
- PEARL | Crystalloid Resuscitation is aimed at maintaining cerebral perfusion*
- Nausea & Vomiting Management Guideline as necessary/indicated
- PEARL | Antiemetic therapy is aimed at reducing intracranial pressure*
- Drug-assisted airway management
-

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Ischemic Stroke BEFAST VAN			
Includes: Thrombotic & Embolic Stroke, Small & Large Vessel Occlusion Stroke, and Transient Ischemic Stroke			
Step 1: Perform FAST Exam			
F	● Face	Unilateral facial droop or palsy?	☐
A	● Arms	Unilateral arm or pronator drift?	☐
S	● Speech	Slurred speech, difficulty speaking, inappropriate words, mute?	☐
T	● Time	Last known well <24 hours (**Last known well time: ____:____**)	☐
Any FAST boxes checked = Stroke Alert			
Step 2: Perform VAN Exam			
V	● Visual	Disturbance, double vision, or new onset blindness?	☐
A	● Aphasia	Expressive, Receptive, or mixed?	☐
N	● Neglect	Forced gaze, inability to track, or ignoring one side?	☐
Any VAN boxes checked = VAN Positive Stroke Alert			

Hemorrhagic Stroke		
Includes: Intracerebral, Intracranial, Epidural, Subarachnoid, and Subdural Hemorrhage		
Perform Hemorrhagic History & Exam		
Hemorrhagic	● Worst headache ever?	☐
	● Severe, new onset, nausea and/or vomiting?	
	● Acute hypertension?	
	● Seizure preceding stroke like or focal neurological deficits?	
	● Sudden, unexplained, altered mental status	
Hemorrhagic box checked = Stroke Alert		

Stroke Alert Destinations

FAST POSITIVE ONLY STROKE ALERTS: Closest Stroke Center

FAST POSITIVE ONLY > 24 HOURS: Closest Comprehensive or Thrombectomy Capable Stroke Center

FAST - VAN POSITIVE STROKE ALERTS: Closest Comprehensive or Thrombectomy Capable Stroke Center

HEMORRHAGIC STROKE ALERTS: Closest Comprehensive or Thrombectomy Capable Stroke Center

Differential Impressions:

- Sepsis – qSOFA
- Septic Shock | Distributive Shock (MAP <65mmHg)
- Pneumosepsis
- Meningeal Sepsis
- Gastro-Intestinal Sepsis
- Septicemia
- Urosepsis
- Skin/Wound Sepsis

Clinical Management Options:E
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- Theory and Operational Guidelines
- General Guidelines
- Complete Sepsis Assessment/Checklist (Adult vs Pediatric)

PEARL | *Sepsis Checklist Positive = Sepsis Alert*

- CPAP 5 – 15cm/H2O PEEP or CPAP with PS (max PS 5)
- Crystalloid Resuscitation 20mL/kg IV, IO over 60 minutes
Pediatric: 20mL/kg IV, IO over 60 minutes

PEARL | *Crystalloid is paramount for survival; do not withhold in normotensive patients*PEARL | *Pressor agent second-line therapy for hypotension/shock after crystalloid infusion*

- Drug-assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

ADULT SEPSIS CHECKLIST

Sepsis 3.1 qSOFA + etCO ₂			
Infection	• Suspected	Fever, UTI/ URI symptoms, etc	☐
	• Known	Taking antibiotics, visible, etc	
Sepsis	• Altered Mental Status	Confused off normal baseline, lethargy, etc	Any 2 ☐ ☐
	• Tachypnea	RR ≥ 22/min	
	• Hypotension	SBP < 100mmHg (consider “lower than normal BP”)	
Capnography	• Hypocapnea	etCO ₂ ≤ 29mmHg	☐
All boxes checked = <i>Sepsis Alert</i>			

Differential Impressions:

- Opiate Ingestion
- Cholinergic Exposure
- Anticholinergic Ingestion
- Sympathomimetic Ingestion
- Antipsychotic Ingestion/ Extrapyrimalal Syndromes
- Poly-Pharmacologic

Clinical Management Options:

- | | | |
|-------------|-----------------------|--|
| E
M
T | M
E
D
I
C | <ul style="list-style-type: none"> • Theory and Operational Guideline • General Guidelines • Contact Poison Control for consultation as necessary/indicated 1-800-222-1222 • Opioid (Narcotic):
 Naloxone 0.4mg IV, IO, IN, IM; repeat PRN; primary goal is to restore spontaneous respiration not mentation
 Pediatric: 0.4mg IV, IO, IN, IM; repeat PRN; primary goal is to restore spontaneous respiration not mentation
 <i>PEARL Basic Progressive Airway, Ventilation, & Oxygenation Management is the key to a safe patient encounter</i>
 <i>PEARL Fentanyl and similar substances may require larger doses of Naloxone.</i>
 <i>PEARL Not appropriate with advanced airway placement and in cardiopulmonary arrest</i> • Consider CPAP 5 – 15cm/H2O PEEP of CPAP with Pressure Support |
| | | <ul style="list-style-type: none"> • Cholinergic:
 Atropine 1mg IV, IO q 5minutes until airway secretions resolved
 Pediatric ≥12 years: 1mg IV, IO; q 5minutes till resolved
 Pediatric <12 years: 0.05mg/kg IV, IO; q 5minutes till resolved
 <i>PEARL For SLUDGEM – No maximum dose</i> • Anticholinergic:
 Sodium Bicarbonate 1mEq/kg IV, IO
 <i>PEARL For Tricyclic Overdose with heart rate >120bpm & QRS >120ms</i> • Sympathomimetic:
 Midazolam 5mg IV, IO or 10mg IM, IN may repeat q 5minutes PRN until heart rate, respiratory rate & blood pressure normalize
 <i>PEARL For hyperadrenergic states with heart rate >120bpm</i> • Dystonic Reactions/Extrapyrimalal Syndrome:
 Diphenhydramine 50mg IV, IO, IM |

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Compensatory Tachycardia
- Stable Tachycardia
- Unstable Tachycardia
- Pre-excitation/Reentry
- Sinus Tachycardia
- Supraventricular Tachycardia (AVRT/AVNRT)
- Atrial Flutter/Fibrillation
- Ventricular Tachycardia

Clinical Management Options:

- E** **M** • Theory and Operational Guidelines
M **E** • General Guidelines
T **D** • Vagal stimulation
I
C

Unstable	Narrow Complex Tachycardia	Wide Complex Tachycardia
	Regular Synchronized Cardioversion 100 joules Pediatric: Refer to Handtevy	Regular Synchronized Cardioversion 200 joules Pediatric: Refer to Handtevy
Stable	Irregular Synchronized Cardioversion 200 joules Pediatric: Refer to Handtevy	Irregular Asynchronous Cardioversion 200 joules
	Narrow Complex Tachycardia	Wide Complex Tachycardia
	SVT Adenosine 12 mg may repeat once IV, IO Pediatric: 1st dose 0.1 mg/kg (max of 12mg) 2nd dose 0.2 mg/kg (max of 12mg)	Monomorphic VT Amiodarone 150 mg in 100 mL over 10 min IV, IO Pediatric: 5 mg/kg (max of 150 mg) in 100 mL over 30 min IV, IO
	Atrial fibrillation/flutter, or refractory SVT Diltiazem 0.25 mg/kg (max of 20 mg) over 2 min IV, IO	Or Lidocaine 1 mg/kg Followed by 0.5 mg/kg if refractory (repeat up to 3mg/kg)
		Torsades de Pointes Magnesium Sulfate 2 g in 100 mL IV, IO over 10 min for
	Suspected Hyperkalemia Calcium 1 g IV, IO	

Medical Control Actions/Orders/Requests:

- Consider medical control for stable refractory tachycardia not responsive to the therapies listed above.

Differential Impressions:

- Acute Coronary Syndrome
- Vasovagal Episode
- Hypoxia
- Hypothermia
- Toxicity
- Sinus Bradycardia
- Junctional Rhythm
- Atrioventricular Block
- Idioventricular Rhythm

Clinical Management Options:

- | | | |
|-------------|-----------------------|---|
| E
M
T | M
E
D
I
C | • Theory and Operational Guidelines |
| | | • General Guidelines |
| | | • Prioritize oxygenation and ventilation |
| | | <ul style="list-style-type: none"> ○ For Neonates and Infants: HR < 100 BPM – continue to oxygenate and ventilate while considering atropine for increased vagal tone ○ For Neonates, Infants, and Pediatrics: HR < 60 BPM with signs of hypoperfusion – move to medical cardiac arrest guideline |
- PEARL | *Pediatrics are defined as someone who is under 13 or has no signs of puberty*

Symptomatic:

- **Atropine 1mg IV/IO (repeat every 3-5min to Max of 3mg)**
Pediatric: 0.02mg/kg (min dose 0.1mg, max dose of 0.5mg) for increased vagal tone.
 PEARL | *Do not repeat Atropine if initial dose is ineffective*
 PEARL | *Consider moving to pacing if vascular access cannot be obtained or is delayed*
 PEARL | *For high degree av blocks, consider moving directly to pacing or epinephrine*

If atropine is ineffective:

- **Transcutaneous Pacing**
 - Consider Procedural Sedation
 - A/P Placement with Rate set to 80 bpm, titrate milliamps to full capture
- or
- **Epinephrine 2 – 10mcg/min IV/IO**
 PEARL | *Push dose epinephrine may be administered while infusion is prepared*
- **Calcium 1g IV, IO** for suspected hyperkalemia, beta blocker, or calcium channel blocker toxicity.

Medical Control Actions/Orders/Requests:

- As needed

Chest Pain, Acute Coronary Syndrome, STEMI

2026

Differential Impressions:

- Chest Pain – Cardiac Pathology
- STEMI (ST Elevation Myocardial Infarction)
- Acute Coronary Syndrome (ACS)
- Unstable Angina Pectoris
- NSTEMI (Non-ST Elevation Myocardial Infarction)

Clinical Management Options:

- E M T**
- Theory and Operational Guidelines
 - General Guidelines
 - **Aspirin 324mg PO (chewable)**
 - **Nitroglycerin 0.4mg SL**; may repeat every 5minutes x 3 times if symptoms persist without evidence of hypoperfusion
 - Continuous Nitrate Therapy
- | Option 1 | OR | Option 2 |
|--|----|--|
| Tridil Infusion 10 – 50mcg/min IV, IO
titrate by 10mcg every 5min PRN | | Nitroglycerin ointment 1 inch
transdermal anterior chest wall |
- Serial 12 Lead ECGs
 - Pain | Anxiety Management Guideline as necessary/indicated
PEARL | Fentanyl is preferred analgesic for chest pain.
 - Nausea & Vomiting Management Guideline as necessary/indicated

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

STEMI		
Monitor Interpretative Statement	• Indicates STEMI; good tracing, stable baseline and free of artifact?	☐
<i>Or</i>		
Paramedic Interpretation	• ECG shows 1mm of ST segment elevation in 2 or more contiguous leads? • QRS width < 120ms (0.12s) or RBBB with ST segment elevation? • ECG good tracing, stable baseline and free of artifact?	☐
Any boxes checked = STEMI Alert		

Differential Impressions:

- Congestive Heart Failure
- Pulmonary Edema
- Right Heart Failure
- Left Heart Failure
- Non-Cardiac Pulmonary Edema (e.g., Drowning)

Clinical Management Options:

- E M T** **M E D I C**
- Theory and Operational Guidelines
 - General Guidelines
 - **CPAP 5 – 15cm/H2O PEEP or CPAP with Pressure Support**
 - **Nitroglycerin 1.2mg (x3 0.4mg) SL; may repeat every 5min PRN for as long as symptoms persist without evidence of hypoperfusion**

- Continuous Nitrate Therapy

Option 1

Tridil Infusion 10 – 100mcg/min IV, IO;
titrate to desired effect in increments of 10mcg/min every 5min

OR

Option 2

Nitroglycerin 1.2mg (x3 0.4mg) SL; may repeat every 5min PRN

OR

Option 3

Nitropaste 1 inch TD (transdermal; chest wall)

- Serial 12 Lead ECGs
- Drug-assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as needed/indicated

Differential Impressions:

- Primary Cardiac Arrest
- Maternal Cardiac Arrest
- Primary Respiratory Arrest
- Pseudo PEA

Clinical Management Options:

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- Theory and Operational Guidelines
- General Guidelines
- Pit Crew Resuscitation Guideline
- Mechanical CPR, ITD, and Neuroprotective CPR when available
- Maternal: Perform Left Lateral Uterine Displacement (LUD)**
- Neonates, Infants, and Pediatrics: Perform chest compressions if HR < 60 with signs of hypoperfusion**

- **Epinephrine 0.5-1mg IV, IO** every 3 -5 min (Max dose 3mg)
Pediatric: 0.01mg/kg IV, IO (Max dose 3mg)
- **Advanced airway management** as needed.
- Identify & treat reversible causes (PEA rhythms) and consider pseudo PEA
- Shockable arrests
PEARL | Shockable arrests benefit from defibrillation and anti-dysrhythmic before administration of epinephrine
 - **Defibrillation (Adults & Pediatrics: Handtevy for energy setting)**
 - **Shock – Anterior/Posterior placement**
 - (Adult Only) After 3rd shock and subsequent shocks - Dual Sequential Defibrillation (DSD) for patients in refractory VF/VT (less than 1 second apart)
PEARL | Refractory VF/VT is defined as a patient that has been defibrillated at least three times and been administered the first dose of anti-dysrhythmic and continues to be in VF/VT
- Administer anti-dysrhythmic medication:

Option 1	OR	Option 2	OR	Option 3
Amiodarone IV, IO • 1st dose: 300mg • 2nd dose: 150mg Pediatric: 5mg/kg (up to 3 doses, not to exceed the total adult dose)		Lidocaine IV, IO • 1st dose: 1.5mg/kg • 2nd dose: 0.5mg/kg Pediatric: 1mg/kg IV, IO		Magnesium Sulfate 2g IV, IO for torsades or refractory VT Pediatric: 50 mg/kg (max: 2g) IV, IO
- PEARL | Lidocaine preferred when arrest is toxicology related*
- PEARL | Blind administration of sodium bicarb or calcium is not recommended*
- PEARL | Naloxone not indicated for cardiac arrest*
- Post resuscitation efforts (as indicated)
- Termination of resuscitation (as indicated)
PEARL | If POCUS is available, scan all PEA rhythms to determine if the rhythm is pseudo-PEA
- **Calcium 1g IV, IO** for suspected hyperkalemia

Medical Control Actions/Orders/Requests:

- As needed

Differential Impressions:

- Traumatic Circulatory Arrest
- Pending Traumatic Circulatory Arrest

Clinical Management Options:

E M T	M E D I C	PEARL <i>Critically ill or injured patients shall receive initial stabilization and resuscitative measures prior to movement.</i> • Theory and Operational Guidelines • General Guidelines • Hemorrhage Control: Direct pressure, pressure dressing, tourniquet, wound packing. • CPR PEARL <i>CPR should not delay hemorrhage control and bilateral decompression (finger thoracostomy or pleural) and should be deferred until procedures are completed.</i> • Prevent hypothermia for all trauma patients. • Pelvic Splinting for all blunt force trauma arrests PEARL <i>Traumatic Circulatory Arrests should be taken to a trauma center if transport time is within 20 minutes</i>
		• Bilateral Pleural Needle Decompression for trauma arrest with suspected thoracic trauma Pediatric: Bilateral Needle Decompression for trauma arrest with suspected thoracic trauma PEARL <i>Bilateral finger thoracostomy is the preferred technique if trained in such.</i> • Epinephrine 0.5 - 1mg IV, IO (max of 3mg) every 3 -5 min Pediatric: 0.01mg/kg IV, IO (Max dose 3mg) • Identify & treat reversible causes (PEA rhythms) ○ Tranexamic Acid 2g IV, IO for moderate/massive hemorrhage Pediatric: 15mg/kg IV, IO max of 1g for moderate/massive hemorrhage • Medical cardiac arrest guideline if shockable arrest presents • Post resuscitation efforts (as indicated) • Termination of resuscitation (as indicated) • Perform chest ultrasound to assess for cardiac activity and for other causes of arrest that may be reversible • Advanced airway management as needed. • Whole blood administration (preferred over crystalloid) PEARL <i>Only indicated if provider-witnessed the arrest</i> • Bilateral finger thoracostomy for trauma arrest with suspected thoracic trauma • Calcium 1g IV, IO for moderate/massive blood loss Pediatrics: 20mg/kg max of 1g for moderate/massive blood loss

Medical Control Actions/Orders/Requests:

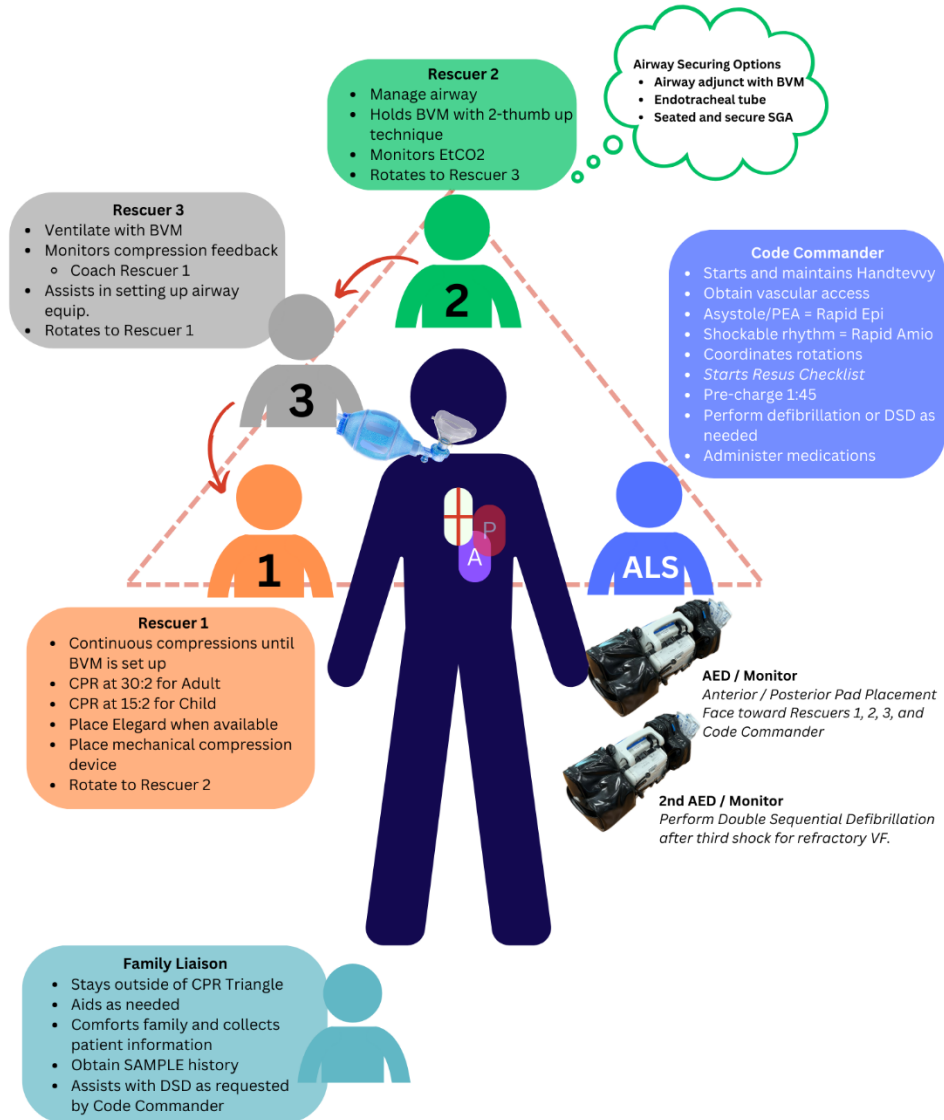
- Consult as necessary/indicated

Clinical Management Options:

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Pit Crew Providers



Differentials:

- Hypoperfusion
- True PEA
- Pseudo PEA

Clinical Management Options:

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C

- Theory and Operational Guideline
- Continue CPR until a definitive palpable pulse is present

PEARL | *Pseudo PEA is a state in which there is an organized rhythm on the monitor without palpable central or peripheral pulses AND the patient has at least two of the following:*

- *Persistent EtCO₂ greater than 30mmHg*
- *Persistent pulse oximetry waveform consistent with pulse rate on the monitor*
- *Coordinated, organized cardiac wall movement confirmed by POCUS*
- *Femoral or carotid pulsation confirmed by POCUS*

PEARL | *If at any point during the pseudo-PEA resuscitation where the patient deteriorates then resume standard resuscitation measures*

PEARL | *Providers should hold standard cardiac arrest epinephrine (0.1mg/mL) and prioritize addressing the profound shock state*

- **Crystalloid Resuscitation 20mL/kg IV,IO** as necessary/indicated
- Pressor therapy

Option 1		Option 2		Option 3
Norepinephrine • 0.1 – 0.5 mcg/kg/min IV, IO • 0.03 – 0.05 mcg/kg/min IV, IO	OR	Epinephrine (use if HR <50) • 0.1 - 0.5 mcg/kg/min IV, IO • 0.03 – 0.2 mcg/kg/min IV, IO	OR	Push dose Epinephrine (0.01mg/mL) • 2mL every 2min PRN • 0.1 mL/kg up to 20 Kg every 2min PRN

PEARL | *Push dose epinephrine may be administered while infusion is prepared*

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Status post-cardiac arrest
- Cardiogenic Shock
- Hypovolemic Shock
- Distributive Shock
- Obstructive Shock

Clinical Management Options:

- | | | |
|-------------|---|--|
| E
M
T | M | • Theory and Operational Guidelines |
| | E | • General Guidelines |
| | D | • Maintain and continuously monitor adequacy airway, ventilation and oxygenation |
| | I | • Transport Medical Post-Arrest to PCI Center |
| | C | • Transport Trauma Post-Arrest to Trauma Center |
-
- Transition to mechanical ventilation
 - Bradycardia guideline as needed (be prepared to initiate CPR)
 - Maintain temp less than 101°F
- Prior to transport (medical post-arrests):
- Patent and ventilated airway
 - 12-lead ECG obtained and transmitted to receiving facility
 - MAP > 65 mmHg
 - HR > 50 bpm
 - Manage secondary injuries or medical issues during transport
 - Prepare pressor agent
- PEARL | Push dose epinephrine should be utilized on patients that are rapidly deteriorating*
- Drug-assisted airway management

Medical Control Actions/Orders/Requests:

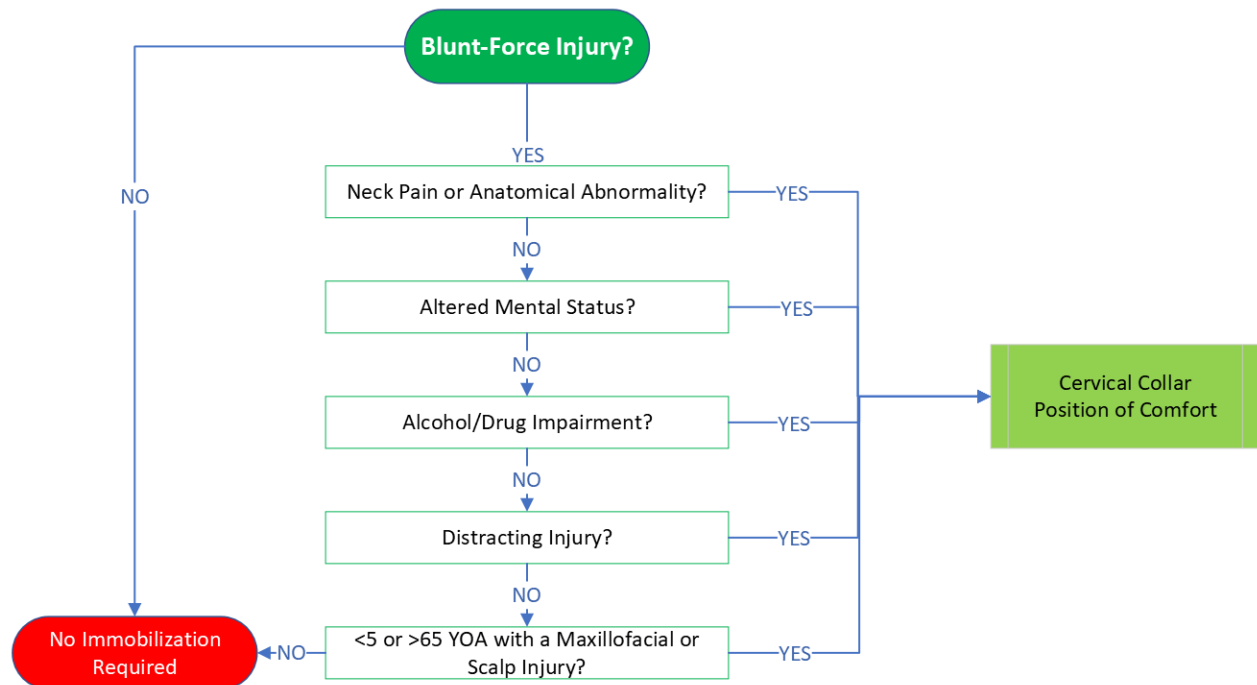
- Consult as necessary/indicated

Goal(s):

- To provide evidence-based and reasoned logic core principles for spinal motion restriction in patients that have sustained injury/trauma
- Rigid Spine Devices are extrication/transfer tools – not a therapeutic intervention

PEARL | *Precautionary spinal immobilization offers no patient value; may result in iatrogenic injury*

General Actions:



Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Falls
- Motor Vehicle Crash
- Pedestrian
- Gunshot wound
- Battery
- Crush Injury
- Other Impact Injury
- Stab Wound or Impalement

Clinical Management Options:

E M T	M E D I C	PEARL Critically ill or injured patients shall receive initial stabilization and resuscitative measures prior to movement • Theory and Operational Guidelines • General Guidelines Trauma Care for Adults & Pediatrics • Spinal Motion Restriction Guideline • Hemorrhage Control: Direct pressure, pressure dressing, tourniquet, wound packing • Thermal preservation for all trauma alerts: Hibler's Method • Pelvic Splinting as necessary/indicated • Extremity Splinting as necessary/indicated • Perform Trauma Triage Criteria & Methodology Assessment
		• Tranexamic Acid 2g IV, IO over 2 min for moderate/massive hemorrhage ○ Upper respiratory tract bleeding: TXA 1gm IN, AT, or topical (soaked gauze) • Pediatric: 15mg/kg max dose of 1g IV, IO slow IV push for moderate/massive hemorrhage • Cefazolin 2g over 15min for open fractures. • Pediatric: 30mg/kg (max of 2g) over 15min. • Pleural Needle Decompression as necessary/indicated PEARL If available, confirm presence of pneumothorax via POCUS prior to decompression PEARL Perfusion target: permissive hypotension to a return of peripheral pulses present • Pain & Anxiety Management Guideline as necessary/indicated • Whole blood administration • Calcium 1g IV, IO for moderate/massive blood loss • Pediatric: 20mg/kg max of 1g for moderate/massive blood loss • Drug-assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
- Sodium bicarbonate for Crush Syndrome

Differential Impressions:

- Isolated Spinal Cord Injury
- Neurogenic Shock
- Spinal Shock
- Keraunoparalysis

Clinical Management Options:

- | | | |
|-------------|---|---|
| E
M
T | M | • Theory and Operational Guidelines |
| | E | • General Guidelines |
| | D | Trauma care for adults & Pediatrics |
| | I | • Spinal Motion Restriction Guideline |
| | C | • Thermal preservation for all patients: Hibler's Method preserves body heat and mitigates Lethal Triad |
| | | • Perform Trauma Triage Criteria & Methodology Assessment |

PEARL | *Pressor agents are first-line therapy for hypotension secondary to Distributive Shock target MAP 70mmHg*

PEARL | *Crystalloids are second-line therapy for hypotension secondary to Distributive Shock*

- **Atropine 1mg IV, IO; repeat x1**
Pediatric: Atropine 0.02mg/kg IV, IO; repeat x1 (minimum dose 0.1mg/maximum dose 0.5mg)

PEARL | *Third-line therapy for hemodynamically significant bradycardia*

- Pain & Anxiety Management Guideline as necessary/indicated
- Nausea & Vomiting Management Guideline as necessary/indicated

PEARL | *Antiemetic therapy is aimed at reducing airway compromise from vomiting*

- Seizure Guideline as necessary/indicated
- Drug assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
Pediatric: Vasopressor agent

Differential Impressions:

- Isolated Closed Head Injury
- Traumatic Brain Injury
- Subdural Hematoma
- Epidural Hematoma
- Intracranial Hemorrhage

Clinical Management Options:

- E M T M E D I C**
- Theory and Operational Guidelines
 - General Guidelines
 - Trauma care for adults & **pediatrics**
 - Spinal Motion Restriction Guideline
 - Hemorrhage Control:
 - Direct Pressure, Pressure Dressing
 - Thermal preservation for all trauma alerts:
 - Cold blood does not clot – Hibler's Method preserves body heat and reduces Lethal Triad
 - Perform Trauma Triage Criteria & Methodology Assessment

PEARL | *Crystalloids are for first-line therapy for hypoperfusion to reduce secondary brain insult*

- Vasopressor agent

Primary option	OR	Option 2
Norepinephrine • 0.1 – 0.5mcg/kg/min IV, IO		Epinephrine • 0.1 – 0.5mcg/kg/min IV, IO

PEARL | *Second-line therapy for hypoperfusion (SBP < 90mmHg or MAP < 65) to reduce secondary brain insult (providers should titrate medication to avoid SBP > 160mmHg)*

- Pain & Anxiety Management Guideline as necessary/indicated
- Procedural Sedation Management Guideline as necessary/indicated
- Nausea & Vomiting Management Guideline as necessary/indicated
- Seizure Guideline as necessary/indicated
- Drug assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
- **Pediatric: Epinephrine 0.1 – 1mcg/kg/min IV, IO infusion**

Differential Impressions:

- Burns (Thermal, Chemical, Electrical, Radiation)
- Electrocution (AC, DC)
- Smoke Inhalation
- Toxic Fume Inhalation

Clinical Management Options:

- | | | |
|-------------|-----------------------|---|
| E
M
T | M
E
D
I
C | <ul style="list-style-type: none"> • Theory and Operational Guidelines • General Guidelines • Thermal preservation for all critical burns |
| | | <p>PEARL Burns – Prevent hypothermia</p> <ul style="list-style-type: none"> • Burn Care (2° and 3° burns) <ul style="list-style-type: none"> Less than 15% BSA – Stop the burning process, moist dressing to patients comfort Greater than 15% BSA – Stop the burning process, moist or dry dressing to patients comfort but not to create hypothermia Remove jewelry and constricting items <p>PEARL Critical Burns: All burns greater than 25% BSA; 3° burns greater than 10% BSA; 2° and 3° burns to the face, eyes, hands, feet, or genitalia; inhalation burns; burns with extremes of age or co-morbidities; electrical burns.</p> |
-
- Perform Trauma Triage Criteria & Methodology Assessment
-
- Pain & Anxiety Management Guideline as necessary/indicated
 - Smoke Inhalation:
 - Reactive Airway Disease Guideline
 - Smoke Inhalation, Carbon Monoxide or Cyanide Toxicity (for patients subject to enclosed fire injuries/illnesses, cardiac arrest secondary to fire, refractory hypotension or AMS)
 - Cyanokit 5g (1 Kit) IV, IO over 15min**
 - Pediatric: 70mg/kg IV,IO over 15min (Max dose: 1 kit)**
 - Drug-assisted airway management
 - Escharotomy of chest or proximal extremities

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Foreign Body/Substance (not embedded)
- Foreign Body (impaled object)
- Corneal Abrasion
- Lacerated Globe
- Global Rupture
- Protruding Eye
- Orbital Fracture
- Retinal Artery Occlusion

Clinical Management Options:

- | | | |
|-------------|--------|---|
| E
M
T | M | • Theory and Operational Guidelines |
| | E | • General Guidelines |
| | D | • Foreign Body/Substance (not embedded) & Corneal Abrasion:
Irrigation 2L or 20 minutes |
| | I
C | <p><i>PEARL Eye irrigation should be a priority in patients with direct eye exposure from chemicals.</i></p> <ul style="list-style-type: none"> • Foreign Body (impaled object), Globe Injury and/or Protruding Eye: <ul style="list-style-type: none"> ○ Shield or cup dress affected eye ○ Consider loose cover to unaffected eye to reduce eye movement ○ Protect loss of fluids: apply saline moistened dressing as necessary ○ Consider C-Collar to reduce head movement ○ Elevate stretcher head • Orbital Fracture <ul style="list-style-type: none"> ○ Shield or cup dress affected eye ○ Consider loose cover to unaffected eye to reduce eye movement ○ Consider C-Collar to reduce head movement ○ Elevate stretcher head |
-
- Pain & Anxiety Management Guideline as necessary/indicated
 - Nausea & Vomiting Management Guideline as necessary/indicated
- PEARL | Antiemetic therapy is aimed at reducing intraocular pressure*

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Adult Trauma Triage Criteria & Methodology

The EMT or paramedic shall assess the condition of those injured persons with anatomical and physiological characteristics of a person sixteen (16) years of age or older for the presence of at least one of the following four (4) criteria to determine whether to transport as a trauma alert. These four criteria are to be applied in the order listed, and once any one criterion is met that identifies the patient as a trauma alert; no further assessment is required to determine the transport destination.

Criteria:

- ☐ 1. Meets color-coded triage system (see below)
- ☐ 2. GCS $\leq$ 12 (Patient must be evaluated via GCS if not identified as a trauma alert after application of criterion 1.)
- ☐ 3. Meets local criteria (specify): _____
- ☐ 4. Patient does not meet any of the trauma criteria listed above but, in the judgment of the EMT or paramedic, should be transported as a trauma alert (document) _____

COMPONENT

AIRWAY	RESPIRATORY RATE OF 30 or GREATER ☐ B	ACTIVE AIRWAY ASSISTANCE ¹ ☐ R
CIRCULATION	SUSTAINED HR OF 120 BEATS PER MINUTE or GREATER ☐ B	LACK OF RADIAL PULSE WITH SUSTAINED HEART RATE (>120) or BP <90 mmHg ☐ R
BEST MOTOR RESPONSE	BMR =5 ☐ B	BMR = 4 or LESS or PRESENCE OF PARALYSIS, or SUSPICION OF SPINAL CORD INJURY or LOSS OF SENSATION ☐ R
CUTANEOUS	SOFT TISSUE LOSS ² or GSW TO THE EXTREMITIES ☐ B	2ND OR 3RD ° BURNS TO 15% or MORE TBSA or AMPUTATION PROXIMAL TO THE WRIST or ANKLE or ANY PENETRATING INJURY TO HEAD, NECK, or TORSO ³ ☐ R
Longbone FRACTURE⁴	SINGLE FX SITE DUE TO MVA or FALL 10' or MORE ☐ B	FRACTURE OF TWO or MORE Longbones ☐ R
AGE	55 YEARS or OLDER ☐ B	
MECHANISM OF INJURY	EJECTION FROM VEHICLE ⁵ or DEFORMED STEERING WHEEL ⁶ ☐ B	

 R = any **one (1)** - transport as a trauma alert

 B = any **two (2)** - transport as a trauma alert

1. Airway assistance beyond administration of oxygen.
2. Major degloving injuries, or major flap avulsion (>5 in.)
3. Excluding superficial wounds in which the depth of the wound can be determined.
4. Longbone (Including humerus, (radius, ulna), femur, (tibia or fibula).
5. Excluding motorcycle, moped, all terrain vehicle, bicycle, or open body of a pickup truck.
6. Only applies to driver of vehicle.

Statutory Definition:

The EMT or Paramedic shall assess the condition of those injured individuals with anatomical and physical characteristics of a person fifteen (15) years of age or younger for the presence of one or more of the following criteria to determine the transport destination per 64E-2.001, Florida Administrative Code, (F.A.C.):

Scorecard

 R = RED, any **one (1)**-transport as a trauma alert

 B = BLUE, any **two (2)** - transport as a trauma alert

Consciousness & Airway	AMNESIA OR LOSS OF CONSCIOUSNESS	AIRWAY ASSISTED OR INTUBATED (1)	ALTERED MENTAL STATUS (2) OR COMA OR PRESENCE OF PARALYSIS OR SUSPICION OF SPINAL CORD INJURY OR LOSS OF SENSATION
	 B	 R	 R
Circulation & Perfusion	CAROTID OR FEMORAL PULSES PALPABLE, BUT THE RADIAL OR PEDAL PULSE NOT PALPABLE	FAINT OR NON-PALPABLE CAROTID OR FEMORAL PULSE	AGE 0-9: SBP LESS THAN 70 AGE 10-15: SBP LESS THAN 90
	 B	 R	 R
Injury	SINGLE CLOSED LONG BONE (3) DEFORMITY (4)	OPEN LONG BONE (3) FRACTURE (4) OR MULTIPLE FRACTURE SITES OR MULTIPLE DISLOCATIONS (4)	SKULL DEFORMITY OR FRACTURE, OR SEVERE FACIAL INJURY OR CHEST WALL INJURY OR DEFORMED CHEST OR AMPUTATION (6) OR ANY PENETRATING INJURY TO HEAD, NECK, OR TORSO (7)
	 B	 R	 R
	SIGNIFICANT MECHANISM OF INJURY (8)	PARTIAL THICKNESS OR FULL THICKNESS BURNS TO $\geq 10\%$ TBSA	MAJOR SOFT TISSUE DISRUPTION (5) OR MAJOR FLAP AVULSION OR SEATBELT SIGN WITH SEVERE ABDOMINAL PAIN
	 B	 R	 R

 Patient does not meet any of the trauma criteria listed above, but the EMT or Paramedic can call a "Trauma Alert" if, in his/her judgment, the trauma patient's condition warrants such action. Must be documented on run report pursuant to FAC 64E-2.013.

Notes

1. Airway assistance includes manual jaw thrust, continuous suctioning, or use of other adjuncts to assist ventilatory efforts.
2. Altered mental states include drowsiness, lethargy, inability to follow commands, unresponsiveness to voice, and total unresponsiveness.
3. Long bones include the humerus, (radius, ulna), femur, (tibia or fibula).
4. Long bone fractures do not include isolated wrist or ankle fractures or dislocations.
5. Includes major degloving injury.
6. Amputation is proximal to wrist or ankle.
7. Excluding superficial wounds where the depth of the wound can be determined
8. Examples of significant Mechanism of Injury (each mechanism is 1 blue criteria)

****This list does not include every significant mechanism****

- **SIGNIFICANT INTRUSION (INCLUDING ROOF) - OVER 12" INCHES ON OCCUPANT SIDE OR 18" ANY OTHER SITE**
- **DEATH IN SAME PASSENGER COMPARTMENT**
- **VEHICLE TELEMETRY DATA CONSISTENT WITH SIGNIFICANT IMPACT**
- **FORCEFUL EJECTION FROM A MOVING VEHICLE**
- **CHILD (AGE 0-9) UNRESTRAINED OR IN AN UNSECURED CHILD SAFETY SEAT**
- **SEATBELT SIGN**
- **SEVERE ABDOMINAL PAIN**
- **FALL FROM HEIGHT GREATER THAN TWICE THE HEIGHT OF THE CHILD**

Differential Impressions:

- Human bite
- Animal bite
- Snake bite/envenomation
- Spider bite/envenomation
- Jellyfish sting
- Stingray/Catfish sting
- Ant, Bee, Wasp bite/envenomation

Clinical Management Options:

- | | | |
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C | • Theory and Operational Guidelines |
| | | • General Guidelines |
| | | • Human & Animal:
Irrigate and dress wounds as necessary/indicated |
| | | PEARL <i>Human Bites are highly infectious are highly infections at 24-48 hours. Strong encouragement for transportation to the hospital</i> |
| | | • Snake & Spider: Make early contact with Venom 1 (412-849-8022)
Immobilize extremity in elevated position |
| | | PEARL <i>No ice, tourniquets, cutting or suctioning of site</i> |
| | | PEARL <i>Do not attempt to catch, kill or bring snake</i> |
| | | • Ant, Bee, Sawfly, Wasp:
Remove/scrape off stingers/venom sacs with a blunt-edge object (e.g., credit card or tongue depressor) |
| | | • Jellyfish:
Remove from skin with tweezers (preferred) or sea water, rinse with vinegar (if available) and immerse in hot, non-scalding water |
| | | • Stingray/Catfish:
Do not remove barb – immerse in hot, non-scalding water |
| • Allergic Reaction & Anaphylaxis Guideline as necessary/indicated | | |
| • Pain & Anxiety Management Guideline as necessary/indicated | | |
| • Nausea & Vomiting Management Guideline as necessary/indicated | | |

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Drowning
- Submersion

Clinical Management Options:

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C | • Theory and Operational Guidelines |
| | | • General Guidelines |
| | | • CPAP 5 – 15cm/H2O PEEP or CPAP with Pressure Support |
| | | • Exposure Emergencies Hypo & Hyperthermia Guideline as necessary/indicated |
| | | • Reactive Airway Disease Guideline as necessary/indicated |

PEARL | Resuscitate cold water drowning until warm – transport

- Drug assisted airway management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
- Pediatric: Epinephrine 0.1 – 1mcg/kg/min IV, IO Infusion**

Differential Impressions:

- Decompression Illness (DCI)
- Nitrogen Narcosis
- Arterial Gas Embolism (AGE)
- Stroke
- Central Retinal Artery Occlusion (CRAO)
- Retinal Detachment
- Sudden Vision Loss

Clinical Management Options:

- E** **M**
M **E**
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- Drowning / Submersion Guideline as necessary and indicated.
 - Consult Diver's Alert Network (DAN) 1(919) 684-9111 or MD1
 - If DAN Supports & Suggests Hyperbaric Chamber Center due to signs and symptoms of AGE, Consider Air Medical for direct transport.
 - High Flow Oxygen Therapy with 100% FiO2
 - PEARL | *High Flow Oxygen Should be administered regardless of SpO2 readings*
 - PEARL | *Avoid transporting patients in Trendelenburg or High Fowler's Position.*
 - PEARL | *If patient utilizes dive computer, it should be transported with patient. Otherwise, obtain record of Depth, Duration, Gas Mix, and number of dives completed < 24 hours.*

Option 1		Option 2		Option 3
Ketorolac 15mg IV, IO, IM Pediatric: 0.5mg/kg IV, IO, IM (Max 15mg)	OR	Fentanyl 1mcg/kg IV, IO, IM every 10min PRN Pediatric: 1mcg/kg IV, IO, IM; every 10min PRN	OR	Ketamine 0.2mg /kg IV, IO, IM every 20min Pediatric: 0.2mg/kg IV, IO, IM every 20min

- Drug-assisted Airway Management

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Pre-Eclampsia
- Eclampsia
- Post-Partum Eclampsia
- AMS
- Hyperreflexia
- Hypertension
- Proteinuria
- Headache
- Edema
- Visual Changes
- Seizure Activity

Clinical Management Options:

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- Theory and Operational Guidelines
- General Guidelines
- Place in left lateral recumbent position
- BP every 5 minutes

- Elevated blood pressure (BP) in pregnancy can indicate severe disease and can result in both maternal and fetal morbidity and mortality. Elevated BP in pregnancy and up to 6 weeks postpartum requires special attention and treatment.
- In clinical situations in which **antihypertensives** and **magnesium sulfate** are indicated, **antihypertensive** medications are the highest priority.
- An elevated BP may be the only abnormal vital sign. Do not ignore asymptomatic high BP!
- Transport the patient to a hospital with obstetric services or the most appropriate local/regional facility if an obstetric facility is not readily available.
- Vital signs of concern in a pregnant or postpartum patient:



SBP 140–159 or DBP 90–109 mmHg: Abnormal, possible preeclampsia



SBP 140–159 or DBP 90–109 mmHg with ANY of the following:

severe headache, blurred vision, or right upper quadrant or epigastric abdominal pain:

Abnormal, preeclampsia with severe features

- Requires treatment with **MAGNESIUM SULFATE**



SBP ≥ 160 or DBP ≥ 110 mmHg or higher: Abnormal, preeclampsia with severe features even without other symptoms

- Requires treatment with **ANTIHYPERTENSIVE AND MAGNESIUM SULFATE**

Administer antihypertensive as outlined below if:



SBP $\geq$ 160 or DBP $\geq$ 110 mmHg is persistent for 10 minutes

EXCLUSIONS: Trauma Patients, HR < 60 bpm, CHF, Heart Block on ECG, Asthma

Labetalol: 20mg IV over 2 min. Allow 10 minutes from the time of administration for the medication to take effect.

- If BP remains severely elevated (SBP $\geq$ 160 or DBP $\geq$ 110 mmHg), give 40mg IV over 2min. Reassess BP in 10 minutes.
- If repeat BP is still SBP $\geq$ 160 or DBP $\geq$ 110 mmHg, give 80mg IV over 2min every 10 minutes up to a maximum cumulative dosage of 300mg.

Target BP: SBP 130-150 AND DBP 80-100 mmHg

- Once target BP is achieved, monitor BP every 5 minutes until arrival at destination.
- During monitoring, if BP elevates back up to SBP $\geq$ 160 or DBP $\geq$ 110 mmHg, readminister **Labetalol** using the same dose that previously achieved the target BP.

Administer magnesium sulfate as outlined below if:



**SBP $\geq$ 160 mmHg or
DBP $\geq$ 110 mmHg**

OR



**SBP 140–159 or DBP 90–109 mmHg
with ANY of the following: severe
headache, blurred vision, or right upper
quadrant or epigastric abdominal pain**

Magnesium Sulfate: 4g in 100 mL of D5W IV/IO over 20 min.

- If IV/IO access cannot be obtained, a 10g IM loading dose of **Magnesium Sulfate** (5g in each buttock) may be administered.

If the patient is actively seizing:

Magnesium Sulfate: 4g in 100mL of D5W IV/IO loading dose, administered wide open until seizure stops

- If IV access cannot be obtained, a 10g IM loading dose of **Magnesium Sulfate** (5g in each buttock) may be administered.
- Refer to the seizure guideline for patients that are refractory to the above treatment.

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Normal Spontaneous Vaginal Delivery
- Complicated Spontaneous Vaginal Delivery
- Stillborn Delivery
- Newborn Distress Delivery

Clinical Management Options:

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C | <ul style="list-style-type: none"> • Theory and Operational Guidelines • General Guidelines • Vaginal Bleeding Guideline as necessary/indicated • Normal Delivery Procedure <ul style="list-style-type: none"> ○ Place the mother on a firm surface and elevate hips ○ Inspect the vaginal area for impending delivery (crowning), or any signs of abnormal presentation – prolapsed amniotic sac, limb presentation, cord presentation, or breech presentation |
| | | <p>PEARL Signs of imminent delivery include: membrane rupture or bloody show, contractions, urge to move bowels and/or urge to push</p> <ul style="list-style-type: none"> ○ Apply gentle palm pressure to the infant's head to prevent explosive delivery and tearing of perineum ○ If amnion is still intact as head delivers, instruct mother to stop pushing and gently tear open membrane ○ Suction mouth, then nose (if needed) ○ Keep newborn warm and dry ○ Stimulate the newborn as necessary/indicated ○ After cord stops pulsating, clamp the cord 6 and 9 inches away from baby and cut between the clamps ○ Document the time of delivery and perform APGAR score at 1 and 5 minutes |
- Complicated Delivery Procedures
 - Nuchal Cord:*
 - As delivery occurs, attempt to slip the umbilical cord over the newborn's head
 - If umbilical cord is too tight to maneuver, immediately clamp and cut
 - Continue with delivery
 - Prolapsed Cord:*
 - Do not delay transport

PEARL | Primary objective: maintain a pulsatile umbilical cord

 - Place the mother in knee-to-chest position
 - Instruct the mother to pant and not push with each contraction
 - Apply upward manual pressure through the vagina lifting the presenting newborn anatomy away from and off the umbilical cord
 - With the umbilical cord now pulsating, maintain that position and transport
 - Limb Presentation:*
 - Do not delay transport
 - Place the mother head down with pelvis elevated position

- Instruct the mother to pant and not push with each contraction
- Maintain that position, do not pull on the exposed limb and transport

Breech Presentation:

- Do not delay transport
- Place the mother head down with pelvis elevated position
- Instruct the mother to pant and not push with each contraction
- Deliver the anterior shoulder in a gentle, controlled fashion, then deliver the posterior shoulder and the remainder of the newborn
- As the newborn's head passes the pubis, apply gentle upward pressure until the mouth appears over the perineum and immediately suction the mouth, then nose
- If the head does not deliver, form a "V" with the index and middle finger on either side of the infant's nose.
- Push the vaginal wall from the face, maintain that position and transport

Postpartum Hemorrhage:

- Massage the uterus/fundus from pubis toward umbilicus
- Vaginal bleeding guideline as necessary/indicated

• Complicated Delivery Procedures

Meconium Aspiration Syndrome:

- If the baby is not vigorous (depressed respiratory effort, poor muscle tone, and/or heart rate <100/min): Direct laryngoscopy, intubate, and suction the ETT for no longer than 5 seconds. If no meconium is retrieved, do not repeat intubation and suction. If meconium is retrieved and no bradycardia is present, reintubate and suction. If heart rate is <100/min, administer positive pressure ventilation and consider suctioning again later.
- If the baby is vigorous (normal respiratory effort, normal muscle tone, and heart rate >100/min): **Do not electively intubate.** Gently clear secretions/meconium from mouth and nose with a bulb syringe.

• Nausea & Vomiting Management Guideline as necessary/indicated

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated

Differential Impressions:

- Abrutio placenta
- Ectopic pregnancy rupture
- Placenta previa
- Inevitable abortion
- Spontaneous abortion
- Therapeutic abortion
- Threatened abortion
- Endometrosis
- Memorrhagia
- Postpartum hemorrhage
- Sexual battery/Vaginal trauma
- Uterine rupture

Clinical Management Options:

- | | | |
|-------------|-----------------------|---|
| E
M
T | M
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D
I
C | • Theory and Operational Guidelines |
| | | • General Guidelines |
| | | • If pregnant and if delivery is not imminent, transport in left lateral recumbent position |
| | | • If postpartum, massage the uterus/fundus and encourage newborn breast feeding |
- PEARL | Do not pack vagina to arrest bleeding**
-
- **Tranexamic Acid 2g IV, IO over 2 min** for moderate/massive hemorrhage
- PEARL | Perfusion target: permissive hypotension; peripheral pulses present – restrict crystalloid**
- **Calcium Chloride 1g IV/IO**
 - **Pediatric: 20mg/kg max dose of 1g**
 - **Whole blood administration**

Medical Control Actions/Orders/Requests:

- Consult as necessary/indicated
- **Epinephrine 0.1 – 0.5mcg/kg/min IV, IO Infusion** as necessary/indicated

PHARMACOLOGY REFERENCE

Definitions

- **Class:** The family of drugs that this medication belongs to
- **Action:** What this medication does.
- **Indications:** Reason to give the medication.
- **Contraindications:** Reason that the medication CAN NOT be given.
- **Precautions/Notes:** Things to consider when administering this medication.
- **Side effects:** Things that this medication might do in addition to its desired effect.

Medication Safety

All medication administrations should be performed (when possible) with a partner. Providers shall perform a medication administration check and partner cross check ensuring that the following six rights of medication administration are confirmed:

1. Right patient
2. Right drug
3. Right dose
4. Right route
5. Right time
6. Right documentation

Weight Based Medication Dosing

Most medications (unless specified differently) in this reference should be dosed (when applicable) based on ideal body weight. Ideal body weight categories can be found in the Handtevy reference where pediatrics is categorized by age and adults are categorized as small adult, large adult.

Pregnancy Class

Category	Risk to Fetus
A	No demonstrated risk. Safe to use.
B	Minimal demonstrated risk. Generally safe to use.
C	Demonstrated risk. Should only be given if benefit to mother outweighs the risk to the fetus
D	Demonstrated risk. Should only be given in life-threatening situations.
X	Demonstrated risk. Contraindicated in pregnancy.

Pediatric Dosing

In general, pediatric dosing should never exceed the adult dose unless specified.

ACETAMINOPHEN

TYLENOL

Class: Non-opioid analgesic, antipyretic

Action: Inhibition of the cyclooxygenase (COX) pathways. Possesses analgesic and antipyretic properties like NSAIDs but lacks peripheral anti-inflammatory effects.

Indications:

- Moderate pain
- Fever from non-environmental source

Contraindications

- Liver failure patients

Precautions

- Pregnancy safety: Category B
- Avoid use in patients presenting with an overdose as it can worsen liver dysfunction when co-ingested

Side effects:

- Nausea, vomiting, constipation, pruritus and abdominal pain

Route: IV, IO, PO

ADENOSINE

ADENOCARD

Class: Antidysrhythmic

Action: Slows conduction through the AV node; can interrupt reentrant AV nodal pathways.

Indications:

- Conversion of re-entry supraventricular tachycardia (SVT)
- Reentry SVT due to Wolff-Parkinson-White syndrome

Contraindications

- None in emergent settings

Precautions

- Pregnancy safety: Category C

Notes

- Not effective in converting atrial fibrillation/flutter or ventricular tachycardia
- Arrhythmias are common during the cardioversion
- Short half-life limits adverse effects in most patients
- Bronchoconstrictive lung disease (asthma, COPD), providers should listen to lung sounds after administration.
- Adenosine should not be used for patients who are hemodynamically unstable

Side effects:

- Headache, dizziness, dyspnea, bronchospasm, dysrhythmias, palpitations, hypotension, chest pain, facial flushing, cardiac arrest, nausea, metallic taste, pain in head or neck, paresthesia, diaphoresis

Route: IV, IO; must be administered rapidly, followed by a flush.

ALBUTEROL

PROVENTIL/VENTOLIN

Class: Sympathomimetic, Bronchodilator

Action: Selective beta-2 (β_2) agonist that stimulates adrenergic receptors of the sympathetic nervous system. This results in smooth muscle relaxation and bronchodilation.

Indications:

- Bronchospasm

Contraindications

- None in emergent settings

Precautions

- Pregnancy safety: Category C
- Symptomatic tachycardia

Notes

- Causes tachycardia so it may precipitate angina and dysrhythmias

Side effects:

- Headache, fatigue, light-headedness, irritability, restlessness, aggressive behavior, pulmonary edema, hoarseness, increased sputum, chest pain, palpitations, dry mouth, and tremors.

Route: Nebulized

AMIODARONE

CORDARONE/PACERONE

Class: Antidysrhythmic (Class III)

Action: Blocks Sodium, Potassium, and Calcium channels. Prolongs the action potential and repolarization. Decreases AV conduction and sinoatrial node function. Decreases peripheral vascular resistance (α and β blockade).

Indications:

- Stable monomorphic ventricular tachycardia
- Stable polymorphic ventricular tachycardia with normal QT interval
- Cardiac Arrest with ventricular fibrillation or ventricular tachycardia
- Atrial tachyarrhythmias (narrow rhythms) refractory to other treatments

Contraindications

- QTC > 500ms

Precautions

- Heart failure
- Torsades de Pointes

Notes

Side effects:

- Dizziness, fatigue, malaise, tremor, ataxia, lack of coordination, ARDS, pulmonary edema, cough, progressive dyspnea, heart failure, bradycardia, hypotension, prolongation of QT interval, nausea, vomiting, Stevens-Johnson syndrome.

Route: IV, IO

ASPIRIN

Class: Platelet inhibitor and anti-inflammatory

Action: Inhibits platelet aggregation.

Indications:

- New chest pain or cardiovascular equivalent suggestive of acute coronary syndrome

Contraindications

- Known anaphylaxis
- Active, uncontrolled bleeding

Precautions

- GI bleeding

Notes

- Administer a full dose of aspirin to patients who have taken a morning dose of 81mg.
- Aspirin administration should be prioritized in patients with acute coronary syndrome symptoms.

Side effects:

- Heartburn, nausea, vomiting, wheezing

Route: Oral

ATROPINE SULFATE

Class: Parasympatholytic (anticholinergic)

Action: Blocks acetylcholine receptors. Increases heart rate. Decreases gastrointestinal secretions.

Indications:

- Hemodynamically significant bradycardia
- Organophosphate poisoning
- Third-line therapy for bradycardia from spinal cord shock

Contraindications

- None when used emergently

Precautions

- Tachycardia
- Hypertension

Notes

- If administered too slowly, reflex bradycardia may occur

Side effects:

- Palpitations, tachycardia, headache, dizziness, anxiety, dry mouth, pupillary dilation, blurred vision, urinary retention (especially older men)

Route: IV, IO

CALCIUM CHLORIDE

Class: Electrolyte

Action: Increases cardiac contractility (positive inotropic effect). May enhance ventricular automaticity.

Indications:

- Hyperkalemia
- Calcium channel blocker toxicity
- Beta-blocker toxicity
- Used in conjunction with whole blood administration

Contraindications

- None when used emergently

Precautions

- Pregnancy safety: Category C
- Ventricular fibrillation (relative)
- Digitalis toxicity

Notes

- Do not use routinely in cardiac arrest, unless the cause of the arrest is a suspected hyperkalemia.
- Triple the potency of calcium gluconate
- Local infiltration results in necrosis

Side effects:

- Arrhythmias (bradycardia/asystole), hypotension

Route: IV, IO

Cefazolin

ANCEF

Class: Antibiotic

Action: A first-generation cephalosporin antibiotic that inhibits bacterial cell wall synthesis by binding to penicillin-binding proteins (PBPs). Bactericidal effect—leads to cell lysis and death, primarily targeting gram-positive organisms with some gram-negative coverage

Indications:

- Open fracture(s)

Contraindications

- Known hypersensitivity to cefazolin, other cephalosporins, or beta-lactam antibiotics
- Caution is warranted in patients with a history of severe penicillin allergy due to possible cross-reactivity

Precautions

- **Renal Impairment:** Dosage adjustment is necessary; monitor renal function
- **Allergy Risk:** Use with caution in patients with penicillin or beta-lactam allergies
- **Gastrointestinal:** Monitor for signs of *Clostridium difficile*-associated diarrhea or colitis
- **Infusion-Related Reactions:** Ensure proper dilution and rate of administration to minimize adverse reactions

Side effects:

- Pain, redness, swelling, or phlebitis at the site of administration
- Nausea, vomiting, and diarrhea, which are typically mild to moderate in intensity

Route: IV, IO, IM

CYANOKIT

HYDROXYCOBALAMIN

Class: Vitamin

Action: An injectable form of vitamin B12 that binds with cyanide from the tissues creating cyanocobalamin which is the urinated out safely

Indications:

Any potential cyanide toxicity

Contraindications

- None in an emergency setting

Precautions

- None

Notes

- Strongly consider for victims of combustion in enclosed space with **any** of following
 - Altered mental status (confusion, disoriented, unresponsive, altered responsiveness)
 - Hypotension
 - Low EtCO₂
 - Cardiac arrest

Side effects:

- Red coloring of skin or urine, warmth sensation under skin, hypertension, vomiting, diarrhea

Route: IV, IO

DEXAMETHASONE

Class: Corticosteroids, anti-inflammatory agents

Action: Decreases inflammation by suppressing migration of polymorphonuclear leukocytes (PMNs) and reducing capillary permeability. Upregulates beta receptors to allow albuterol to work better.

Indications:

- Reactive airway disease

Contraindications

- None in emergent setting

Precautions

- Pregnancy safety: Category C

Side effects:

- Nausea, vomiting, dizziness, restlessness

Route: IV, IO, PO

DEXTROSE

D50, D25, D12.5, D10, D5W, Oral Glucose

Class: Carbohydrate, antihypoglycemic

Action: Rapidly elevates serum glucose levels. Short-term osmotic diuresis.

Indications:

- Hypoglycemia

Contraindications

- None in emergent setting

Precautions

- Pregnancy safety: Category C

Notes

- Extravasation (infiltration) may cause tissue necrosis with higher concentrations

Side effects:

- Extravasation leads to tissue necrosis, cerebral hemorrhage, cerebral ischemia, pulmonary edema, warmth, pain, burning from IV infusion, hyperglycemia.

Route: IV, IO, PO (Oral Glucose Gel only)

DIAZEPAM

VALIUM

Class: Tranquilizer (benzodiazepine)

Action: Anticonvulsant, skeletal muscle reactant, and sedative

Indications:

- Major motor seizures
- Status epilepticus

Contraindications

- None

Precautions

- Venous irritation

Notes

- Short duration of effect

Side effects:

- Drowsiness, hypotension, respiratory depression, apnea

Route: IV, IO, IM

DILTIAZEM

CARDIZEM

Class: Calcium channel blocker, Antidysrhythmic (Class IV)

Action: Inhibits extracellular calcium ion influx across membranes of myocardial cells and vascular smooth muscle cells, resulting in inhibition of cardiac and vascular smooth muscle contraction and thereby dilating main coronary and systemic arteries; no effect on serum calcium concentrations; substantial inhibitory effects on the cardiac conduction system, acting principally at the AV node, with some effects at the SA node.

Indications:

- Stable, narrow complex tachycardia (refractory to adenosine and vagal maneuvers)
- Atrial fibrillation or atrial flutter with rapid ventricular response

Contraindications

- Wide-complex tachycardia (may cause ventricular fibrillation)
- Presence of Wolff-Parkinson-White syndrome

Precautions

- Underlying cause of tachycardia – sepsis, drug use, etc.
- Pregnancy safety: Category C
- Hypotension / cardiogenic shock
- Renal or hepatic dysfunction

Notes

- Dysrhythmias may present during pharmacologic cardioversion

Side effects:

- Dizziness, headache, dyspnea, cough, dysrhythmias, heart failure, peripheral edema, bradycardia, hypotension, AV blocks, syncope, VF/VT, cardiac arrest, chest pain, nausea, and vomiting

Route: IV, IO

DIPHENHYDRAMINE

BENADRYL

Class: Antihistamine

Action: Blocks histamine receptors. Has some sedative side effects.

Indications:

- Allergic/anaphylactic reactions
- Extrapyrarnidal symptoms

Contraindications

- None in emergency setting

Precautions

- Asthma
- Nursing mothers
- Hypotension

Notes

- Diphenhydramine is not the primary treatment for anaphylactic reactions

Side effects:

- Sedation, dries bronchial secretions, blurred vision, headache, palpitations

Route: IV, IO, IM

DROPERIDOL

Class: Antipsychotic and Antiemetic

Action: Butyrophenone antipsychotic causes CNS depression; antiemetic effect is from dopamine blocking. Main actions from potent dopamine (2) receptor antagonism.

Indications:

- Hyperactive delirium
- Mild to moderate agitation
- Nausea
- Vomiting
- Gastroparesis abdominal pain / cyclic vomiting

Contraindications

- None

Precautions

- Known long QT interval

Side effects:

- Restlessness, anxiety, dystonic reactions, drowsiness

Route: IV, IO, IM

EPINEPHRINE

1 mg/mL (1:1000)

Class: Alpha and Beta receptor agonist

Action: Increases heart rate and automaticity. Increases cardiac contractile force. Increases myocardial electrical activity. Increases systemic vascular resistance. Increases blood pressure. Causes bronchodilation.

Indications:

- Anaphylactic reactions
- Bronchial asthma
- COPD exacerbation
- Severe reactive airway disease
- Croup

Contraindications

- None

Precautions

- Patients with cardiovascular disease
- Hypertension
- Pregnancy
- Patients with tachyarrhythmia
- Should protect from light

Side effects:

- Palpitations, anxiety, tachycardia, tremulousness, nausea, and vomiting

Route: IM, AT

EPINEPHRINE

0.01 mg/mL (1:100,000)

Class: Alpha and Beta receptor agonist

Action: Increases heart rate and automaticity. Increases cardiac contractile force. Increases myocardial electrical activity. Increases systemic vascular resistance. Increases blood pressure. Causes bronchodilation.

Indications:

- Pressor agent

Contraindications

- None

Precautions

- Should be protected from light
- Can be deactivated by alkaline solutions (sodium bicarb)

Side effects:

- Palpitations, anxiety, tremulousness, nausea, and vomiting

PEARL | 1:100,000 concentration is made by taking 1mL of 0.1mg/mL (1:10,000) and adding it to 9mL of saline.

Route: IV, IO

EPINEPHRINE

0.1 mg/mL (1:10,000)

Class: Alpha and Beta receptor agonist

Action: Increases heart rate and automaticity. Increases cardiac contractile force. Increases myocardial electrical activity. Increases systemic vascular resistance. Increases blood pressure. Causes bronchodilation.

Indications:

- Cardiac arrest

Contraindications

- None

Precautions

- Should be protected from light
- Can be deactivated by alkaline solutions (sodium bicarb)

Side effects:

- Palpitations, anxiety, tremulousness, nausea, and vomiting

Route: IV, IO

EPINEPHRINE INFUSION

1 mg/250 mL, 1 mg/500 mL, 1mg/100mL

Class: Alpha and Beta receptor agonist

Action: Increases heart rate and automaticity. Increases cardiac contractile force. Increases myocardial electrical activity. Increases systemic vascular resistance. Increases blood pressure. Causes bronchodilation.

Indications:

- Symptomatic bradycardia
- Cardiogenic shock
- Sepsis-induced hypotension
- Anaphylactic shock (when IM epinephrine is insufficient)

Contraindications

- None in emergency settings when clinically indicated.

Precautions

- Patients with cardiovascular disease
- Hypertension
- Pregnancy
- Patients with tachyarrhythmia
- Should protect from light

Side effects:

- Palpitations, anxiety, tachycardia, tremulousness, nausea, and vomiting

Route: IV, IO

ETOMIDATE

AMIDATE

Class: General anesthetic

Action: Ultrashort-acting, non-barbiturate hypnotic intravenous anesthetic agent

Indications:

- Drug-assisted airway management

Contraindications

- None

Precautions

- Caution in sepsis patients

Side effects:

- None

Route: IV, IO

FENTANYL

SUBLIMAZE

Class: Narcotic

Action: Central nervous system depressant. Decreases sensitivity to pain.

Indications:

- Pain management

Contraindications

- None

Precautions

- Respiratory depression
- Hypotension
- Nausea

Notes

- EtCO₂ monitoring should be continually performed post administration
- Intubation can be a painful procedure, Fentanyl may be used as a pain management tool post intubation

Side effects:

- Dizziness, altered mental status, bradycardia, rigid chest wall syndrome

Route: IV, IO, IN, IM

HYDROCORTISONE

Class: Corticosteroids, Glucocorticoid

Action: Replaces deficient cortisol in adrenal insufficiency. Provides anti-inflammatory and immunosuppressive effects. Helps restore vascular tone and blood pressure by enhancing responsiveness to catecholamines.

Indications:

- Adrenal insufficiency Crisis (Addisonian Crisis)

Contraindications

- None in emergent setting

Precautions

- Pregnancy safety: Category C

Side effects:

- Headache, hypertension, hyperglycemia, sodium/fluid retention, hypokalemia

Route: IV, IO, IM

HYDROMORPHONE

DILAUDID

Class: Narcotic

Action: Central nervous system depressant. Decreases sensitivity to pain.

Indications:

- Pain management

Contraindications

- None

Precautions

- Respiratory depression
- Hypotension
- Nausea

Notes

- EtCO₂ monitoring should be continually performed post administration

Side effects:

- Dizziness, altered mental status, bradycardia

Route: IV, IO, IM, IN

IPRATROPIUM

ATROVENT

Class: Anticholinergic

Action: Causes bronchodilation. Dries respiratory tract secretions.

Indications:

- Refractory bronchospasm

Contraindications

- None in emergent situation

Precautions

- Should not be used as primary treatment for bronchospasm

Side effects:

- Palpitations, dizziness, anxiety, headache, nervousness

Route: Nebulized

KETAMINE

KETALAR

Class: Sedative/hypnotic analgesic

Action: Causes dissociative state

Indications:

- Pain management
- Sedation
- Hyperactive delirium with agitation

Contraindications

- Children less than 3 months old

Precautions

- Catecholamine depleted states
- Eye impalement

Notes

- Hallucinations may occur
- Ketamine (100mg/mL) IV/IO should be diluted with an equal volume of Normal Saline

Side effects:

- Hallucinations, increased skeletal muscle tone

Route: IV, IO, IM, IN

KETOROLAC

TORADOL

Class: Nonsteroidal anti-inflammatory drugs

Action: Anti-inflammatory and pain reliever. Works by blocking prostaglandins that lead to pain, fever, and inflammation pain management

Indications

- Pain management

Contraindications

- Age > 75 y/o
- Active peptic ulcer disease
- Recent gastrointestinal bleeding
- Known kidney failure or dysfunction
- Suspected kidney failure

Precautions

- None

Notes

- Highly effective for use in mild to moderate pain related to kidney stones and muscle skeletal pain
- May have serious side effects in the elderly (renal impairment)

Side effects:

- Nausea, vomiting, dizziness, bleeding

Route: IV, IM, IO

LABETALOL

Normodyne, Trandate

Class: Non-selective Beta-Blocker, Alpha-1 Blocker, Antihypertensive

Action: Labetalol is a competitive alpha-1 and non-selective beta-adrenergic receptor blocker that results in decreased heart rate, myocardial contractility, and peripheral vascular resistance. It effectively lowers blood pressure without significantly reducing cardiac output or causing reflex tachycardia.

Indications:

- Pregnancy-induced hypertension (pre-eclampsia/eclampsia)

Contraindications

- Asthma or other significant bronchospastic disease
- Bradycardia
- Hypotension

Precautions

- Pregnancy safety: Category C
- Use caution in patients with COPD or reactive airway disease
- May mask symptoms of hypoglycemia in diabetic patients
- Should be used cautiously in patients with heart failure

Notes

- Monitor blood pressure frequently; excessive hypotension may occur
- May cause bronchospasm, especially in patients with asthma
- Has a more pronounced effect on diastolic blood pressure compared to systolic

Side effects:

- Dizziness, fatigue, headache, hypotension, bradycardia, nausea, vomiting, bronchospasm, orthostatic hypotension

Route: IV Push, IV Infusion

LEVALBUTEROL

Xopenex

Class: Sympathomimetic, Beta agonist, Short-acting bronchodilator

Action: Levalbuterol A is a beta₂-adrenergic agonist indicated for the treatment or prevention of bronchospasm in patients 4 years of age and older with reversible obstructive airway disease.

Indications:

- Bronchospasm

Contraindications

- None in emergent settings

Precautions

- Pregnancy safety: Category C
- Symptomatic tachycardia
- Paradoxical bronchospasm
- Hypokalemia

Notes

- Causes tachycardia so it may precipitate angina and dysrhythmias

Side effects:

- Headache, fatigue, light-headedness, irritability, restlessness, aggressive behavior, pulmonary edema, hoarseness, increased sputum, chest pain, palpitations, dry mouth, and tremors.

Route: Nebulized

LEVETIRACETAM

KEPPRA

Class: anticonvulsants

Action: The mechanism of action of Keppra is not known, but it is thought to bind to a protein called synaptic vesicle protein 2A (SV2A).SV2A regulates the release of hormones and signaling molecules in the brain, notably gamma-aminobutyric acid (GABA). GABA reduces the excitability of nerve cells, or the signals sent by nerve cells during normal signaling. Binding to SV2A is thought to make it more active, increasing the amount of GABA that is secreted, which reduces or “tunes down” nerve signaling and reduces seizure activity.

Indications:

- Seizures
- Status epilepticus

Contraindications

- None in emergent situation

Precautions

- Pregnancy Safety: Category C

Side effects:

- Ectopy

Route: IV, IO

LIDOCAINE HYDROCHLORIDE

XYLOCAINE

Class: anesthetic, antidysrhythmic

Action: *Local anesthetic:* Inhibits transport of ions across the neuronal membrane, blocking conduction of normal nerve impulses. *Antidysrhythmic:* Lidocaine is specifically indicated in the acute management of ventricular arrhythmias.

Indications:

- Local anesthetic post IO placement
- Ventricular Arrhythmia

Contraindications

- None in emergent situation

Precautions

- Pregnancy Safety: Category B

Notes

- Monitor for central nervous system toxicity
- Preferred antidysrhythmic if toxin caused dysrhythmia

Side effects:

- Ectopy

Route: IV, IO

LORAZEPAM

ATIVAN

Class: Benzodiazepine

Action: Anticonvulsant, sedative

Indications:

- Motor seizures
- Status epilepticus

Contraindications

- None in emergent situation

Precautions

Notes

- Dilute with normal saline (1:1) prior to intravenous administration
- Respiratory depression common

Side effects:

- Drowsiness, hypotension, amnesia, respiratory depression, apnea

Route: IV, IO, IM

MAGNESIUM SULFATE

Class: Electrolyte, anti-inflammatory

Action: Reduces striated muscle contractions and blocks peripheral neuromuscular transmission by reducing acetylcholine release at the myoneural junction. Induces uterine relaxation. Can cause bronchodilation after beta-agonists and anticholinergics have been administered. Is a smooth muscle relaxer.

Indications:

- Eclampsia
- Torsades de Pointes
- Hypomagnesemia
- Severe bronchospasm refractory to other treatments (specifically in asthma)

Contraindications

- Hypotension/hypoperfused
- Heart block

Precautions

- Pregnancy safety: category D
- Caution in patients receiving digitalis
- Use with caution in patients with renal failure

Side effects:

- Drowsiness, CNS depression, respiratory depression, respiratory tract paralysis, abnormal ECG, AV block, hypotension, vasodilation, hyporeflexia

Route: IV, IO, IM

METHYLPREDNISOLONE

SOLU-MEDROL

Class: Steroid

Action: Anti-inflammatory. Suppresses immune response. Upregulates beta receptors.

Indications:

- Severe anaphylaxis
- Asthma / COPD

Contraindications

- None in the emergency setting

Precautions

Notes

- Must be reconstituted
- Onset of action may be 2-6 hours

Side effects:

- Hyperglycemia
- Sleep disturbances

Route: IV, IM, IO

MIDAZOLAM

VERSED

Class: Benzodiazepine

Action: Hypnotic/sedative

Indications:

- Severe motor seizures
- Need for dissociation
- Acute severe anxiety
- Sympathomimetic toxicity
- Drug-Assisted Airway Management

Contraindications

- None in emergent situation

Precautions

Notes

- Emergency resuscitative equipment must be available
- Respiratory depression common
- Can have an augmented effect when administered with opioids

Side effects:

Drowsiness, hypotension, amnesia, respiratory depression, apnea

Route: IV, IO, IM, IN

MORPHINE

Class: Narcotic

Action: Central nervous system depressant. Causes peripheral vasodilation. Decreases sensitivity to pain.

Indications:

- Pain management

Contraindications

- None in emergent situation

Precautions

- Respiratory depression
- Hypotension
- Nausea

Side effects:

- Dizziness, altered level of consciousness

Route: IV, IO, IM, IN

NALOXONE

NARCAN

Class: Narcotic antagonist, antidote

Action: Competitive inhibition at narcotic receptor sites. Reverses respiratory depression secondary to opiate/opioid drug use. Inhibits the effect of opiate/opioid medications.

Indications:

- Narcotic overdoses

Contraindications

- Cardiac arrest

Precautions

- Pregnancy safety: Category C

Notes

- Not intended to return patient to baseline mental state.
- Assist ventilations prior to administration to avoid sympathetic stimulation
- May cause withdrawal effects in patients with narcotic dependence
- Short acting: half-life of Naloxone is often shorter than the narcotic.
- Not beneficial in cardiac arrest

Side effects:

- Restlessness, seizures, dyspnea, pulmonary edema, tachycardia, hypertension, dysrhythmias, cardiac arrest, nausea, vomiting, withdrawal symptoms in narcotic addicts, diaphoresis.

Route: IV, IO, IM, IN

NITROGLYCERIN

NITROGLYCERIN SPRAY, TRIDIL

Class: Antianginal

Action: Smooth muscle relaxant. Reduces cardiac work. Dilates coronary arteries. Dilates systemic arteries.

Indications:

- Acute coronary syndrome
- Congestive heart failure

Contraindications

- Erectile dysfunction medications taken in last 72 hours
- Hypotension / hypoperfusion

Precautions

- Constantly monitor blood pressure
- Right ventricular infarct
- Syncope

Side effects:

- Headache, dizziness, hypotension

Route: IV, PO, Sublingual, Transdermal

NOREPINEPHRINE

LEVOPHED

Class: Sympathomimetic, vasopressor

Action: Potent alpha-agonist resulting in intense peripheral vasoconstriction while also having mild-moderate Beta activity (positive chronotropic).

Indications:

- Cardiogenic shock
- Septic shock
- Hypotension/hypoperfusion

Contraindications

- None

Precautions

- Pregnancy safety: C
- Hemorrhagic shock

Side effects:

- Headache, anxiety, dizziness, restlessness, dyspnea, bradycardia, hypertension, dysrhythmias, chest pain, peripheral cyanosis, cardiac arrest, nausea, vomiting, urinary retention, renal failure, decreased blood flow to GI tract, kidneys, skeletal muscle, and skin, tissue necrosis from extravasation

Route: IV, IO

ONDANSETRON HYDOCHLORIDE

ZOFRAN

Class: Serotonin receptor antagonist, antiemetic

Action: Blocks the serotonin receptors. Serotonin is a natural substance that causes nausea and vomiting.

Indications:

- Nausea, vomiting

Contraindications

- None in emergent situations

Precautions

- Pregnancy safety: Category B

Notes

- Constantly monitor blood pressure
- Use with caution during pregnancy or while breastfeeding
- Use with caution in patients on serotonin reuptake inhibitors

Side effects:

- Headache, malaise, wheezing, bronchospasm, atrial fibrillation, abnormal ECG, prolonged QT interval, ST segment depression, 2nd degree AV block, constipation, diarrhea, hives, skin rash.

Route: IV, IO, IM, PO

ORAL GLUCOSE

Dextrose / Oral Glucose Gel

Class: Carbohydrate, antihypoglycemic

Action: Elevates serum glucose levels. Short-term osmotic diuresis.

Indications:

- Hypoglycemia

Contraindications

- Inability to protect the airway
- Absent gag reflex
- Unable to swallow safely
- Unconscious/Unresponsive
- Active vomiting

Precautions

- Pregnancy safety: Category C
- Risk of aspiration
- Consider complex carbohydrates after return to normal mentation.

Notes

- Use only when the patient can protect their airway and safely swallow. Consider complex carbohydrates after correction of hypoglycemia and return to normal mentation. If hypoglycemia resolves but the blood glucose drops again within 10 minutes, consider sulfonylurea toxicity/overdose, which can be especially dangerous in pediatric patients.

Side effects:

- Hyperglycemia. Nausea or vomiting may occur. Risk of aspiration if administered to a patient who cannot protect their airway.

Route: PO

PROMETHAZINE

PHENERGAN

Class: Phenothiazine, antiemetic, antihistamine

Action: H-1 receptor antagonist; blocks action of histamine; possesses sedative, anti-motion, antiemetic, and anticholinergic activity; potentiates the effects of narcotics to induce analgesia

Indications:

- Nausea
- Vomiting

Contraindications

- None in emergent situations

Precautions

- Pregnancy safety: Category C
- Use caution in patients with asthma, peptic ulcer, or bone marrow suppression.
- Extravasation (infiltration) can cause tissue damage

Side effects:

- Headache, dizziness, drowsiness, confusion, restlessness, wheezing, chest tightness, thickening of bronchial secretions, palpitations, bradycardia, reflex tachycardia, QT prolongation, postural hypotension, diarrhea, nausea, vomiting

Route: IV

RACEMIC EPINEPHRINE

11.25mg/0.5mL (2.25%)

Class: Alpha and Beta receptor agonist

Action: Stimulates alpha-adrenergic receptors causing vasoconstriction, leading to decreased mucosal edema in the upper airway. Stimulates beta-adrenergic receptors causing bronchodilation. Results in improved airflow in patients with upper airway obstruction.

Indications:

- Suspected croup

Contraindications

- Hypersensitivity
- Epiglottitis

Precautions

- Use caution in patients with underlying cardiac disease due to potential beta-adrenergic effects
- Monitor for rebound airway obstruction after initial improvement
- Should be protected from light

Side effects:

- Tachycardia, palpitations, anxiety, tremors, hypertension

Route: AT

ROCURONIUM

ROCURONIUM BROMIDE

Class: Nondepolarizing neuromuscular blocker

Action: Antagonizes acetylcholine at the motor end plate producing skeletal muscle paralysis.

Indications:

- Drug assisted airway intubation

Contraindications

- None in an emergency setting

Precautions

- Pregnancy safety: Category B
- Airway obstruction (partial/full)

Side effects:

- Bronchospasm, respiratory depression, apnea, dysrhythmias, nausea, vomiting

Route: IV, IO

SODIUM BICARBONATE

Class: Systemic hydrogen ion buffer, alkalizing agent

Action: Buffers metabolic acidosis and lactic acid buildup in the body caused by anaerobic metabolism secondary to severe hypoxia by reacting with hydrogen ions to form water and carbon dioxide.

Indications:

- Tricyclic antidepressant, Aspirin, and Phenobarbital overdose
- Hyperkalemia
- Crush injuries

Contraindications

- None in emergent situations

Precautions

- Pregnancy safety: Category C
- Electrolyte imbalance due to severe vomiting or diarrhea

Notes

- Repeat as needed with TCA overdoses until QRS complex narrows
-
- Avoid contact with other medications (may deactivate)
- Heart failure and renal disease

Side effects:

- Hyponatremia, alkalosis, tissue sloughing, cellulitis, necrosis at injection site, seizures, fluid retention, hypokalemia, electrolyte imbalance, tetany, sodium retention, peripheral edema.

Route: IV, IO

SUCCINYLCHOLINE

Class: Depolarizing neuromuscular blocker

Action: Succinylcholine chloride is a short-acting depolarizing neuromuscular blockade producing skeletal muscle paralysis

Indications:

- Drug assisted airway intubation

Contraindications

- None in an emergency setting

Precautions

- Major burns
- Major trauma
- History of malignant hyperthermia (self or family)

Side effects:

- Hyperkalemia, malignant hyperthermia

Route: IV, IO

TRANEXAMIC ACID

TXA

Class: Anti-fibrinolytic

Action: Competitively inhibits multiple plasminogen binding sites, decreasing plasmin formation and fibrinolysis

Indications:

- Significant blood loss or hypotension, tachycardia, and time of injury < 3 hours

Contraindications

- None in emergent situations

Precautions

- Must be administered within 3 hours of injury
- Hypotension has been observed with rapid IV administration.
- Pregnancy class C

Notes

Side effects:

- Hyponatremia, alkalosis, tissue sloughing, cellulitis, necrosis at injection site, seizures, fluid retention, hypokalemia, electrolyte imbalance, tetany, sodium retention, peripheral edema.

Dosage

Route: IV, IO, Nebulized, Oral, IN

PROCEDURE

BLS or ALS

Indications:

- When should the procedure be performed

Contraindications

- When the procedure shall not be performed

Precautions

- Side effects
- Points of caution

Equipment Needed

-

Procedure

1. Stepwise approach to successfully complete the procedure

12-Lead Electrocardiography

ALS procedure

Indications:

- Chest Pain/Discomfort
- Epigastric Pain
- Syncope/Dizziness
- Unexplained Diaphoresis
- Dyspnea
- Weakness
- Dysrhythmia
- Atraumatic thoracic pain

Contraindications

- None

Precautions

- None

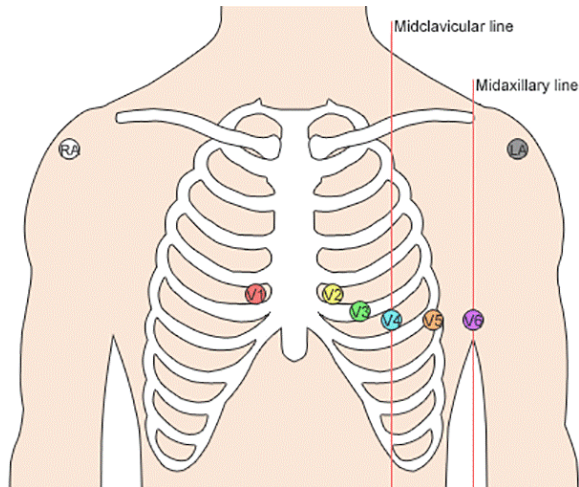
Equipment Needed

- Monitor/ defibrillator
- Electrodes
- Razor

Procedure

1. Place patient in Semi-Fowler's position
2. Shave excess hair and dry skin
3. Remove shirt/clothing covering the chest while protecting modesty
4. Place limb lead electrodes on deltoids and thighs
5. Place precordial lead electrodes
6. Assure that wires are not over the arms to reduce artifact
7. Enter patient age and gender then obtain tracing

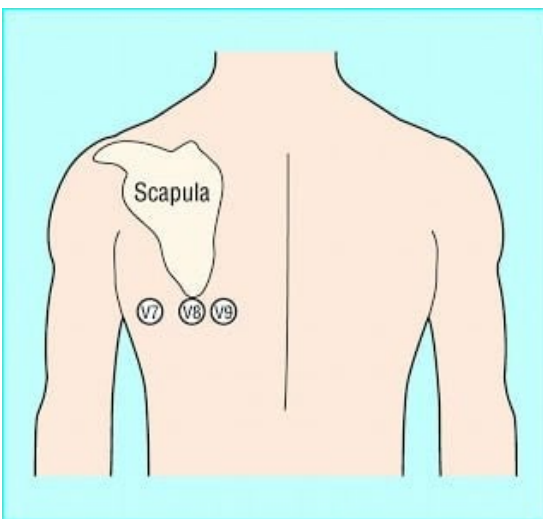
Standard Placement



- V1:** 4th interspace right parasternal border
- V2:** 4th interspace left parasternal border
- V3:** Diagonally between V2 and V4
- V4:** 5th interspace at Lt. mid-clavicular line
- V5:** Lt. anterior-axillary line, same plane as V-4
- V6:** Lt. mid-axillary line, same plane as V-4

V3 & V4 may be mirrored to Right chest for a "right-sided 12-lead"

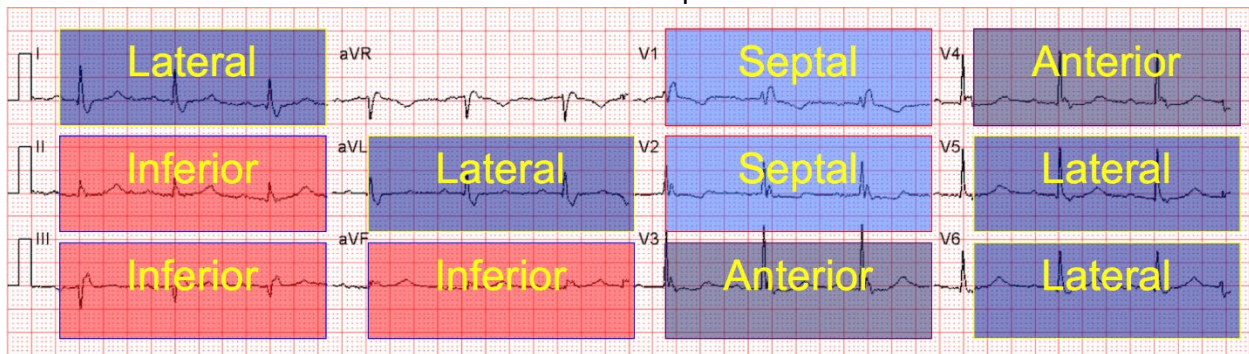
Posterior Placement



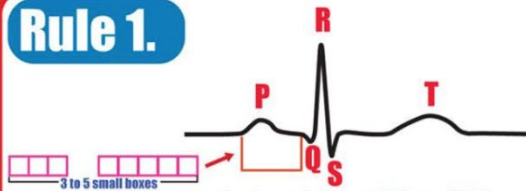
Electrodes wrap around to back at same level of standard V6 placement:

- V7:** Move V4 to posterior axillary line
- V8:** Move V5 to mid-scapular line
- V9:** Move V6 to Left paraspinal border

Anatomical Perspectives

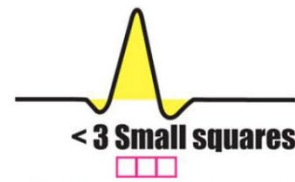


Rule 1.



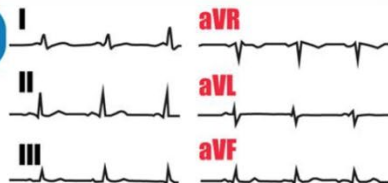
PR interval should be 120 to 200 milliseconds or 3 to 5 little squares.

Rule 2.



The width of the QRS complex should not exceed 110 ms, less than 3 little squares.

Rule 3.



The QRS complex should be dominantly upright in leads I and II.

Rule 4.



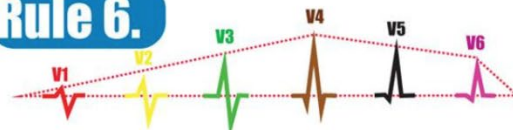
QRS and T waves tend to have the same direction in the limb leads.

Rule 5.



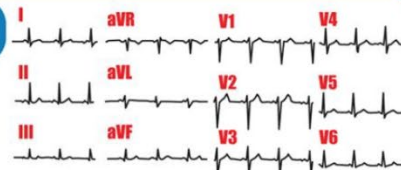
All waves are negative in lead aVR

Rule 6.



**The R wave must grow from V1 to at least V4
The S wave must grow from V1 to at least V3
and disappear in V6**

Rule 7.



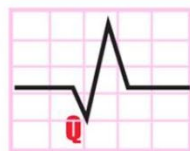
The ST segment should start isoelectric except in V1 and V2 where it may be elevated.

Rule 8.



The P waves should be upright in I, II, and V2 to V6

Rule 9.



There should be no Q wave or only a small q less than 0.04 seconds in width in I, II, V2 to V6.

Rule 10.



The T wave must be upright in I, II, V2 to V6

Cardiac Monitoring

ALS procedure

Indications:

- ALS Patients

Contraindications

- None

Precautions

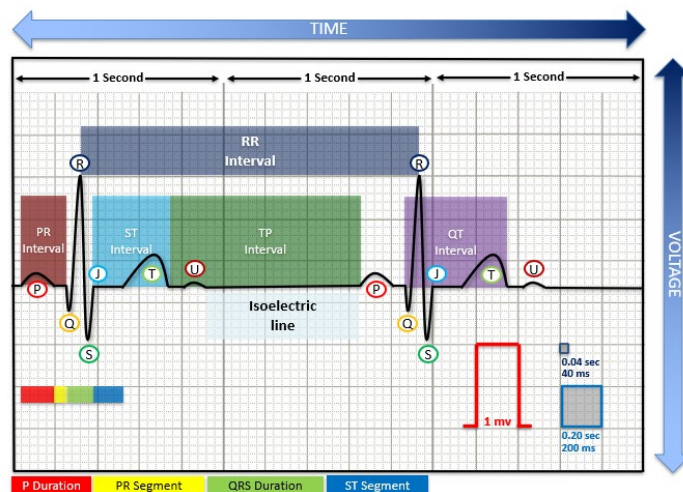
- None

Equipment Needed

- Monitor/ defibrillator
- Electrodes
- Razor

Procedure

1. Analyze the rate (six-second or triplicate method)
2. Analyze the rhythm (regular, irregular, pattern)
3. Analyze the P-waves (present, regular, upright, or inverted)
4. Analyze the P-R interval (normal duration 120 - 200 ms)
5. Analyze the QRS complex (normal duration 40 - 120 ms)



STEP 1 Check ECG machine & paper settings

1mm (small square)
= 0.04 sec or 0.1 mV

5mm (big square)
= 0.2 sec or 0.5 mV

VOLTAGE

0.5
mV
=
5 mm

TIME

Standard paper speed
is 25mm/second.

Calibration is
usually set to 1mV
= 10 small squares.

*ECG's should be recorded at 0.05 – 150 Hz

STEP 2 Check Heart Rate & Rhythm

Normal Heart Rate: 60 - 100 bpm

Abnormal:

<60 Bradycardia

> 100 Tachycardia

Normal: Regular

Confirm heart rate with rhythm strip = 300 divided the number of large squares between 2 complexes.

Abnormal: Irregular

Confirm heart rate with rhythm strip = The number of complexes in 30 large squares (6 secs) multiplied by 10.

STEP 3 P wave analysis

Normal: Are they all sinus, same morphology & present and conducting followed by QRS, none hidden?
Abnormal: Non sinus P waves, absent, inverted, or not conducting. P wave is typically biphasic in V1.

Left Atrial Enlargement

II, III, aVF
Bifid P wave with
> 40 ms between
the two peaks
P wave duration
> 110 ms

Right Atrial Enlargement

II, III, aVF
> 2.5 mm in the
inferior leads
> 1 small box

Bifid P wave

Amplitude ≥ 2.5mm
Duration ≥ 120 ms

Bifid P wave

Initial positive deflection ≥ 1.5mm tall
Terminal negative deflection ≥ 1mm deep
Terminal negative deflection ≥ 40 ms duration

Other causes of P wave abnormalities include:

Multifocal Atrial Tachycardia, Atrial Tachycardia, Atrial Ectopics, Accelerated Junctional Rhythm (AJR).

*P waves should be upright in I, II & V2 to V6

STEP 4 PR interval analysis

Normal: 0.12 - 0.20 secs 3 to 5 small boxes

Abnormal:

Short PR: WPW, ectopic atrial pacemaker or AV junctional rhythm.
<0.12 secs (3-small boxes)

Prolonged PR: First degree AV block
>0.20 secs (5-small boxes)

PR Depression: Causes include Acute Pericarditis, Atrial ischemia.

First Degree AV Block

Second Degree AV Block

Mobitz Type I (Wenckebach)

Second Degree AV Block

Mobitz Type 2

3rd Degree AV Block

*Consider AV conduction blocks or delays.

STEP 5 QRS analysis

Normal: QRS wave < 3 Small squares

Abnormal: > 3 Small squares

High or low voltages

Cardiac axis

Lead I, Lead aVF, Normal, Left, Right, Extreme Rt

QRS/QT duration Narrow, Wide, Short QT, Long QT

Pathological Q wave

> 1mm wide
> 2mm deep
> 25% of depth of QRS complex

Fragmented QRS

*QRS should be dominantly upright in I & II

STEP 6 ST segment & T wave analysis

Normal: ST segment

isoelectric except V1

Abnormal: Elevation

and depression of ST segment.

Normal T waves are

asymmetrical

Symmetrical

Asymmetrical

Biphasic

Peaked

*T wave is always upright in leads I, II, V3-6, and always inverted in lead aVR.

STEP 7 R wave progression in the precordial leads

*The R wave must grow from V1 to at least V4.
*The S wave must grow from V1 to at least V3 and disappear in V6.

V1 V2 V3 V4 V5 V6

Early R wave progression from V1-V6

Normal R wave progression from V1-V6

Typically the tallest R wave peaks at V4 or V5.

Late R wave progression from V1-V6

STEP 8 Consider MI & ACS

NSTEMI vs acute MI

STD **TWI** **STE**

STEMI mimics

BER, LBBB, Brugada, Pericarditis, LV aneurysm, vasospasm, Osborn wave

STEMI equivalents

De Winter ST-T, Isolated Posterior STEMI, Wellens, New RBBB with LAFB, Sgarbossa Criteria

STEP 9 Consider other issues & abnormalities

U wave

Abnormal electrolytes

HYPOKALEMIA **HYPERKALEMIA** **HYPOCALCEMIA** **HYPERCALCEMIA**

Drugs & medication related
i.e. antiarrhythmic drugs

*Quinidine, phenothiazines and tricyclic antidepressants
*Verapamil *Digoxin toxicity *Amiodarone *Phenytoin

ECG machine/operator errors checks

Lead misplacement and reversal, artefact issues etc.

CPAP: Z-Vent

ALS procedure

Indications:

- Dyspnea
- Pulmonary Edema
- Cardiogenic Pulmonary Edema
- Hypoxia
- Acute Respiratory Distress Syndrome
- Bronchospasm

Contraindications

- Altered Mental Status
- Hypotension
- Upper airway obstruction
- Respiratory Failure
- Pneumothorax
- Vomiting

Equipment Needed

- Z-Vent
- CPAP Circuit
- CPAP Mask
- Oxygen Cylinder
- Cardiac Monitor
- Capnography

Procedure

1. Position patient in sitting or high Fowler's
2. Continuously monitor BP, ECG, EtCO₂, and SpO₂
3. Connect Appropriate CPAP Circuit
4. Turn on Z-Vent
5. Select *Mask CPAP* on device menu
6. Perform Circuit Check
7. Witness *Patient Disconnect & High Pressure Alarms*
8. Set PEEP to 5 cmH₂O
9. Set PS to 5 cmH₂O
10. Complete the Confirmation Checklist
11. Apply mask with circuit attached to patient
12. Reassess the patient
13. Confirm Tidal Volume (V_t) and Minute Volume (V_{min})

Cricothyrotomy: Quicktrach

ALS procedure

Indications:

- Unable to oxygenate or ventilate

Contraindications

- Capable of providing less invasive means of ventilation and oxygenation.
- Inability to identify anatomical landmarks
- Tracheal Dissection

Precautions

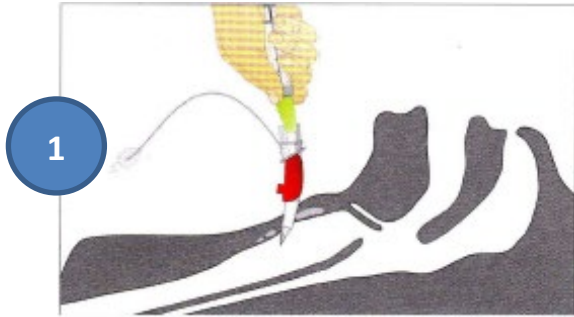
- Anatomical abnormalities

Equipment Needed

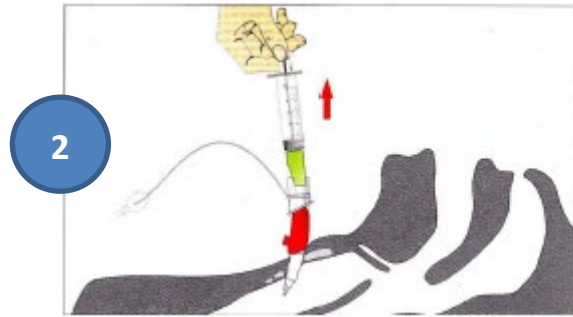
- | | |
|-----------------------------------|-------------------------|
| • Antiseptic | • BVM/Ventilator |
| • Quicktrach/Quicktrach II device | • Suction with catheter |
| • 10 mL syringe | • Capnography |

Procedure

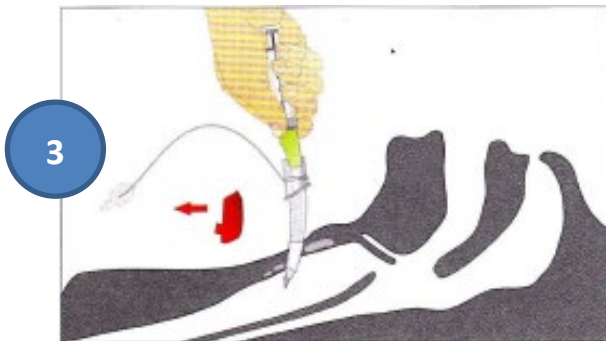
1. Hyperextend the head.
2. Evacuate the cuff completely.
3. Locate the cricothyroid membrane.
4. Puncture cricothyroid membrane.
5. Because of the sharp tip and conical shape of the needle, an incision is not necessary.
6. The opening of the trachea is obtained by dilating the skin.
7. Insert the Quicktrach further towards the trachea up to the stopper.
8. Aspirate air with the syringe to determine the position of the cannula.
9. Remove the stopper from the plastic cannula.
10. Push the plastic cannula forward with the thumb until the safety clip audibly clicks.
11. Further insert the Quicktrach until the flange rests on the neck.
12. The metal needle can now be removed.
13. Inflate the cuff with the prepared syringe (10mL).
14. Secure the plastic cannula with the foam necktape.
15. Ventilate the patient.



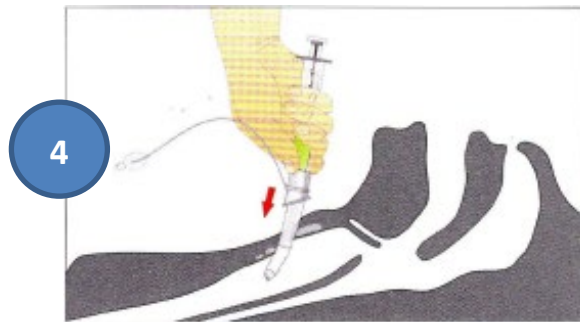
1
Hyperextend the head. Evacuate the cuff completely. Locate the cricothyroid membrane by palpation of the depression between the thyroid and cricoid cartilage and puncture. Because of the sharp tip and conical shape of the needle, an incision is not necessary. The opening of the trachea is obtained by dilating the skin which reduces the risk of bleeding. Insert the Quicktrach further towards the trachea up to the stopper. The stopper prevents the needle from being inserted too deep and therefore avoids perforation of the posterior tracheal wall.



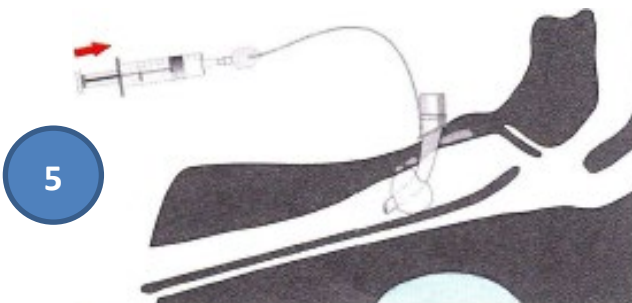
2
Aspirate air with the syringe to determine the position of the cannula. If this is possible, the needle is in the trachea.



3
Remove the stopper from the plastic cannula.



4
Push the plastic cannula forward with the thumb until the safety clip audibly clicks into position. This indicates that the tip of the metal needle is covered by the plastic cannula to prevent trauma. Further insert the Quicktrach until the flange rests on the neck. The metal needle can now be removed.



5
Inflate the cuff with the prepared syringe (10ml).



6
Secure the plastic cannula with the foam neck tape. Ventilate the patient via the 15mm standard connector.

Cricothyrotomy (Needle)

ALS procedure

Indications:

- Unable to oxygenate or ventilate
- Recommended technique for pediatric patient

Contraindications

- Capable of providing less invasive means of ventilation and oxygenation.
- Inability to identify anatomical landmarks
- Tracheal Dissection

Precautions

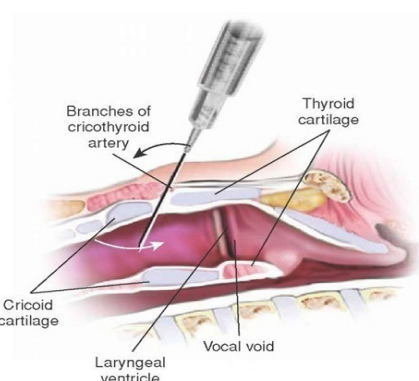
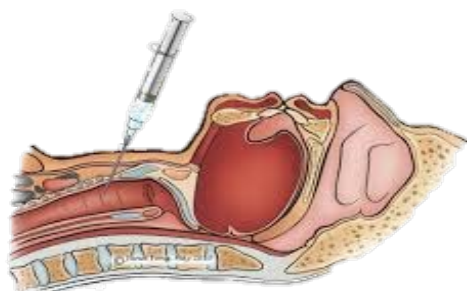
- Anatomical abnormalities

Equipment Needed

- | | |
|---------------------------|-------------------------|
| • Antiseptic | • BVM/Ventilator |
| • 14 gauge catheter (ARS) | • Suction with catheter |
| • 3 mL syringe | • Bulky dressing |
| • 3 mm ET tube adapter | • Capnography |

Procedure

1. Position patient Supine with head and neck midline in the neutral position
2. Locate the Cricothyroid Membrane
3. Connect 14 ga needle/catheter to the 3 mL syringe
4. Insert needle/catheter toward the midline, caudally, and posteriorly at 45-degree angle
5. Aspirate air to confirm tracheal placement
6. Advance the catheter and remove the needle after placement is confirmed
7. Connect the 3 mm adapter and attach BVM
8. Ventilate with capnography attached



Cricothyrotomy (Surgical)

ALS procedure

Indications:

- Unable to oxygenate or ventilate

Contraindications

- Capable of providing less invasive means of ventilation and oxygenation.
- Inability to identify anatomical landmarks
- Tracheal Dissection
- Pediatric patients under 10 y/o

Precautions

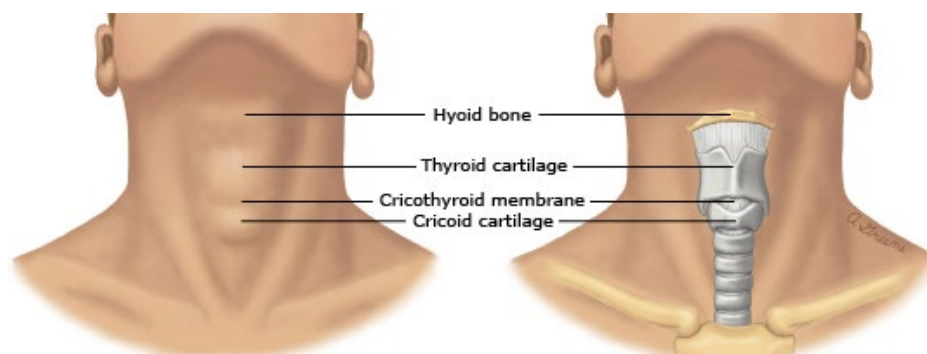
- Anatomical abnormalities

Equipment Needed

- | | |
|-------------------------|-------------------------|
| • Antiseptic | • Bougie |
| • Scalpel | • BVM/Ventilator |
| • 10 mL syringe | • Suction with catheter |
| • Forceps/Introducer | • Bulky dressing |
| • 5.5 Endotracheal Tube | • Capnography |

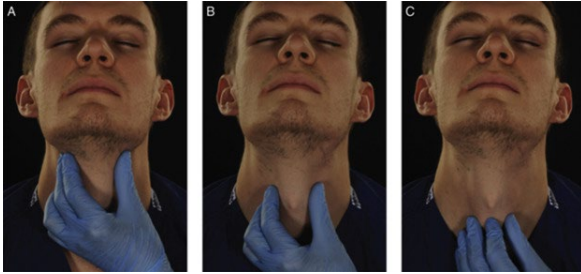
Procedure

1. Position patient Supine with head and neck midline in the neutral position
2. Locate the Cricothyroid Membrane using “Laryngeal Handshake”
3. Make a 3 cm vertical incision in the skin with the scalpel
4. Puncture the exposed membrane horizontally with the scalpel
5. Insert handle of scalpel/forceps/finger to dilate the opening
6. Introduce the bougie through the stoma until the cuff is in the trachea
7. Introduce the ET tube over the bougie until the cuff is in the trachea
8. Inflate the cuff, remove the bougie and secure the tube
9. Ventilate with capnography attached



Surgical Cricothyrotomy Job Aid

STEP 1 "Laryngeal Handshake"



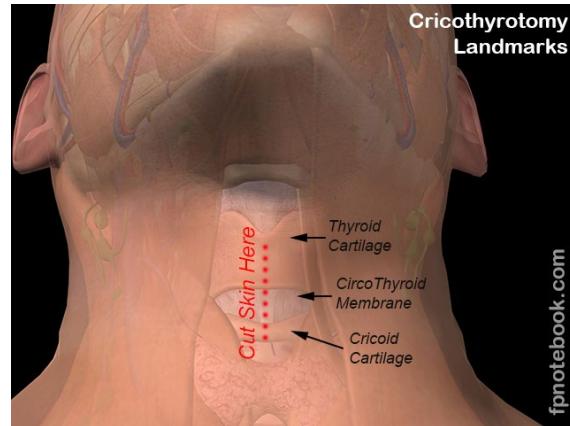
(A) The index finger and thumb grasp the top of the larynx (the greater cornu of the hyoid bone) and roll it from side to side. The bony and cartilaginous cage of the larynx is a cone, which connects to the trachea. **(B)** The fingers and thumb slide down over the thyroid laminae. **(C)** Middle finger and thumb rest on the cricoid cartilage, with the index finger palpating the cricothyroid membrane.

STEP 2

Prep the site with aseptic technique.

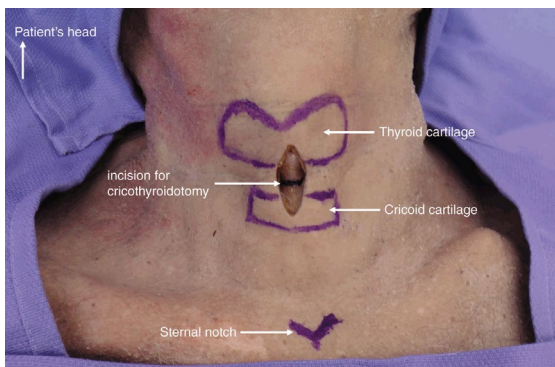
Step 3

Make vertical incision.



Step 4

Make horizontal puncture through cricothyroid membrane.



Step 5

Dilate the opening & insert tube.



Double Sequential Defibrillation (DSD)

ALS procedure

Indications:

- Refractory Ventricular Fibrillation
- Refractory Ventricular Tachycardia

Contraindications

- None

Precautions

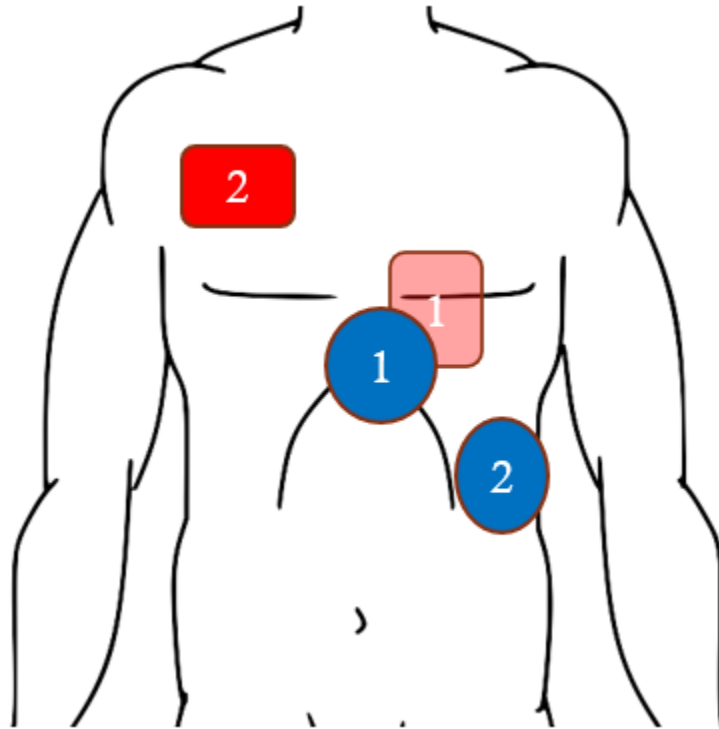
- Make sure patient is dry

Equipment Needed

- Monitor/ defibrillator
- Second monitor / defibrillator
- Defibrillator pads
- Razor

Procedure

1. Apply pads for both monitors
 - a. Monitor 1: Anterior / Posterior placement
 - b. Monitor 2: Anterior / Lateral (traditional) placement
2. One paramedic performs the following
 - a. Charge monitors
 - b. Verbalize "CLEAR"
 - c. Use the same finger for the following:
 - i. Press the "SHOCK" button on monitor 1
 - ii. Press the "SHOCK" button on monitor 2
3. Reassess patient



Manual Defibrillation

ALS procedure

Indications:

- Ventricular Fibrillation
- Pulseless Ventricular Tachycardia

Contraindications

- None

Precautions

- Make sure patient is dry

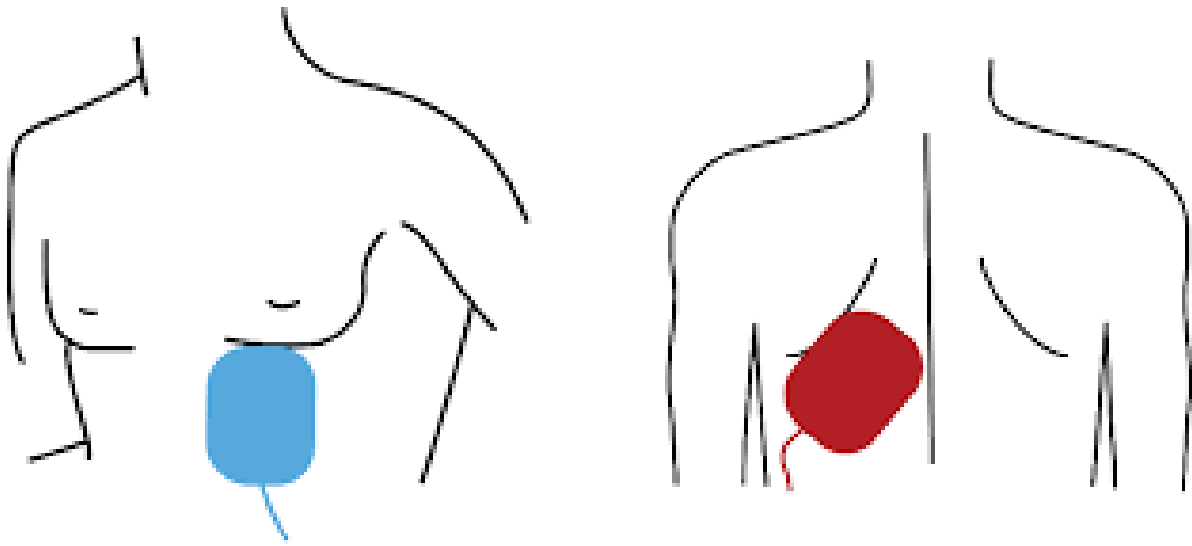
Equipment Needed

- Monitor/ defibrillator
- Defibrillator pads
- Razor

Procedure

1. Apply pads
 - a. Vector 1: Anterior / Posterior placement
 - b. Vector 2: Anterior / Lateral placement
2. One paramedic performs the following
 - a. Charge monitor to manufacture recommended settings
 - b. Confirms shockable rhythm
 - c. Verbalize "CLEAR"
 - d. Press the "SHOCK" button
3. Continue CPR

Anterior/Posterior Pad Placement



Anterior / Lateral Placement



Endotracheal Intubation: Direct Laryngoscopy

ALS procedure

Indications:

- Patient unable to maintain airway
- Prolonged artificial ventilation
- High risk of aspiration

Contraindications

- None when indicated

Precautions

- Severe trauma
- Cervical spinal injury
- Assess for difficult airway

Equipment Needed

- | | |
|---|---------------------------------|
| • Proper size endotracheal tube | • Suction available |
| • Laryngoscope and properly sized blade | • Magill Forceps within reach |
| • 10 mL syringe | • Rescue airway (BIAD) prepared |
| • Proper size stylet for ET tube | • Nasal Cannula |
| • Bougie if available | • BVM or mechanical ventilator |
| • Waveform capnography | • Stethoscope |
| • ET tube securing device | • Lubricant |

Procedure

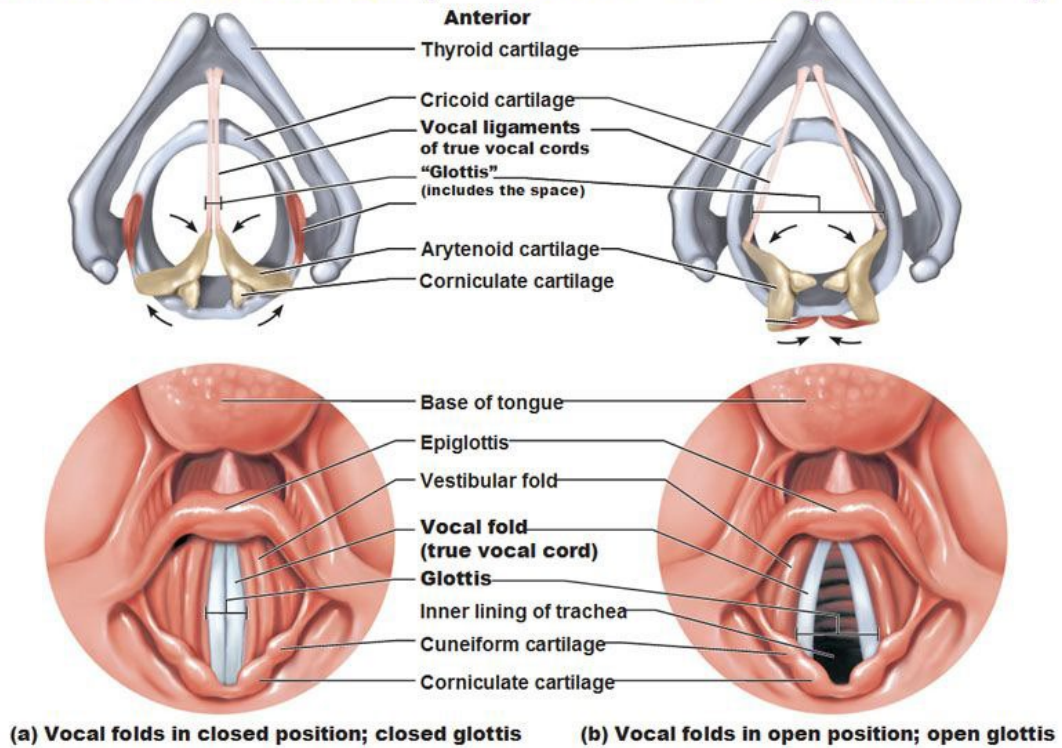
Prepare:

1. Don gloves and eye protection
2. Monitor EtCO₂, ECG, SpO₂, and BP
3. Position patient with Airway Axis Alignment
4. Place patient on high flow Nasal Cannula
5. Pre-Oxygenate and Pre-Ventilate
6. Suction within reach
7. Assign Team Roles
8. Rescue airway prepared
9. "Failed Airway Plan" verbalized to team
10. Test ET Tube cuff & leave syringe attached
11. Place tube holder in position
12. Perform LEMON Exam

Perform:

1. Remove OPA if present
2. Grip the laryngoscope handle in Left hand
3. Use Right hand to open mouth
4. Direct blade downward and insert into mouth
5. Progress blade slowly to view glottic opening
a. Do not make contact with teeth
6. Suction as needed
7. Utilize bougie if available
8. Introduce ET tube and pass through vocal cords
9. Assure the cuff is about ½ " past the glottis
10. Remove stylet (or bougie)
11. Inflate cuff with 1 mL of air per 1 mm tube size
12. Ventilate patient while auscultating
13. Confirm placement with capnography
14. Secure the ET tube with appropriate device

True Vocal Cords (= “Folds” or “Ligaments”)



Endotracheal Intubation: Video Laryngoscopy

ALS procedure

Indications:

- Patient unable to maintain airway
- High risk of aspiration

Contraindications

- None when indicated

Precautions

- Assess for difficult airway

Equipment Needed

- | | |
|---------------------------------|---------------------------------|
| • Proper size endotracheal tube | • Suction available |
| • Video laryngoscope | • Magill Forceps within reach |
| • 10 mL syringe | • Rescue airway (BIAD) prepared |
| • Proper stylet for VL device | • Nasal Cannula |
| • Bougie | • BVM or mechanical ventilator |
| • Waveform capnography | • Stethoscope |
| • ET tube securing device | • Lubricant |

Procedure

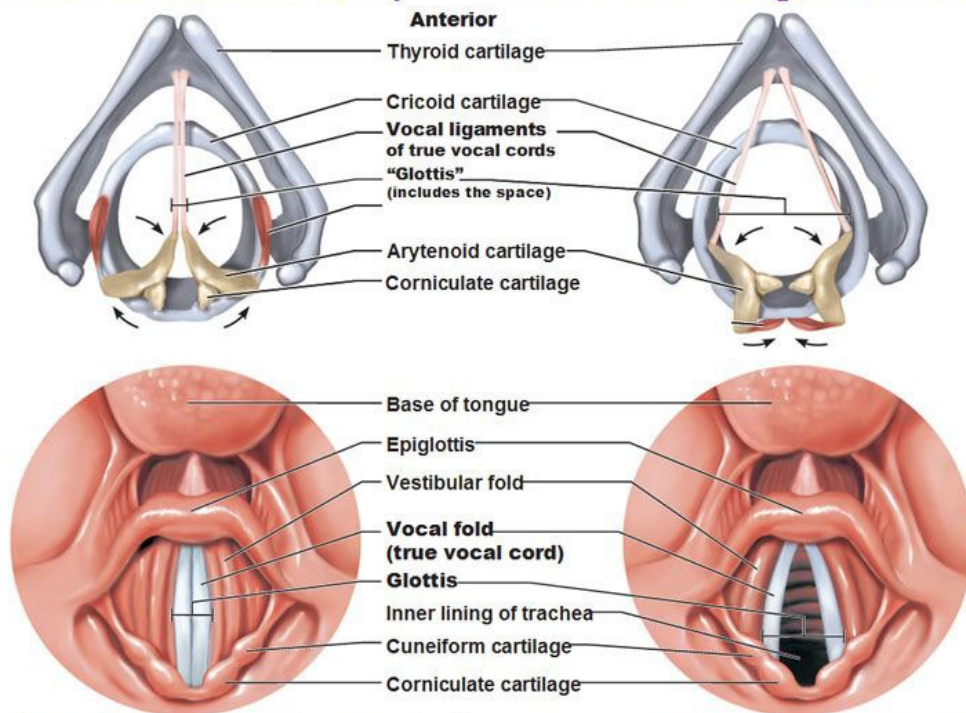
Prepare:

1. Don gloves and eye protection
2. Monitor EtCO₂, ECG, SpO₂, and BP
3. Position patient with Airway Axis Alignment
4. Place patient on high flow Nasal Cannula
5. Pre-Oxygenate and Pre-Ventilate
6. Suction within reach
7. Assign Team Roles
8. Rescue airway prepared
9. "Failed Airway Plan" verbalized to team
10. Test ET Tube cuff & leave syringe attached
11. Place tube holder in position
12. Perform LEMON Exam

Perform:

1. Airway Axis Alignment
2. Hold device in left hand and introduce blade into oropharynx
3. Watch the screen while gliding down the surface of the tongue
4. Manipulate device to visualize the palate, uvula, then epiglottis
5. Advance tip of blade into vallecula and gently lift the tip to expose glottis
6. Optimize view
7. Insert the pre-curved stylet loaded with an ET tube on the Right side of mouth
8. Visualize the tube on the screen
9. Angle and rotate tube until tip is aligned with glottis
10. Advance the ET tube and withdraw the stylet
11. Confirm placement
12. Withdraw VL device
13. Inflate cuff of ET to appropriate pressure
14. Turn off VL device

True Vocal Cords (= “Folds” or “Ligaments”)



(a) Vocal folds in closed position; closed glottis

(b) Vocal folds in open position; open glottis



Escharotomy

ALS procedure

Indications:

Eschar compressing or potentially compressing tissue in, or surrounding burn area Compressed tissue is identified by any of the following:

- Absent distal arterial flow as determined by pulses, skin color, and capillary refill.
- An SpO₂ < 95% in the distal end of the extremity as shown by pulse oximetry in the absence of systemic hypoxia.
- Impending or established respiratory compromise due to circumferential torso or neck burns.

Contraindications

- None

Precautions

Complications from the procedure include:

- Bleeding
- Infection
- Damage to neurovascular structures
- Inadvertent fasciotomy

Equipment Needed

- Cleansing solution, such as povidone-iodine or chlorhexidine
- Scalpel
- Sterile gauze

Procedure

Relevant Anatomy

- Eschars occur in full-thickness burns or, less commonly, deep partial-thickness burns through at least the dermis.
- A properly executed escharotomy releases the eschar to the depth of subcutaneous fat only. This release results in minimal bleeding, which can be controlled with local pressure or electrocautery.
- Major neurovascular structures to avoid when incising at the following sites include:
 - Elbow: Ulnar nerve
 - Wrist: Radial nerve
 - Fibular head: Superficial peroneal nerve
 - Ankle: Posterior tibial artery
 - Neck: Jugular veins

General

- Cleanse the site with povidone-iodine or chlorhexidine solution.
- If burns are particularly painful, utilize pain management guideline.
- Unsedated patients may benefit from procedural sedation.

Limbs

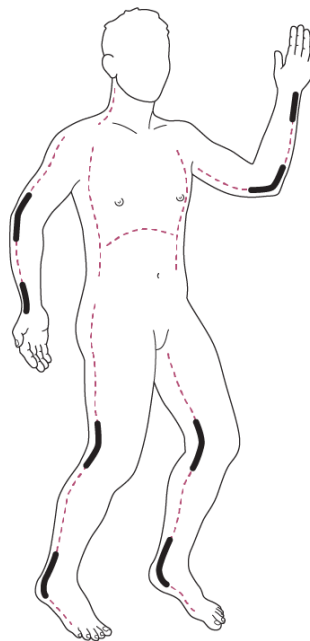
1. Using sterile technique, incise the lateral and medial aspects of the involved extremity with a scalpel from 1 cm proximal to the burned area to 1 cm distal to the involved area of constricting burn.
2. Avoid vital structures such as major arteries and nerves (eg, ulnar nerve at the elbow, the radial nerve at the wrist, the superficial peroneal nerve near the fibular head, the posterior tibial artery at the ankle).
3. Carry the incision only through the full thickness of skin. Incisions should cross joints. This incision should result in immediate separation of the constricting eschar to expose subcutaneous fat.
4. In circumferential burns of the hands, extend the incisions to the thenar and hypothenar aspects of the hand.
5. In circumferential burns of the feet, extend the incision to the great toe medially and the little toe laterally.
6. Reassess perfusion: A properly done escharotomy results in near-immediate softening of the tissue, improved distal tissue perfusion, sensation, Doppler flow signal strength, and oximetry values. If perfusion fails to improve after the procedure, reassess the escharotomy depth and location and reincise any insufficiently deep incisions.

Chest

1. Using sterile technique, incise the chest wall from the clavicle to the costal margin in the anterior axillary line bilaterally; avoid breast tissue in females (see figure Escharotomy incision sites). Consider joining this by transverse incisions to result in a chevron-shaped subcostal incision.
2. Assess response: Increased airway pressure or an inability to ventilate is evidence of the need to reincise the eschar.

Escharotomy incision sites

The dashed lines are the preferred escharotomy incision sites. The bold lines are areas where vascular structures and nerves may be damaged by escharotomy incisions.



Gastric Decompression

ALS procedure

Indications:

- Gastric decompression in unresponsive or intubated patients
- Removal of gastric contents (air, blood, toxins as appropriate)

Contraindications

- Known esophageal varices
- Recent esophageal or gastric surgery
- Nasogastric route contraindicated in patients with severe facial trauma with suspected basal skull fracture

Precautions

- Increased risk of aspiration—have suction ready
- Do not force the tube if resistance is met
- Confirm placement before use

Equipment Needed

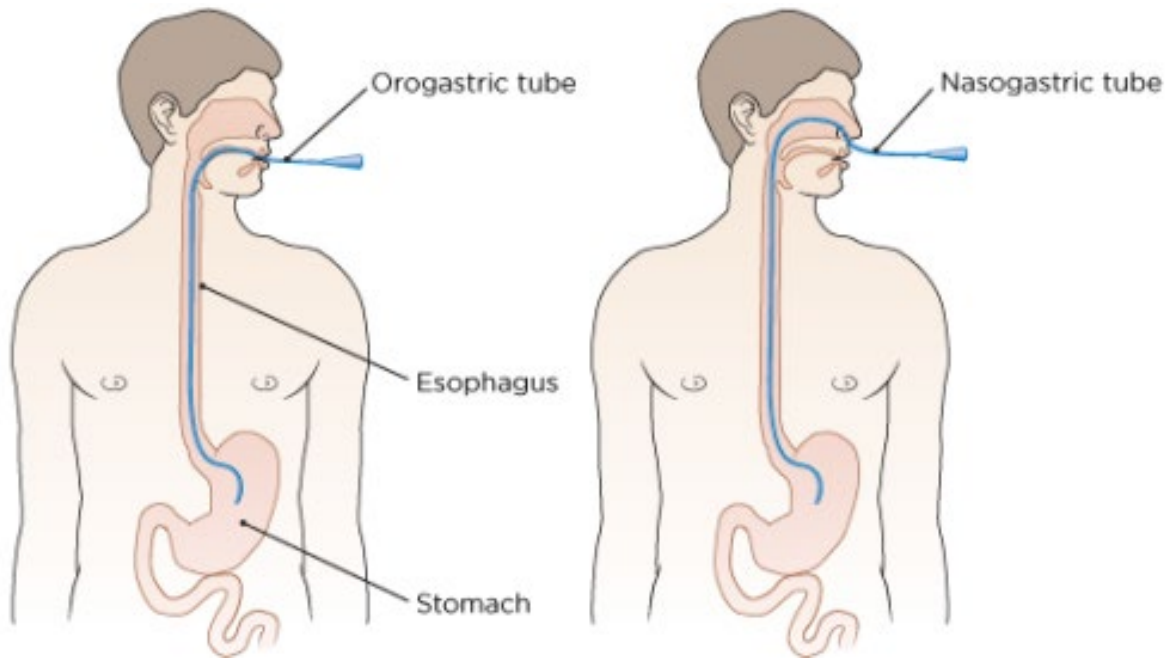
- Appropriate size OG tube (e.g., 10-14 Fr adult, 8-10 Fr pediatric)
- Lubricant (water-soluble)
- 60 mL syringe
- Stethoscope
- Suction setup
- Tape or securing device
- PPE (gloves, eye protection)

Procedure

1. **Prepare**
 - a. Don PPE (gloves, eye protection)
 - b. Position unresponsive patient: supine with head in neutral position (or slightly elevated if not contraindicated)
 - c. Measure OG tube: from mouth to earlobe, then down to xiphoid; mark this length
2. **Insert Tube**
 - a. Lubricate distal tip of OG tube generously
 - b. Insert tube into mouth, guiding along tongue and posterior pharynx
 - c. Advance gently; encourage swallowing if responsive
 - d. Stop if resistance is met or patient coughs/chokes excessively
3. **Confirm Placement**
 - a. Inject 20-30 mL air while auscultating epigastrium for "whoosh" sound

4. Secure

- a. Tape tube at insertion point to prevent migration
- b. Connect to suction if indicated
- c. Document placement, confirmation method, and patient response



Meconium Aspiration

ALS procedure

Indications:

- Newborn with meconium staining

Contraindications

- None

Precautions

- May cause bradycardia

Equipment Needed

- Meconium Aspirator
- Suction tubing
- Suction device
- Appropriate ET Tube (likely a 2.5)

Procedure

1. Connect barbed end of meconium aspirator to suction line
2. Set the suction pressure at 80 mmhg or less
3. Intubate patient
4. Connect the ET tube to the other end of meconium aspirator
5. Block thumb port with thumb to begin suctioning
6. Suction for 2 seconds or less at a time until meconium is fully removed
7. Continue to suction while ET tube is withdrawn
8. Discard after use



Push Dose Pressor (Epinephrine)

ALS procedure

Indications:

- Hypoperfusion – as a bridge to vasopressor infusion
- Post-arrest hypotension

Contraindications

- None

Precautions

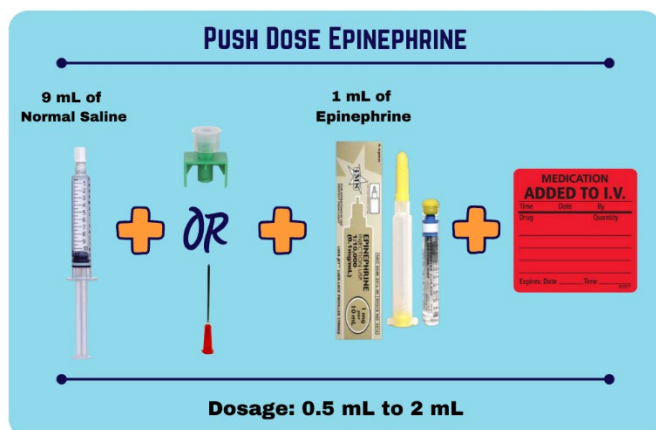
- Cross-check the correct concentration.

Equipment Needed

- 1mg in 10mL Epinephrine (1:10,000)
- Medication Transfer Device (or needle)
- 10 mL Saline Flush
- Medication label

Procedure

1. Open saline flush and eject 1 mL, leaving 9 mL in the syringe.
2. Attach medication transfer device (or appropriate needle)
3. Open epinephrine vial and aspirate 1 mL into the saline syringe
 - a. Concentration is now 0.1mg of epinephrine in 10mL.
4. Perform safety cross-check.
5. Fill out medication label and attach to syringe.
6. Complete the 6 Patient Rights prior to administration.
7. Dosage: 0.5 to 2 mL (repeat as needed until infusion is in place)



Sapphire IV Pump (Infusion)

ALS procedure

Indications:

- Administration of medication infusion

Contraindications

- None

Precautions

- None

Equipment Needed

- Medication to be infused
- Sapphire administration set
- Sapphire IV pump

Procedure

1. Power ON
2. Ensure correct drug library
 - a. AP Medic vs. Paramedic
3. Select New Infusion
4. From Drug Name screen find medication to be infused
5. If weight-based medication, insert the appropriate kg of the patient
6. Depending on time-based or weight-based infusion, enter the appropriate parameters for the medication to be infused
7. Perform cross-check during the 'Confirm' screens
8. Press Start; ensure green LED light running light is on

Suction Assisted Laryngeal Airway Decontamination

ALS procedure

Indications:

- Contaminated Larynx
- Patient unable to maintain physiologic airway

Contraindications

- None

Precautions

- Severe oral trauma
- Suspected cervical injury

Equipment Needed

- | | |
|---|--------------------------------|
| • Proper size endotracheal tube w/ stylet | • Suction available |
| • Laryngoscope and properly sized blade | • Ducanto Suction Catheter |
| • Video-laryngoscope if available | • BVM or mechanical ventilator |
| • 10 mL syringe | • Oxygen supply |
| • Bougie if available | • Stethoscope |
| • Waveform capnography | • Gloves and eye protection |
| • ET tube securing device or tape with bite block | • Lubricant |

Procedure

1. Position equipment
2. Hold laryngoscope handle with blade in Left Hand
3. Hold suction catheter in Right Hand with overhand grip
4. Visualize oropharynx
5. Begin suctioning as tip of laryngoscope blade is entered
6. Push and lift suction catheter to move tongue out of way
7. Advance laryngoscope blade and position suction catheter
8. Remove catheter and place on Left side of laryngoscope
9. Perform "SALAD Poke"
10. Obtain view of glottic opening
11. Perform endotracheal intubation

Synchronized Cardioversion

ALS procedure

Indications:

- Hemodynamic Instability caused by an organized tachycardia

Contraindications

- None

Precautions

- Usually unnecessary if HR is less than 150 bpm
- Assure the patient is dry
- Cardioversion of chronic atrial fibrillation to an organized rhythm may cause an ischemic stroke

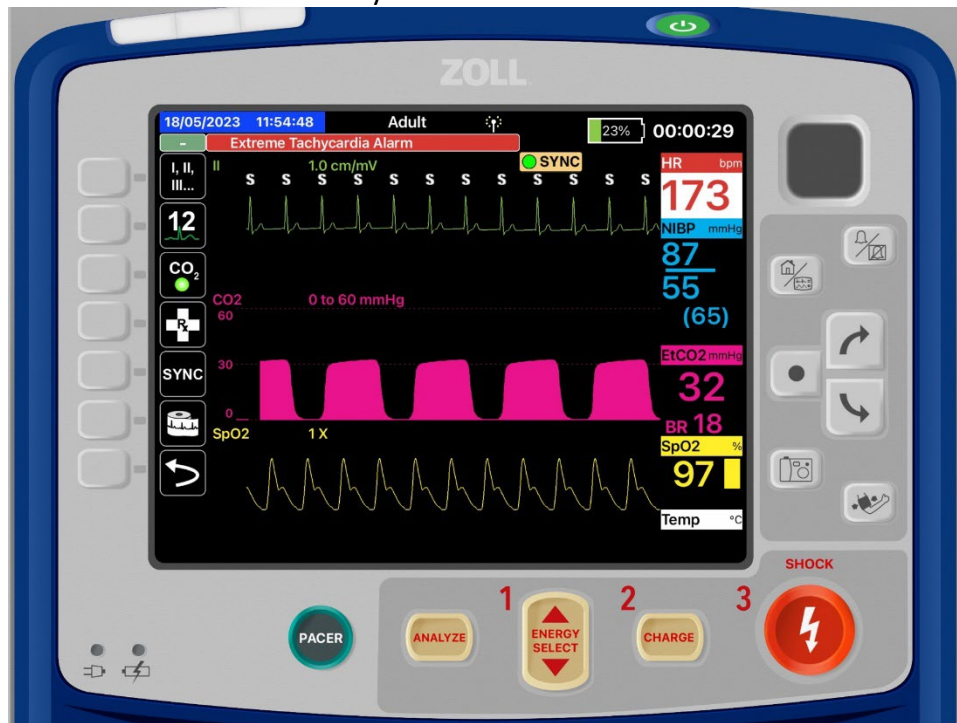
Equipment Needed

- Monitor/ defibrillator
- Defibrillator pads
- Razor

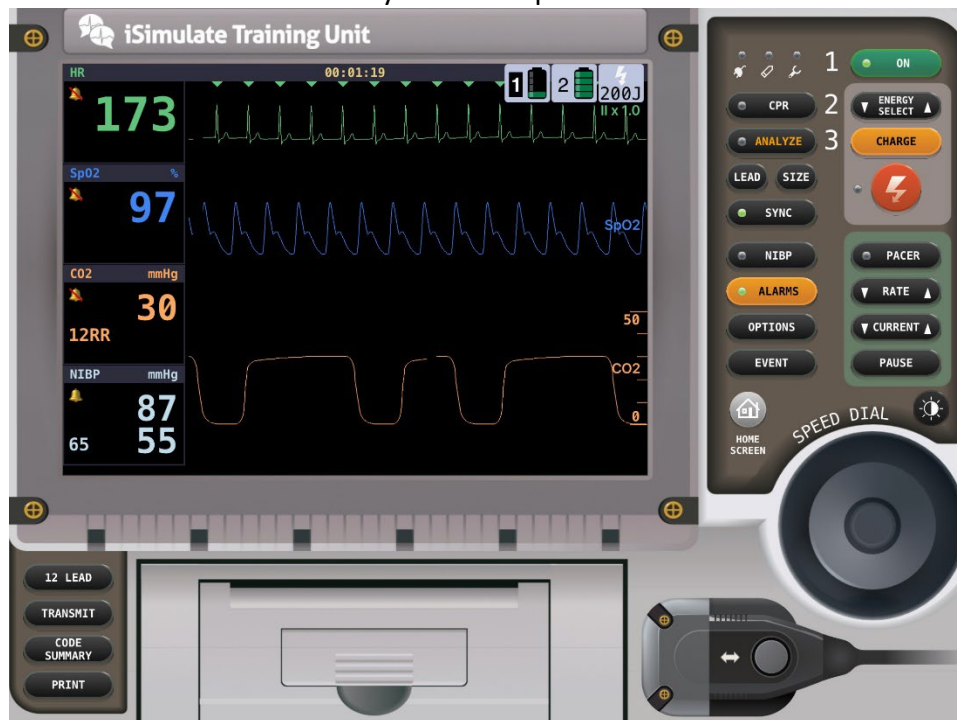
Procedure

1. Provide sedation if responsive and time permits
2. Attach pads with A/P placement
3. Select ECG/EKG lead with most prominent QRS complex
4. Press "SYNC" button
5. Recognize R wave indicators on monitor
6. Select appropriate energy:
 - a. Narrow & Regular: 100 Joules
 - b. Narrow & Irregular: 200 Joules
 - c. Wide & Regular: 200 Joules
 - d. Wide & Irregular: Perform defibrillation at 200 Joules
 - Pediatric: Refer to Handtevy
2. Charge monitor
3. Verbalize "CLEAR"
4. Press and hold the "SHOCK" button
5. Reassess patient

“Sync” on Zoll X Series



“Sync” on Lifepak 15



Finger Thoracostomy

ALS procedure

Indications:

- Traumatic Arrest

Contraindications

- Simple pneumothorax

Precautions

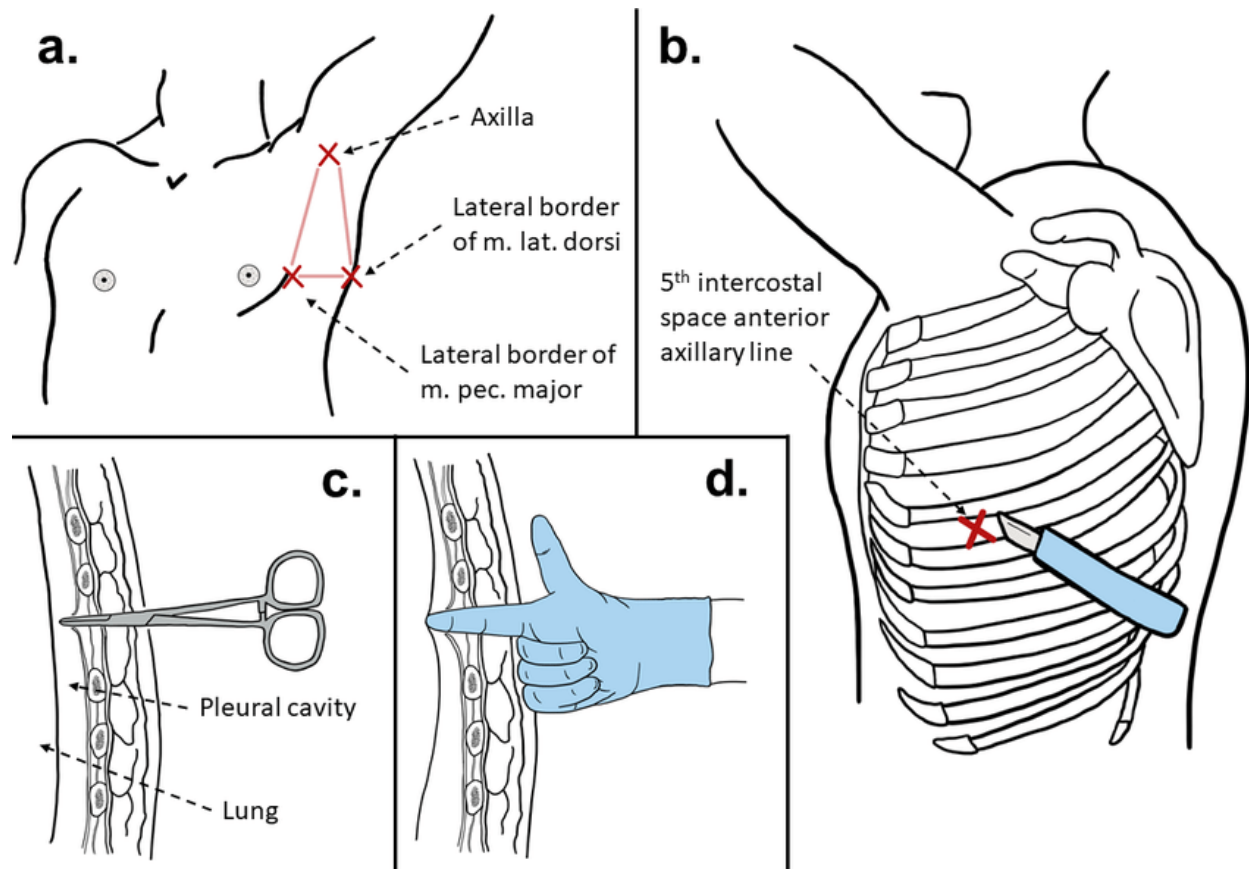
- Enter skin above ribs
- May create a pneumothorax

Equipment Needed

- Sterile Gloves
- Scalpel
- Kelly Clamp
- Betadine
- Lidocaine, Syringe & Needle

Procedure

1. Position patient supine or in semi-Fowler's
2. Recognize possible tension physiology.
3. Continuous monitoring (ECG, SpO2, EtCO2)
4. Locate 5th intercostal space at anterior/mid-axillary line.
5. Clean with betadine or chlorhexadine.
6. Consider lidocaine/local anesthetic
7. Don sterile gloves.
8. Using a scalpel, make a 4cm cut through skin over and parallel to the superior border of the inferior rib.
9. Using Kelly clamps, quickly blunt dissect through subcutaneous tissue and muscle just over the superior border of the inferior rib.
10. With closed Kelly clamps, puncture through the parietal pleura.
 - a. the "give" of the parietal pleura indicates access of the pleural space
 - b. if hemopneumothorax is present, you may hear air or note swift return of blood
11. Remove the Kelly clamps from the tract and insert your full gloved finger into the space. Next, rotate your finger 180 degrees clockwise and counterclockwise feeling inside of the ribs
 - a. intrapleural palpation confirms access of the pleural space
 - b. re-expansion of the lung parenchyma may be palpated, especially if patient is receiving positive pressure ventilation (e.g. intubated)
12. If lung is already expanded on palpation and there is no forceful air/fluid release, it may be possible to close the thoracic wound with occlusive dressing only.



Needle Thoracostomy (Anterior axillary)

ALS procedure

Indications:

- Tension Pneumothorax

Contraindications

- None

Precautions

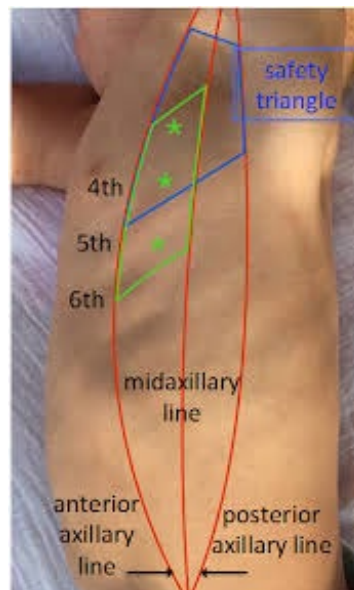
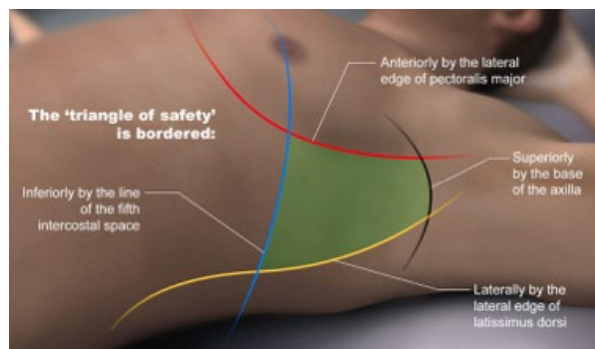
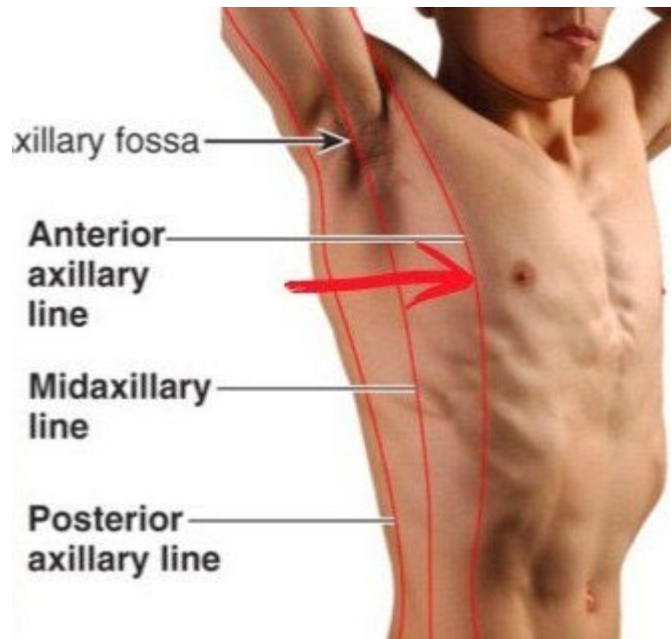
- Hemothorax
- Simple Pneumothorax
- Pediatric Patients

Equipment Needed

- Alcohol or Betadine swab
- ARS Needle or 14 or 16ga IV catheter/ minimum 2" length for adult patients
- 10 mL syringe
- Children may require needle smaller than 2"

Procedure

1. Identify the 4th or 5th intercostal space, anterior-axillary line.
2. Cleanse site
3. Remove the red cap with a twisting motion
4. Attach 10 mL syringe
5. Remove the ARS from the case
6. Insert the ARS into the skin over the superior border of the fifth or sixth rib, anterior axillary line, and direct it into the intercostal space at a 90-degree angle to the chest wall. Ensure ARS entry into the chest is not medial to the nipple line and not directed toward the heart
7. Insert the ARS into the pleural space. Listen for the sudden escape of air as the tension pneumothorax is decompressed
8. If frank blood appears, remove needle and discontinue procedure
9. Remove the needle portion of the ARS and secure the catheter in place
10. Monitor closely for recurrence of respiratory distress
11. Reassess patient



Needle Thoracostomy (Anterior)

ALS procedure

Indications:

- Tension Pneumothorax

Contraindications

- None

Precautions

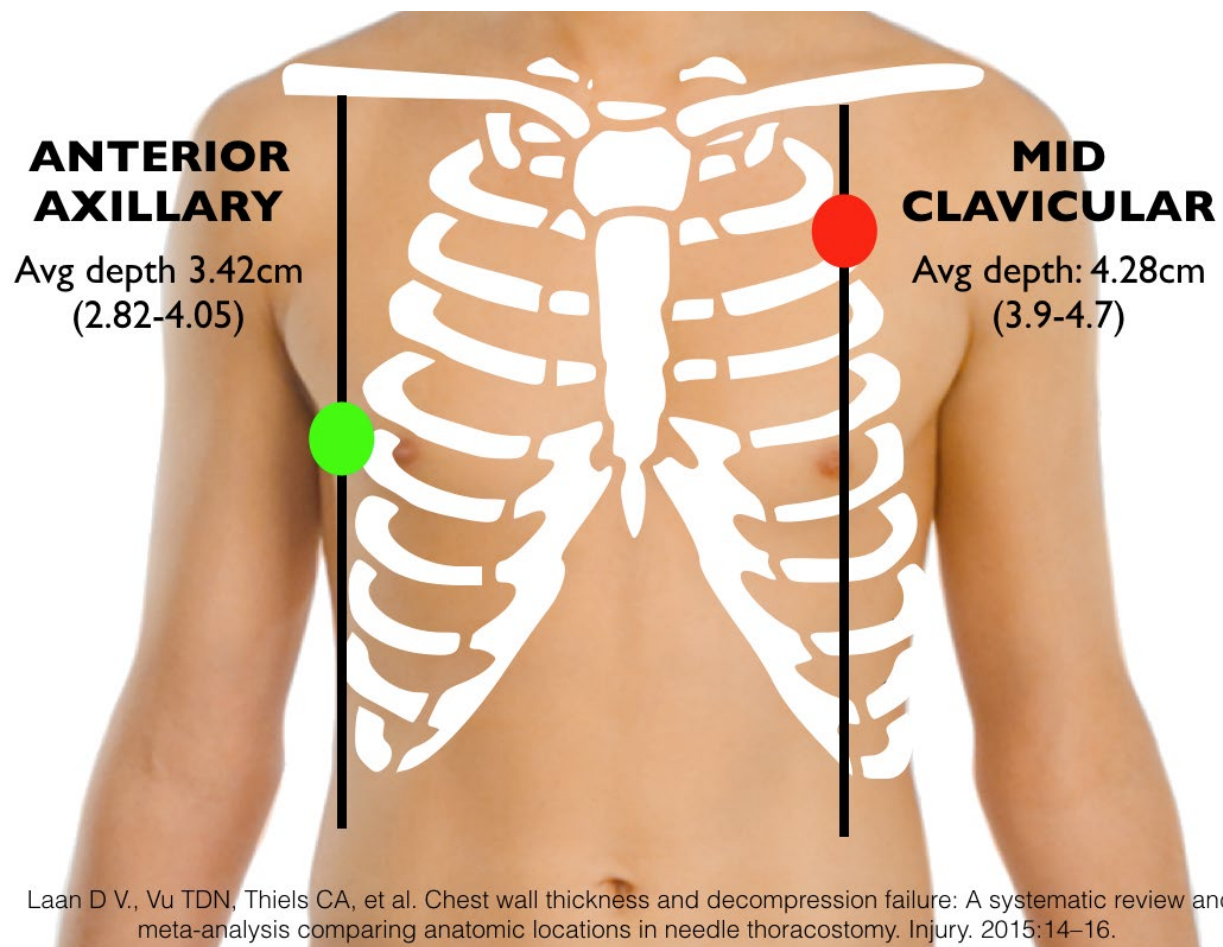
- Hemothorax
- Simple Pneumothorax
- Pediatric Patients

Equipment Needed

- Alcohol or Betadine swab
- ARS Needle or 14 or 16ga IV catheter/ minimum 2" length for adult patients
- 10 mL syringe
- For patients smaller than adult size, use only the length of needle that is necessary to decompress the chest, usually much less than 2"

Procedure

1. Identify the 2nd or 3rd intercostal space on the anterior chest at the midclavicular line on the same side as the injury
2. Cleanse site
3. Remove the red cap with a twisting motion
4. Remove the ARS from the case
5. Insert the ARS into the skin over the superior border of the third or fourth rib, midclavicular line, and direct it into the intercostal space at a 90-degree angle to the chest wall. Ensure ARS entry into the chest is not medial to the nipple line and not directed toward the heart
6. Insert the ARS into the pleural space. Listen for the sudden escape of air as the tension pneumothorax is decompressed
7. If frank blood appears, remove needle and discontinue procedure
8. Remove the needle portion of the ARS and secure the catheter in place
9. Monitor closely for recurrence of respiratory distress
10. Reassess patient



Transcutaneous Pacing

ALS procedure

Indications:

- Symptomatic Bradycardia

Contraindications

- Cardiac Arrest

Precautions

- Severe Hypothermia

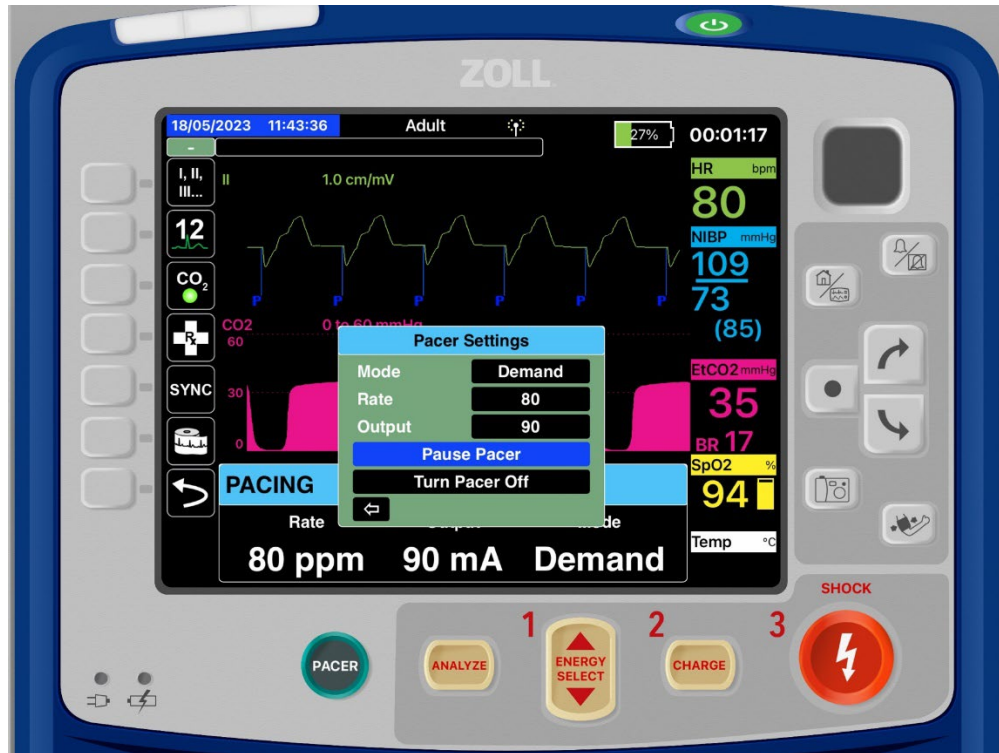
Equipment Needed

- Monitor/ defibrillator
- Defibrillator pads
- Razor

Procedure

1. Treat patient per Bradycardia guideline
2. Identify rhythm on the cardiac monitor
3. If patient is conscious and aware of situation during pacing, consider sedation
4. Apply pads anterior/posterior as well as 4-lead
5. Set the pacemaker to 80 beats per minute (monitor defaulted to 80)
6. Set the output setting to 30 (monitor defaulted to 30)
7. Turn on the pacer
8. Quickly increase output to 60 ma while observing for capture
9. Slowly increase the output until electrical capture is detected
 - a. Electrical capture: Spike, loss of intrinsic rhythm, large QRS, & discordant T-wave
10. Palpate for femoral pulse (mechanical capture)
11. Reassess the vital signs. BP, SpO2, EtCO2
12. Adjust as necessary to maintain perfusion

Pacing on Zoll X Series



Pacing on Lifepak 15



Ultrasound: Cardiac Exam

ALS procedure

Indications:

- Hemodynamically Unstable Trauma Patient
- Abdominal & Thoracic Trauma
- Previously stable trauma patient with acutely worsening clinical status

Contraindications

- None

Precautions

- Doesn't localize injured organ.
- Views may be limited with subcutaneous emphysema.
- Views may be limited in patients who have a hollow-viscus injury with free air.

Equipment Needed

- Ultrasound device
- Transducer – Phased Array (or cardiac probe)

Procedure

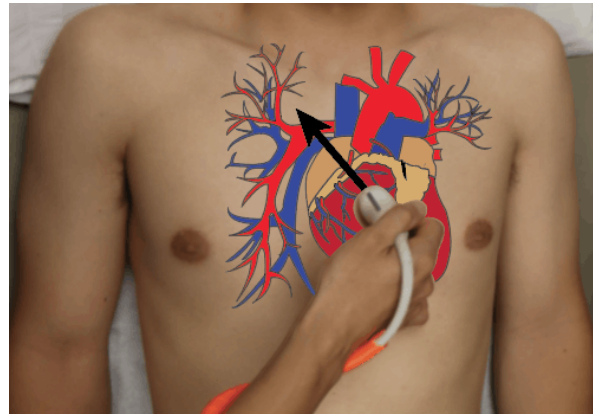
1. **Patient laying supine**
 - a. *If you are unable to visualize the heart in the supine position, consider repositioning the patient into the Left lateral decubitus.*
2. **Select Preset: Cardiac**
 - a. *Make sure the U/S Image Indicator Marker is on the RIGHT side of the U/S screen.*
3. **Obtain the preferred view:**
 - a. **Parasternal Long Axis (PSLA)**
 - b. **Parasternal Short Axis (PSSA)**
 - c. **Apical View**
 - d. **Subxiphoid (Subcostal)**
 - e. **Inferior Vena Cava (IVC)**

Parasternal Long Axis View

The **Parasternal Long Axis View** is often abbreviated as PSLA or PLAX.

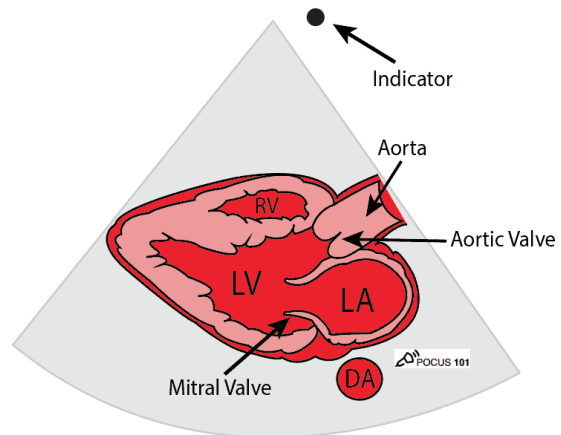
It is usually the first cardiac ultrasound view obtained and will give you an immediate assessment of the general condition of the heart including ejection fraction and overall left and right ventricular sizes.

- Point the probe indicator toward the patient's right shoulder
- Place the probe at the 4th intercostal space which is approximately the nipple line for males or the inframammary fold to the sternum (females).



Structures to identify in the Parasternal Long Axis (PSLA) View:

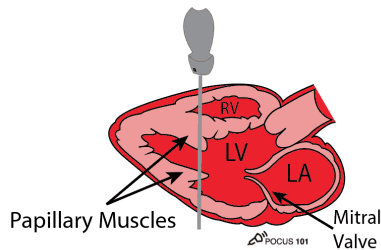
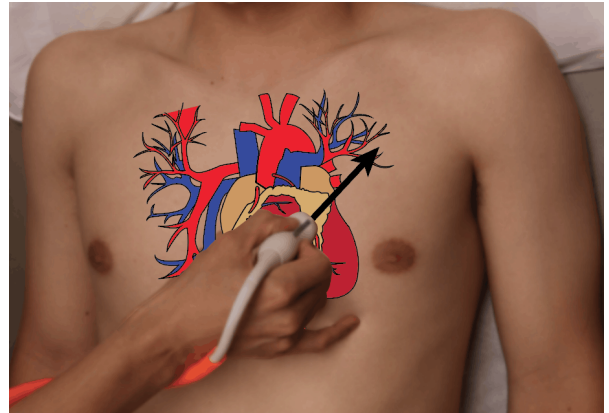
- RV: Right Ventricle
- LV: Left Ventricle
- LA: Left Atrium
- AV: Aortic Valve
- MV: Mitral Valve
- AO: Aorta
- DA: Descending Aorta
- Pericardium



Parasternal Short Axis View

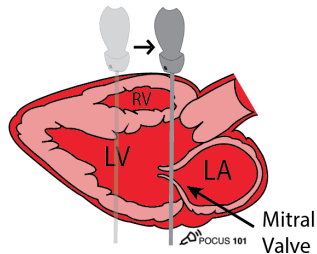
The **Parasternal Short Axis view** is often abbreviated as PSSA or PSAX.

- From the Parasternal Long Axis view, rotate your probe 90 degrees clockwise so that the indicator is now pointing towards the patient's left shoulder.



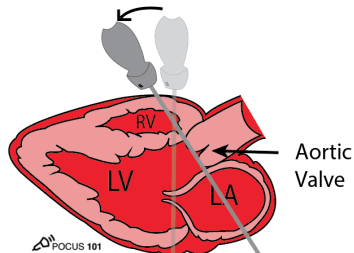
Mid-Papillary Level

Slide the Probe Towards the Mitral Valve

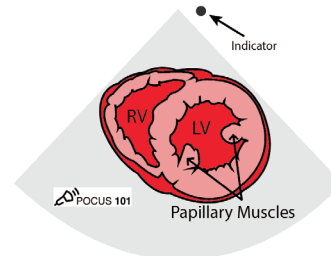


Mitral Valve Level

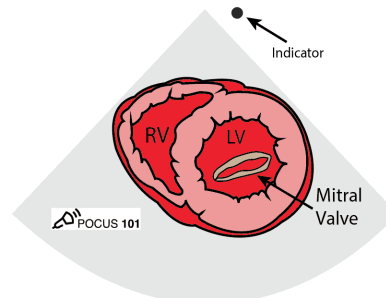
Tilt the Probe Towards the Base of the Heart



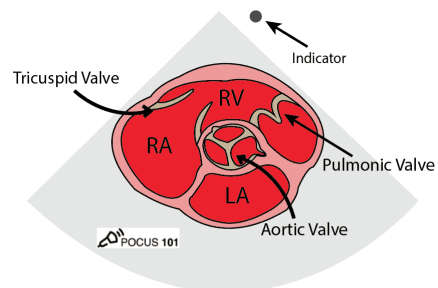
Aortic Valve Level



Mid-Papillary Level



Mitral Valve Level ("Fish Mouth" View)

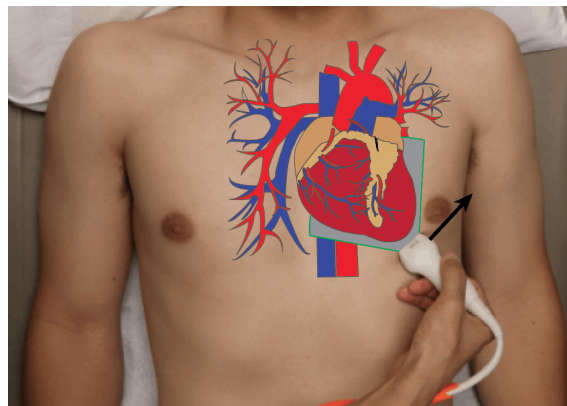


Aortic Valve Level ("Mercedes Benz" View)

Apical View

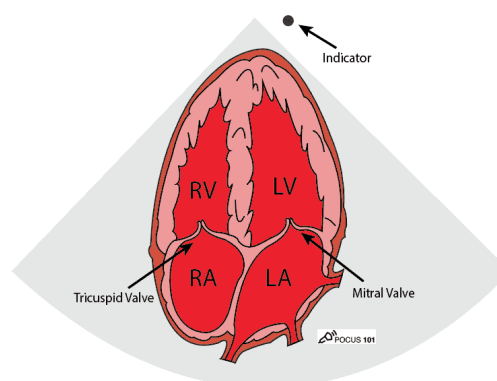
The Apical Views are some of the most important views to be able to obtain when doing hemodynamic assessment of the heart. This includes looking at diastolic dysfunction, valvular regurgitation, cardiac output, etc.

Unfortunately, it can be one of the most challenging views to obtain when first starting out. In this section, we will show you exactly how to get the main apical views including the apical 4 chamber view, 5 chamber view, and the coronary sinus view.



Structures to identify in the **Apical 4 Chamber (A4C)** View:

- LV: Left Ventricle
- RV: Right Ventricle
- LA: Left Atrium
- RA: Right Atrium
- TV: Tricuspid Valve
- MV: Mitral Valve

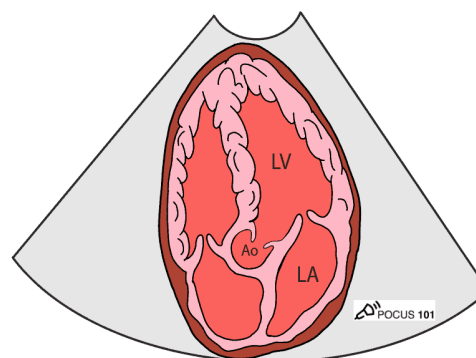


Apical 5 Chamber (A5C) View

Of course, there are not really 5 chambers in the heart but in echocardiography, the “5th chamber” is when you can see the appearance of the aortic valve and the left ventricular outflow tract.

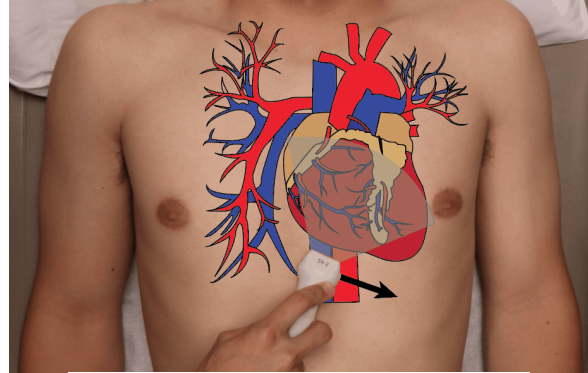
Sometimes you will want to intentionally obtain this view in order to calculate the cardiac output of the left heart.

From the Apical 4 Chamber view, slightly tilt the tail of your probe towards the patient’s feet to get the Apical 5 chamber view.



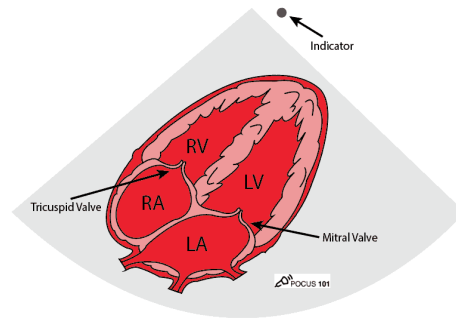
Subxiphoid View

The **Subxiphoid** or “**Subcostal**” view allows you to see similar structures as the Apical 4 Chamber view but just approached from a different angle. This Subxiphoid view is useful when you are having difficulty getting adequate parasternal views (i.e. COPD patients) or when you are evaluating a trauma patient when doing the eFAST scan.



Structures to identify in the Subxiphoid (Subcostal) View:

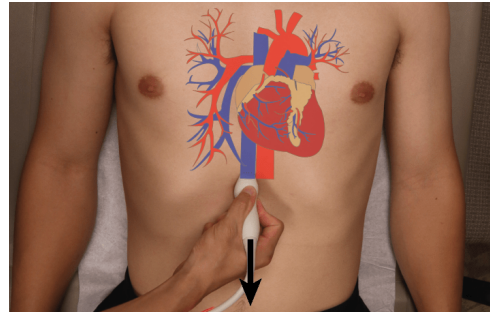
- RV: Right Ventricle
- LV: Left Ventricle
- TV: Tricuspid Valve
- MV: Mitral Valve
- LA: Left Atrium
- RA: Right Atrium
- Pericardium



Inferior Vena Cava View

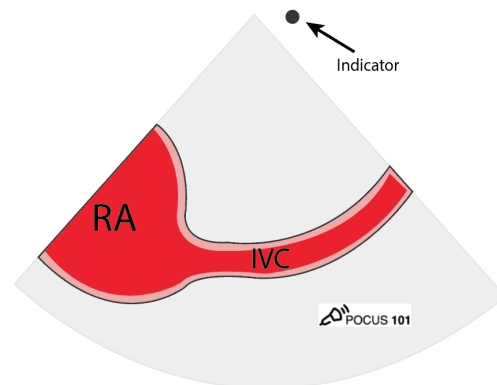
Evaluating the Inferior Vena Cava (IVC) with ultrasound is one of the most used Point of Care Ultrasound (POCUS) applications. It is a great way to estimate central venous pressure (CVP) and fluid tolerance.

- From the Subxiphoid view with the patient still in the supinated position with knees bent, keep the right atrium in view.
- Locate the Inferior Vena Cava within the Liver and center it on the ultrasound screen.
- Once the IVC is centered on the screen, rotate the transducer clockwise (to your right) to bring the indicator down toward the feet (play the instructional video).
- This will give a longitudinal view of the Inferior Vena Cava entering the Right Atrium.



Structures to identify in the Inferior Vena Cava (IVC) View:

- IVC: Inferior Vena Cava
- RA: Right atrium
- HV: Hepatic Vein (sometimes seen)



Ultrasound: eFAST Exam

ALS procedure

Indications:

- Hemodynamically Unstable Trauma Patient
- Abdominal & Thoracic Trauma
- Previously stable trauma patient with acutely worsening clinical status

Contraindications

- None

Precautions

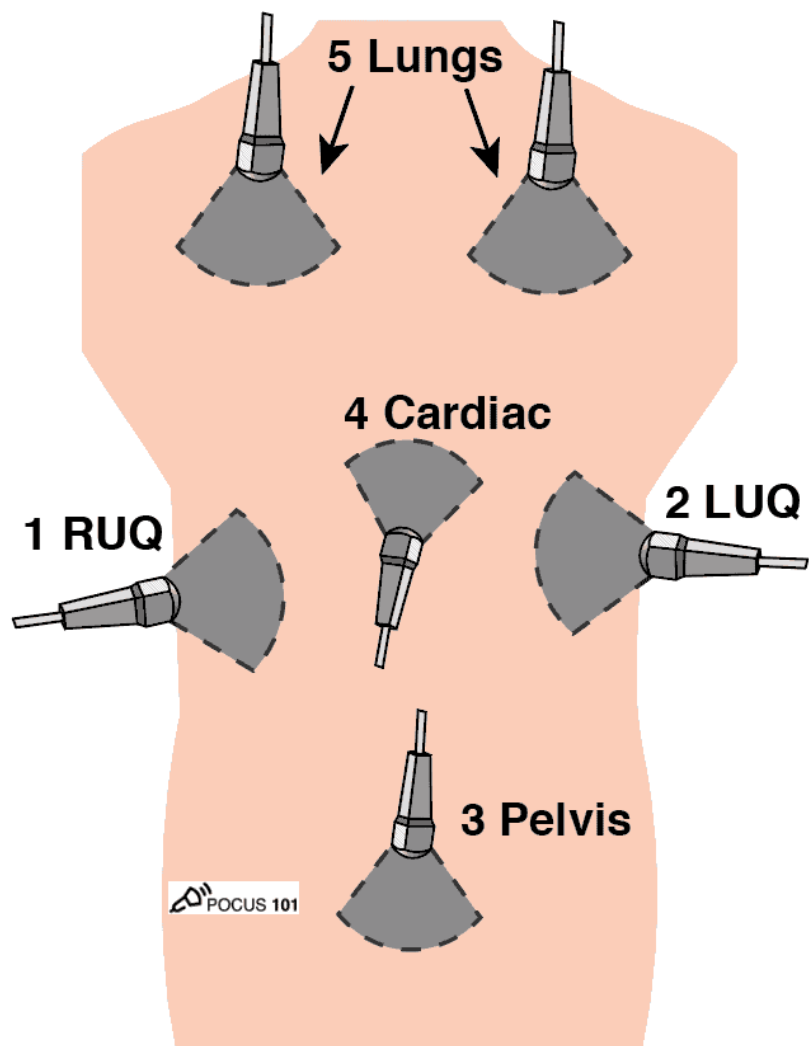
- Doesn't localize injured organ.
- Views may be limited with subcutaneous emphysema.
- Views may be limited in patients who have a hollow-viscus injury with free air.

Equipment Needed

- Ultrasound device
- Transducer – Phased Array (or cardiac probe)

Procedure

1. Patient laying supine
2. Select Preset: FAST Exam or Abdominal Exam
3. Obtain the following views:
 - a. Right Upper Quadrant (RUQ)
 - b. Left Upper Quadrant (LUQ)
 - c. Pelvic
 - d. Cardiac (Parasternal Long Axis or Subxiphoid)
 - e. Lungs (Right & Left)
4. Assess the following:
 - a. Is there free fluid in the Abdomen?
 - b. Is there free fluid in the Thorax?
 - c. Is there cardiac movement?
 - d. Is there fluid in the Pericardium?
 - e. Is there a Pneumothorax?



Intraosseous Access: EZ-IO®

ALS procedure

Indications:

- IO Medication administration
- IV access difficult/impossible

Contraindications

- Fracture of the targeted bone
- Previous, significant orthopedic procedures at insertion site (e.g. prosthetic limb or joint)
- IO in the targeted bone within the past 48 hours

Precautions

- Avoid unrecognized extravasation

Equipment Needed

- EZ-IO® Power Driver
- EZ-IO® Needle Set and EZ-Connect® Extension Set
- EZ-Stabilizer® Dressing
- Non-sterile gloves
- Cleansing agent of choice
- Luer-lock syringe with flush (5-10 mL for adults, 2-5 mL for infant/child)
- Pressure infusion bag
- Lidocaine (2%) for local anesthesia

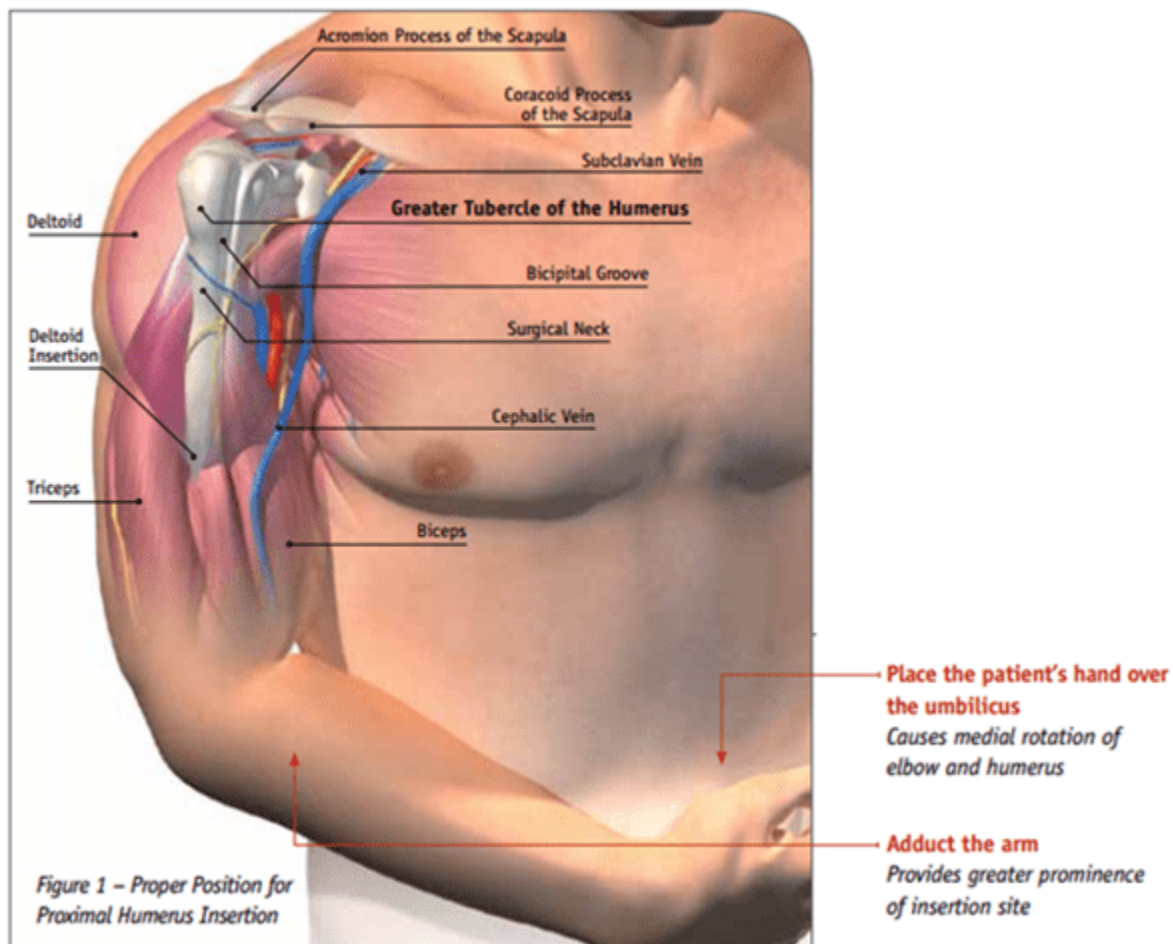
Procedure

1. Use a clean, “no touch” technique, maintaining asepsis
2. Prepare supplies
3. Prepare the site by using antiseptic of your choice; stabilize the extremity
4. Remove the needle set cap
5. Follow site-specific instructions for insertion

INSERTION SITE IDENTIFICATION

Proximal Humerus (Adult/Pediatric)

1. Place the patient's hand over the abdomen (elbow adducted and humerus internally rotated)
2. Place your palm on the patient's shoulder anteriorly; the "ball" under your palm is the general target area
 - a. You should be able to feel this ball, even on obese patients, by pushing deeply
3. Place the ulnar aspect of your hand vertically over the axilla and the ulnar aspect of your other hand along the midline of the upper arm laterally
4. Place your thumbs together over the arm; this identifies the vertical line of insertion on the proximal humerus
5. Palpate deeply up the humerus to the surgical neck
 - a. This may feel like a golf ball on a tee – the spot where the "ball" meets the "tee" is the surgical neck
 - b. The insertion site is 1 to 2 cm above the surgical neck, on the most prominent aspect of the greater tubercle



Proximal Tibia (Adult)

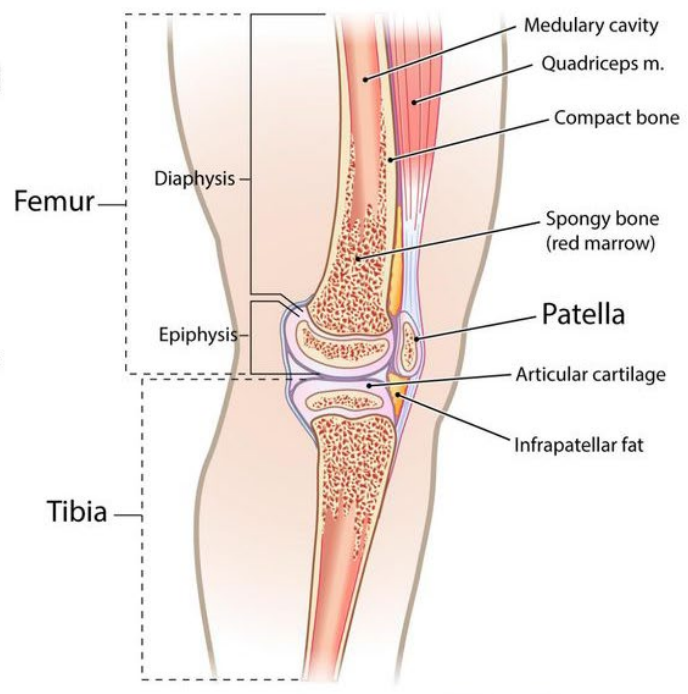
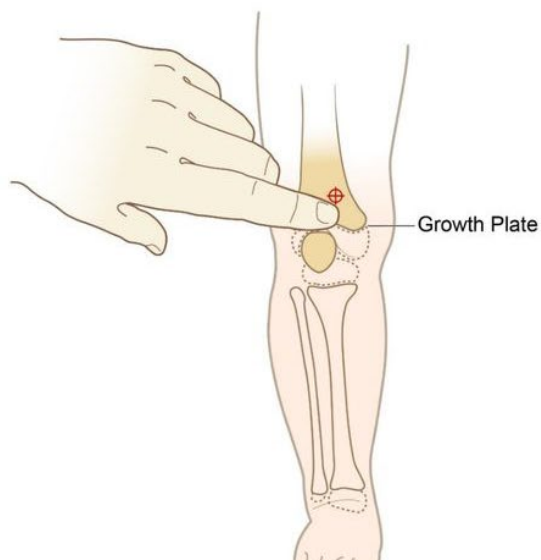
1. Extend the leg.
2. Insertion site is approximately 2 cm medial to the tibial tuberosity, or approximately 3 cm below the patella and approximately 2 cm medial, along the flat aspect of the tibia.



Distal Femur (Adult/Pediatric)

1. Secure the leg outstretched to ensure the knee does not bend.
2. Identify the patella by palpation. The insertion site is just proximal to the patella (maximum 1 cm) and approximately 1-2 cm medial to midline.

- Secure the leg out-stretched
- Ensure the knee does not bend



INSERTION TECHNIQUE

Proximal Humerus

1. Aim the needle set at a 45-degree angle to the anterior plane and posteromedial
2. Push the needle set tip through the skin until the tip rests against the bone
 - a. The 5 mm mark must be visible above the skin for confirmation of adequate needle set length
3. Gently drill into the humerus approximately 2 cm or until the hub is close to the skin; the hub of the needle set should be perpendicular to the skin

Tibia

1. Aim the needle set at a 90-degree angle to the bone
2. Push the needle set tip through the skin until the tip rests against the bone
 - a. The 5 mm mark must be visible above the skin for confirmation of adequate needle set length
3. Gently drill, advancing the needle set approximately 1-2 cm after entry into the medullary space or until the needle set hub is close to the skin

Femur

1. Aim the needle set at a 90-degree angle to the bone
2. Push the needle set tip through the skin until the tip rests against the bone
 - a. The 5 mm mark must be visible above the skin for confirmation of adequate needle set length
3. Gently drill, immediately release the trigger when you feel the loss of resistance as the needle set enters the medullary space; avoid recoil – do NOT pull back on the driver when releasing the trigger.

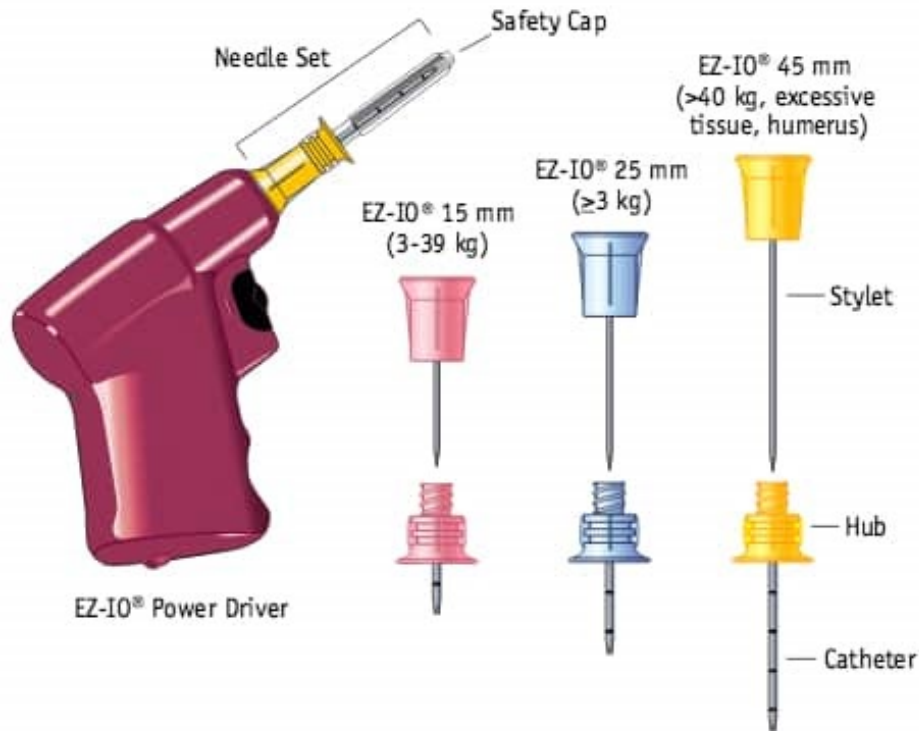
INSERTION COMPLETION

1. Hold the hub in place and pull the driver straight off; continue to hold the hub while twisting the stylet off the hub with counter clockwise rotations; catheter should feel firmly seated in the bone (1st confirmation of placement);
 - a. Dispose of all sharps and biohazard materials using standard biohazard practices and disposal containers.
2. If using the NeedleVISE® 1 port sharps block, place on stable surface and use a one-handed technique.
3. Place the EZ-Stabilizer® Dressing over the hub
4. Attach a primed extension set to the catheter hub, firmly secure by twisting clockwise
5. Pull the tabs off the dressing to expose the adhesive, apply to the skin

EZ-IO® NEEDLE SETS: DESCRIPTION

- Comprised of Catheter with Luer-lock connection, Stylet, Safety Cap.
- 15 gauge, 304 stainless steel in 15 mm, 25 mm and 45 mm lengths.
- Sterile, non-pyrogenic, in protective packaging.
- Intended for use with EZ-IO® Power Driver.

EZ-IO® Power Driver and Needle Sets: Description



NEEDLE SET SELECTION

Select EZ-IO® Needle Set based on patient weight, anatomy and clinical judgment. The EZ-IO® Catheter is marked with a black line 5 mm proximal to the hub. Prior to drilling, with the EZ-IO® Needle Set inserted through the soft tissue and the needle tip touching bone, adequate needle length is determined by the ability to see the 5 mm black line above the skin.

- EZ-IO® 45 mm Needle Set (yellow hub) should be considered for proximal humerus insertion in patients 40 kg and greater and patients with excessive tissue over any insertion site
- EZ-IO® 25 mm Needle Set (blue hub) should be considered for patients 3 kg and greater
- EZ-IO® 15 mm Needle Set (pink hub) should be considered for patients approximately 3-39 kg

Intravenous Access: Peripheral IV

ALS procedure

Indications:

- Administration of IV fluids/medications

Contraindications

- None

Precautions

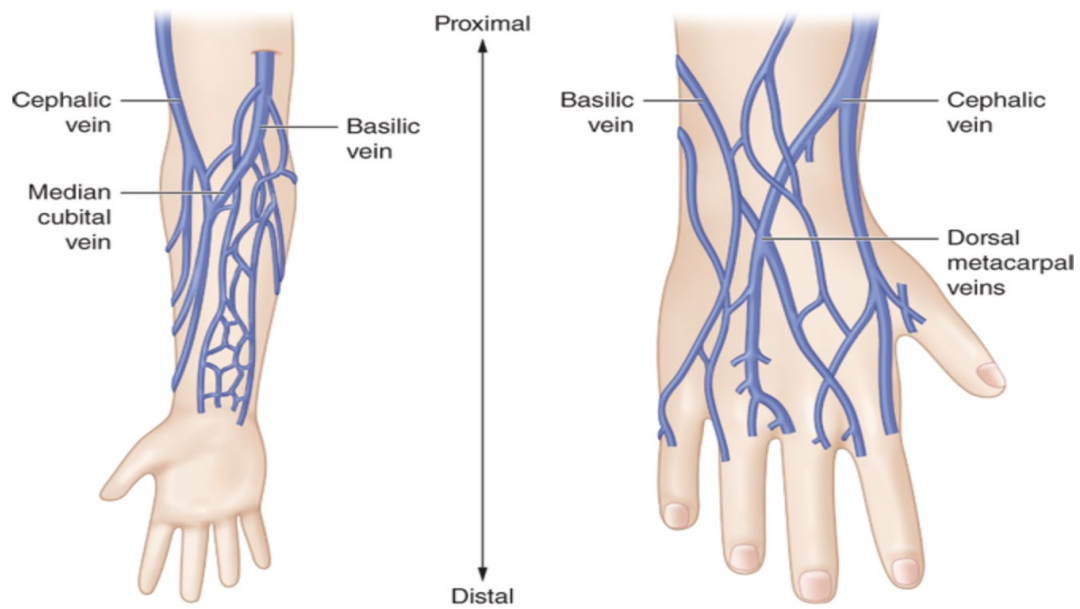
- Avoid unrecognized extravasation
- Proximal fracture

Equipment Needed

- | | |
|-----------------------------------|------------------------------------|
| • IV fluid | • Alcohol or Iodine prep |
| • Appropriately sized IV catheter | • Gauze & tape |
| • Administration set | • Veniguard / dry sterile dressing |
| • Saline flush | • Venous constricting band |

Procedure

1. Explain the procedure to the patient
2. Apply constricting band proximally
3. Locate vein of choice (bouncy, straight, non-rolling)
4. Clean site with aseptic technique
5. Open IV catheter
6. Stabilize vein by holding pressure distally
7. Enter with bevel up
8. Observe for flash
9. Advance catheter
10. Remove constricting band
11. Apply pressure proximal to IV catheter to prevent bleeding
12. Retract needle and place in sharps container
13. Attach administration set to catheter
14. Release pressure
15. Flush, aspirate, flush
16. Assess for signs of infiltration/extravasation
17. Secure site with Veniguard or dry sterile dressing



Automated External Defibrillator (AED)

BLS procedure

Indications:

- Cardiac Arrest

Contraindications

- Do not place AED on patients with a pulse

Precautions

- Remove patient from standing water and wipe water from surface of chest
- Do not place a defibrillation paddle or electrode directly over an implanted pacemaker/defibrillator
- Remove transdermal medication patches and wipe area clean
- Utilize pediatric pads when appropriate

Equipment Needed

- Automated External Defibrillator (AED)

Procedure

1. Determine if patient is unresponsive and pulseless
2. Perform CPR until defibrillator is available
3. Turn on AED
4. Position patches on chest at sternum-apex
5. Follow voice prompts
6. Quickly clear the patient, shock patient if advised by AED and switch compressors
7. Perform CPR for 2 minutes
8. Continue to follow AED voice prompts

Bag Valve Mask Ventilation

BLS procedure

Indications:

- Patient requiring positive pressure ventilation
- Patient in respiratory arrest
- Patient in severe respiratory distress

Contraindications

- None

Precautions

- Inflate only to chest rise.
- Ensure proper chest rise if pop off valve activates (peds only)

Equipment Needed

- Bag-valve-mask with reservoir
- Oxygen tubing
- Oxygen bottle with regulator and flow meter
- Assorted masks.

Procedure

Procedure should be performed with two providers when available

1. Open the airway with jaw thrust or head tilt / chin lift
2. Insert an airway adjunct (oral or nasal airway)
3. Select proper bag: adult, child, infant
4. Select appropriate size mask
5. Connect reservoir and oxygen tubing
6. Create proper mask-to-face seal with the "EC" clamp technique
7. Ventilate adult patient once every 5-6 seconds and every 3-5 seconds for children/infants.
8. Adjust oxygen liter flow to ensure reservoir bag stays inflated

CPAP: Flow-Safe

BLS procedure

Indications:

- Dyspnea
- Hypoxia
- Pulmonary Edema
- Cardiogenic Pulmonary Edema
- Acute Respiratory Distress Syndrome
- Bronchospasm

Contraindications

- Altered Mental Status
- Hypotension
- Respiratory Failure
- Pneumothorax
- Vomiting
- Upper airway obstruction

Equipment Needed

- CPAP Mask and Circuit
- Oxygen Cylinder with regulator
- Cardiac Monitor
- Capnography

Procedure

1. Position patient in sitting or high Fowler's
2. Continuously monitor BP, ECG, EtCO₂, and SpO₂
3. Connect CPAP Circuit to O₂ regulator
4. Turn on oxygen to desired output
5. Reassure and calm the patient
6. Hold the mask to the patient's face and assure seal
7. When the patient tolerates the mask, secure the head straps
8. Continuously monitor patient and repeat VS every 5 minutes

CPAP/PEEP (cm H ₂ O)	Flow (LPM)
5.0	8 - 9
7.5	10 - 12
10.0	13 - 14

Epinephrine Auto-Injector

BLS procedure

Indications:

- Anaphylaxis

Contraindications

- None

Precautions

- May result in tachycardia, pallor, diaphoresis, dizziness, chest pain, headache, nausea

Equipment Needed

- Auto-Injector

Procedure

1. Obtain patient's auto-injector
2. Verify the six Rights and perform partner cross-check
3. Remove the safety cap
4. Place at patient's lateral thigh midway between knee and hip
5. Push injector firmly against the thigh and hold for 10 seconds
6. Reassess the patient



Glucometer

BLS procedure

Indications:

- To obtain a blood glucose level

Contraindications

- None

Precautions

- Capillary or venous sample

Equipment Needed

- Glucometer
- Test strips
- Alcohol prep
- Lancet or blood-letting device
- 2x2 or 4x4
- Bandage

Procedure

1. Prepare test strip and glucometer
2. Cleanse site with alcohol prep
3. Pierce desired site with lancet
4. Wipe site with clean 2x2 or 4x4
5. Compress proximally to puncture site and work blood towards the site
6. Hold test strip to collect sample
7. Allow sample to be drawn into strip
8. Apply bandage
9. Dispose of lancet into sharps bin

I-Gel Laryngeal Airway

BLS procedure

Indications:

- Patient requiring positive pressure ventilation
- Rescue airway for failed ETI attempt

Contraindications

- Intact gag reflex
- FBAO

Precautions

- The presence of emesis/fluids may indicate a need for a more secure airway

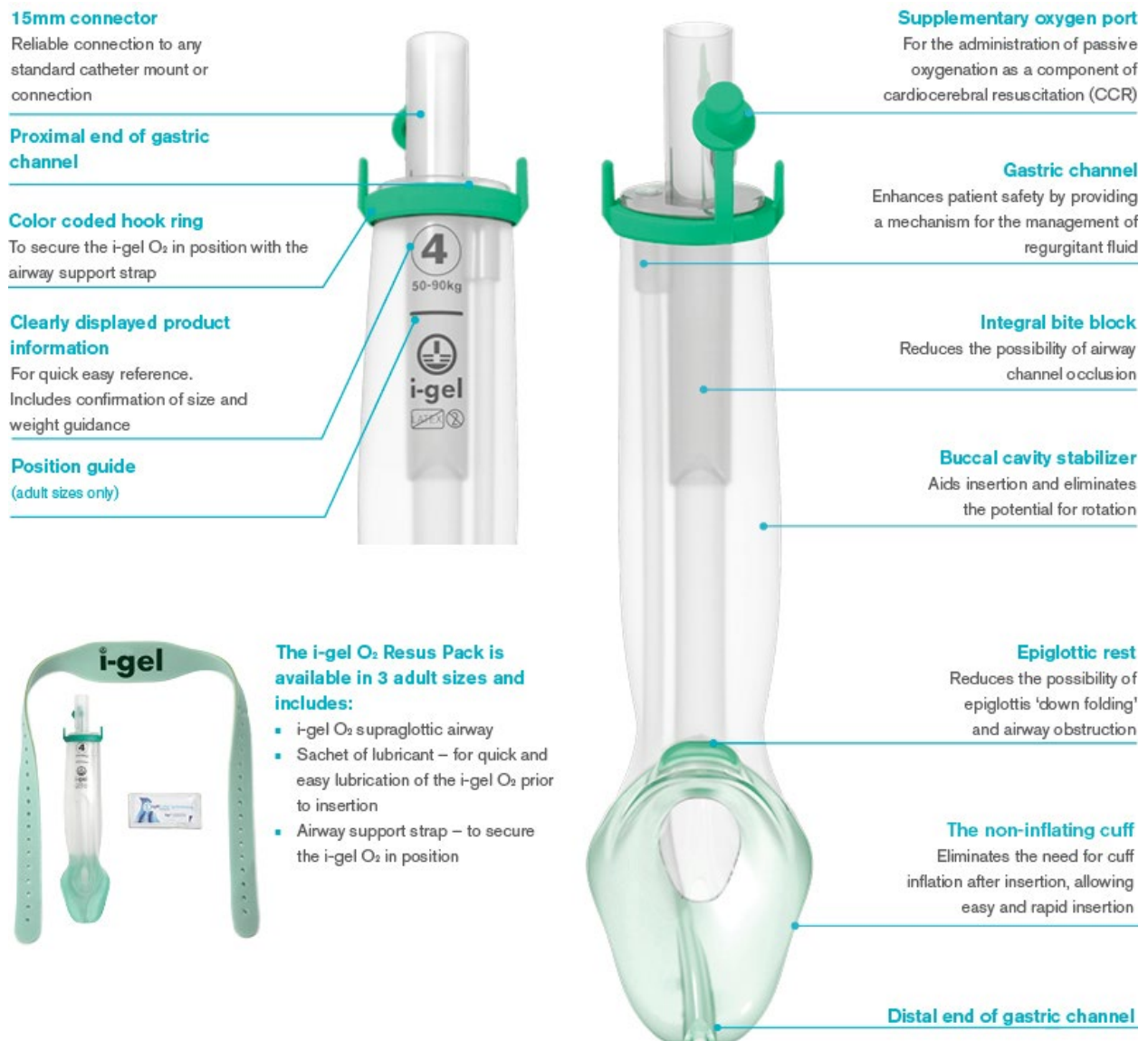
Equipment Needed

- | | |
|---------------------|---------------------------|
| • I-Gel | • Capnography |
| • Lubricant | • Suction |
| • Proprietary strap | • Gastric Tube (optional) |

Procedure

1. Confirm the patient is being properly ventilated with high flow oxygen.
2. Select appropriate I-Gel.
3. *Optional* - Measure the appropriate depth for the gastric tube insertion, typically from the patient's ear to the umbilicus and then to the end of the I-Gel.
4. *Optional* - Insert the gastric tube into the gastric channel of the i-gel. Stop the insertion before the tube extends beyond distal tip of I-Gel.
5. Lubricate posterior portion of device and NOT the mask itself
6. Place head in neutral position or slightly extended (sniffing)
7. Insert device downward along hard palate. Stop when it is felt to “pop” into place or when resistance is felt.
8. Attach Capnography and ventilate via BVM/mechanical ventilator
9. Confirm placement with chest rise and fall, lung sounds and capnography

10. **Optional** - Carefully advance the gastric tube while rotating until predetermined length is achieved or resistance is met.
11. **Optional** – Attach suction to gastric tube to provide decompression.
12. Assess for air leakage. If leakage occurs, reposition or remove the I-Gel if necessary
13. Secure tube with strap or tape



Inhaler Administration

BLS procedure

Indications:

- Bronchospasm with a prescribed inhaler

Contraindications

- Unable to follow commands

Precautions

- Maximum dose

Equipment Needed

- Patient's prescribed metered-dose inhaler (with spacer if available)

Procedure

1. Shake canister and mouthpiece well
2. Hold the device upright and close to the patient's mouth
3. Advise patient to exhale, pushing as much air from lungs as possible
4. Place mouthpiece in patient's mouth and instruct patient to close his / her lips loosely around the mouthpiece with tongue underneath
5. Advise patient to inhale deeply, press down on canister quickly then release it
6. Instruct patient to hold his / her breath for 5 to 10 seconds before exhaling
7. Monitor patient for desired effects

Intramuscular Injection

BLS procedure

Indications:

- The administration of intramuscular medication

Contraindications

- Medications that aren't approved for the IM route

Precautions

- Accidental intravenous administration
- IM injections may be painful

Equipment Needed

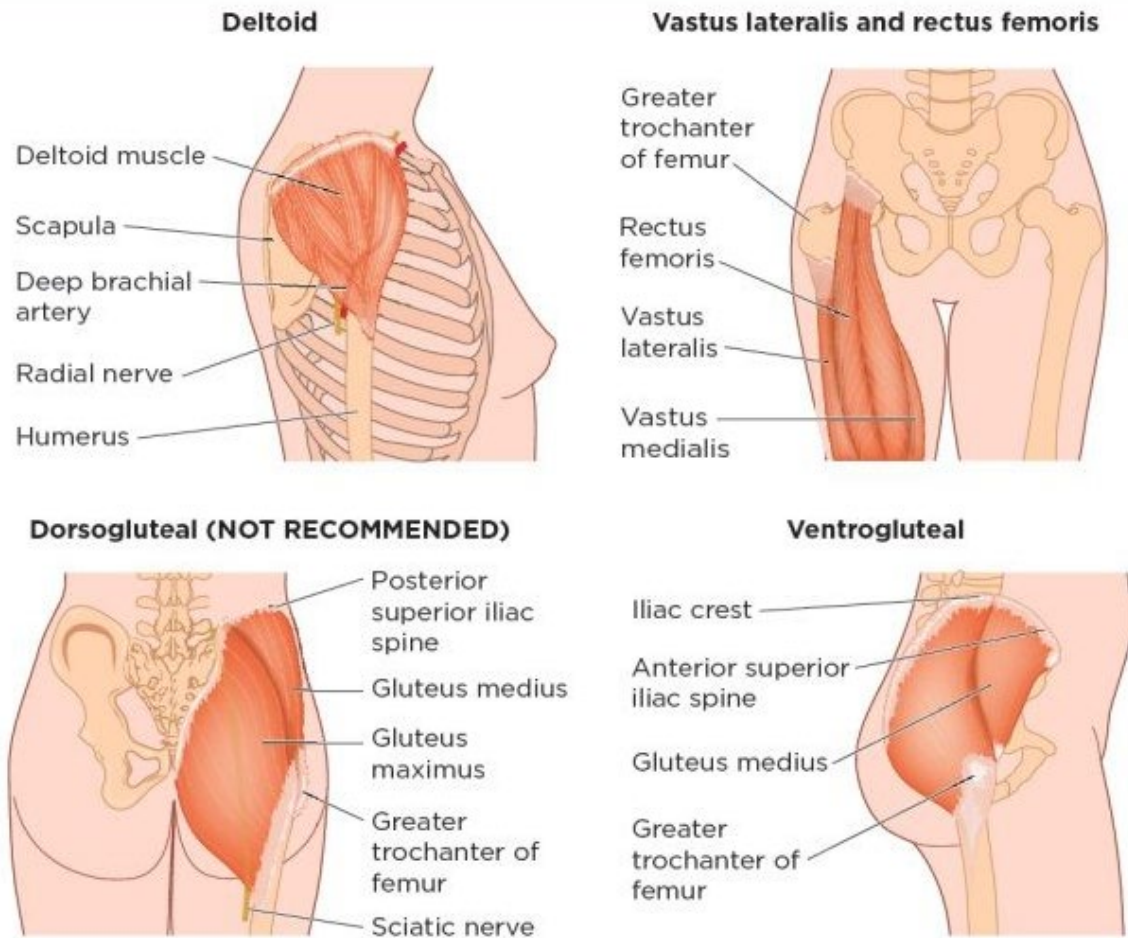
- Appropriate syringe
- 18 g 1.5" – 22g 1-1.5" Needle
- Alcohol prep
- Bandage

Procedure

1. Prepare equipment, medication to be given
2. Perform 6 Rights and partner cross-check
3. Explain procedure to patient
4. Select proper injection site
 - a. (deltoid / ventrogluteal / vastus lateralis)
5. Clean site with alcohol swap, starting with small circles and working into larger ones
6. Hold skin taut
7. Puncture the skin and enter the muscle at a 90 degree angle
8. Inject medication
9. Cover with bandage
10. Dispose of needle/ syringe in sharps container
11. Reassess the patient
12. Document drug given, time given, route, effects and person administering drug



IM Injection Sites



Intravenous Fluid Preparation

BLS procedure

Indications:

- Patient requires IV fluids.

Contraindications

- None

Precautions

- IV administration is an ALS skill
- Pulmonary edema should be considered (auscultate lung sounds)

Equipment Needed

- IV fluids
- Administration set

Procedure

1. Examine IV solution for type, color, clarity, sedimentation, and expiration date.
2. Assure proper administration set and remove from wrapper
3. Clamp off administration set and hold distal end at level of IV fluids
4. Attach the administration set (spike the bag)
5. Squeeze the drip chamber until half full
6. Open the flow of the administration set to flush the line
7. Assure there is no air bubbles in the administration set

KangooFix Neonatal Immobilizer

BLS procedure

Indications:

- To immobilize neonates and infants (0–6 months) during ambulance transport

Contraindications/Precautions

- Infants older than 6 months or above device weight limit
- Obvious deformities or injuries requiring specialized spinal immobilization
- Severe respiratory distress requiring prone positioning

Equipment Needed

- KangooFix Neonatal Immobilizer
- Ambulance stretcher or incubator with restraint points
- Infant appropriately dressed for temperature conditions
- Optional: receiving blanket for padding or swaddling

Procedure

1. Preparation

- Inspect the KangooFix for integrity and cleanliness.
- Confirm that the stretcher or incubator has compatible anchor points.
- Prepare the infant with minimal clothing to prevent overheating, or with thermal support as appropriate.



2. Positioning the Infant

- Lay the KangooFix flat on the stretcher or incubator.
- Center the infant on the KangooFix, ensuring:
 - The head is in the designated head support area.

- The infant is aligned with the midline of the device.
- Arms are at the sides or gently restrained by soft fabric flaps, depending on presentation.



3. Securing the Infant

- Fasten the shoulder straps over the infant's chest, ensuring snug but not constricting fit.
- Secure the pelvic strap across the hips.
- If present, use leg support straps to reduce limb movement.
- Confirm that straps are flat, not twisted, and Velcro is fully engaged.



4. Attaching to the Stretcher or Incubator

- Use the four-point stretcher harness to attach KangooFix to the ambulance stretcher, gurney, or incubator.
- Confirm firm fixation by:
 - Gently attempting to move the KangooFix laterally and longitudinally.
 - Rechecking all buckle closures and Velcro engagements.

5. Monitoring and Reassessment

- Reassess strap tension every 15 minutes or after patient movement.
- Continuously monitor infant's airway, breathing, and circulation.
- Adjust straps as needed if infant's status changes.

Documentation:

- Device use and condition upon application
- Infant condition before and after placement



- **Time of application and securing**
- **Reassessments and any modifications**

Mechanical CPR (AutoPulse)

BLS procedure

Indications:

- The AutoPulse is a consistent and effective delivery method for chest compressions in the pulseless adult patient.
- The AutoPulse is intended for use in adult patients that meet the following parameters:

• Patient Parameter	• AutoPulse Specification
• Chest Circumference	• 30 to 56 in.
• Max Patient Weight	• 400 lbs
• Compression Rate	• 80 compressions per minute
• 30:2 Ratio	• 30 compressions w/ 3s ventilation pause

Contraindications/Precautions

- The AutoPulse must be used only in cases that manual CPR would normally be initiated.
- Failure to properly position the LifeBand at the patient's armpit line may cause injury to the patient.
- If a system error occurs, immediately revert to manual CPR.
- Do not use the AutoPulse in the presence of an oxygen rich atmosphere (>25%), flammable anesthetics, or other flammable agents.

Equipment Needed

- AutoPulse Device

Procedure

Apply the AutoPulse after 8 minutes (4 cycles) of high quality, manual compressions.

1. Remove all clothing from torso to ensure skin-to-platform contact
2. Align armpits onto the yellow line on the platform
3. Ensuring proper alignment will prevent most CPR stoppage
4. Power up the AutoPulse
5. Close chest bands ensuring they are at a 90° angle to the platform and not twisted.
6. Press CONTINUE (green button). Do NOT touch the patient or the LifeBand during this phase.
7. Press START (green button) to begin compressions.
8. To pause or stop operation press STOP (red button)

Troubleshooting

For Fault/User Advisory:

- Lift up and fully extend both chest bands
- Check both lateral and vertical patient placement

Unlock light:

- Usually indicates pin is not in correctly

Caution Light:

- Power off and back on

If you cannot rectify the problem, immediately open the chest bands and revert to manual CPR.

Notes

- When CPR is indicated, it should start immediately and should not be postponed.
- Do not place or position the patient on the AutoPulse platform in either a facedown orientation or on their side.
- Do not touch the patient's thorax while the AutoPulse is analyzing the patient's size.
- If you must move or realign the patient, you must press the Stop/Cancel button before the adjustment
- Operating the AutoPulse on a patient for extended periods of time may result in minor skin irritation to the patient.
- Check the patient's chest rise while ventilating during active operation.

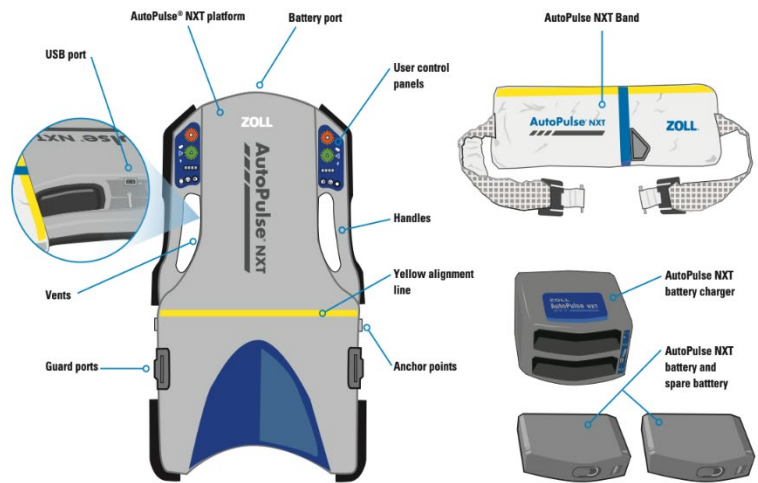
Tips, Battery life and maintenance:

- Rotate batteries daily
 - Battery in platform goes in the charger
- Battery in the charger is used as the spare
- Spare battery goes inside platform
- Power the platform on
- Confirm all green lights
- Power off
- Charge time 2 hours
- Minimum run time, 30 minutes
- Will beep every 30 seconds when battery is low
- Assign roles
 - Provider who changes battery
 - Provider who operates buttons
 - Provider lifts LifeBand to home position
 - This accomplishes 2 things... you can resume manual chest compressions if it is taking too long to change battery or make a defibrillation decision and eliminates an error code when the device is powered back on after the battery change.

Documentation

Include the following (minimum)

- Indications
- Complications (or lack thereof)
- Clinical outcome



Mechanical CPR (LUCAS)

BLS procedure

Indications:

- Viable Medical Cardiac Arrest

Contraindications/Precautions

- Traumatic cardiac arrest
- If it is not possible to position the LUCAS device safely or correctly on the patient's chest.
 - Too small patient: if the LUCAS device alerts with 3 fast signals when lowering the Suction Cup and you cannot enter the PAUSE mode or ACTIVE mode.
 - Too large patient: If you cannot lock the Upper Part of the LUCAS device to the Back Plate without compressing the patient's chest

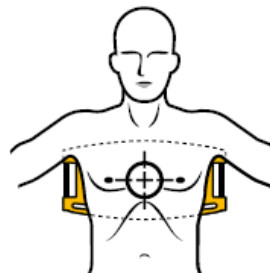
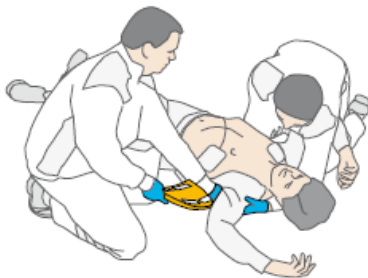
Equipment Needed

- LUCAS 3
- Battery (charged)
- Neck Strap
- Back Plate

Procedure

1. Preparation:

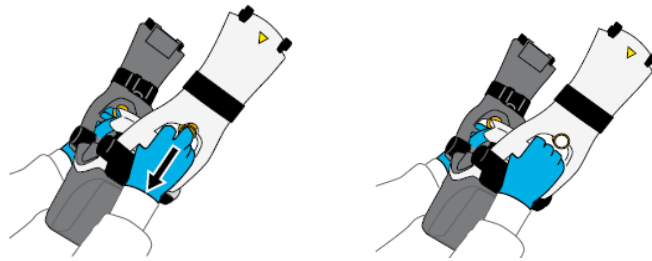
- **Begin manual compressions immediately.**
- **Confirm cardiac arrest: apnea, pulselessness, unresponsiveness.**
- **Attach ECG leads, establish airway and vascular access as per Medical Cardiac Arrest protocol.**



2. LUCAS Deployment:

Minimize interruptions in compressions

- Prepare the LUCAS device while compressions continue.
- Position the backplate behind the patient's chest:
 - Slide or log-roll technique as appropriate.
- Align the compression module:

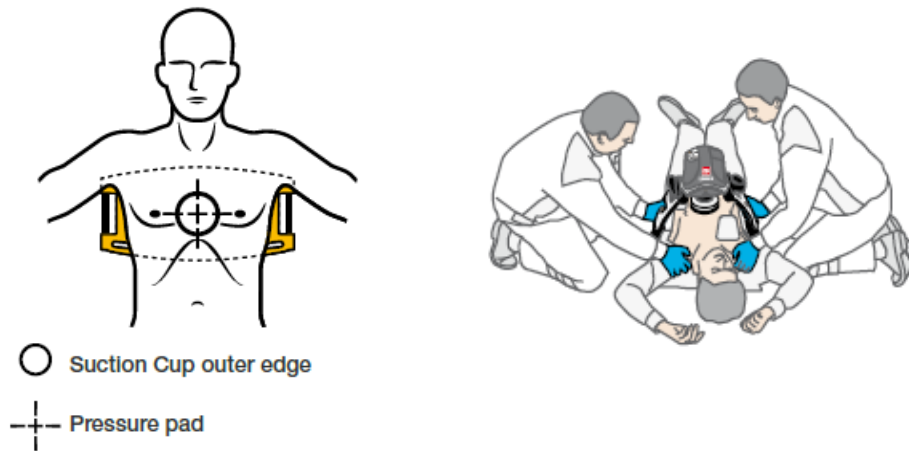


- Pull the release rings once to make sure that the claw locks are open then let go of release rings.



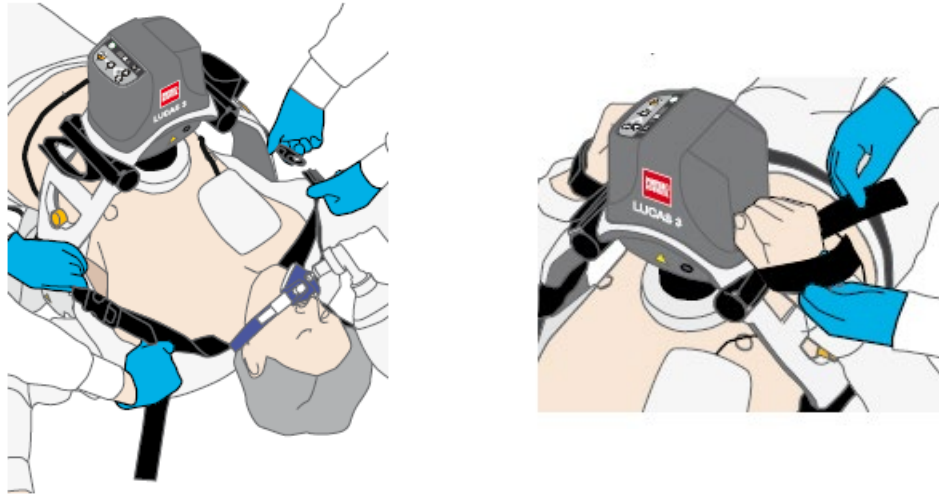
- Coordinate the placement with your team prior to placing.
- During ongoing manual chest compressions, attach the support leg that is nearest to you to the back plate.
- Confirm positioning on the lower half of the sternum (center of the chest).
- Stop manual CPR while attaching the other support leg to the Back Plate, so that the two support legs lock against the Back Plate.
- Listen for a click. Pull up once to make sure that the parts are correctly attached.





- Use your finger to ensure that the lower edge of the Suction Cup is immediately above the end of the sternum.
- Adjust the height of the Suction Cup to set the Start Position.
 - Make sure that the LUCAS device is in the ADJUST mode.
 - Push the Suction Cup down with two fingers until the pressure pad touches the patient's chest without compressing the chest.
 - Push PAUSE to lock the Start Position. Check for the proper position. If not, push ADJUST, pull up the Suction Cup to readjust the central and/or height position for a new Start Position. Push PAUSE.
 - Push ACTIVE (continuous) OR ACTIVE (30:2) to start the compressions.
- * The compression point should be at the same spot as for manual CPR and according to guidelines.
- * When the pressure pad in the Suction Cup is in the correct position, the lower edge of the Suction Cup is immediately above the end of the sternum.
- Confirm compression depth and rate.
- Remove the neck strap, which is a part of the Stabilisation Strap, from the Carrying Bag (the support legs strap of the Stabilisation Strap should already be attached to the support legs).
- Extend the neck strap fully at the buckles.
- Carefully lift the patient's head and put the cushion behind the patient's neck. Position the cushion as near the patient's shoulders as possible.

- Connect the buckles on the support leg straps with the buckles on the neck strap. Make sure that the straps are not twisted.
- Hold the LUCAS support legs stable and tighten the neck strap tightly.
- When you move the patient, you can secure the patient's arms with the Patient Straps on the LUCAS device. This makes it easier to move the patient.



3. Monitoring During Use:

- Continuously assess capnography, chest recoil, and device function.
- Monitor for changes in rhythm and return of spontaneous circulation (ROSC).
- Avoid prolonged interruptions (>10 seconds) when applying or adjusting the device.

4. Transport Considerations:

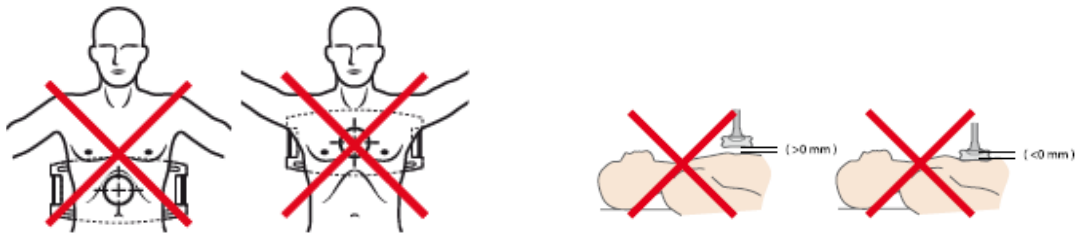
- Secure the device for transport.
- Monitor the patient and device during transit.
- Coordinate early notification to receiving facility.

5. Post-Use Care:

- Document:
 - Start and stop time of mechanical CPR.
 - Device effectiveness (compression fraction, EtCO₂ trends).
 - Complications or dislodgement.
- Remove LUCAS upon ED handoff, unless otherwise directed.

Special Notes & Safety Considerations:

- *Do not use in wet or explosive environments (e.g., oxygen-rich atmospheres) due to electrical hazard risk .*
 - *Do not adjust or move the patient by pulling on the device.*
 - *If signs of chest wall injury or ineffective compressions arise, switch to manual CPR.*
-



Mucosal Atomization Device (MAD)

BLS procedure

Indications:

- To administer intranasal medication

Contraindications

- None

Precautions

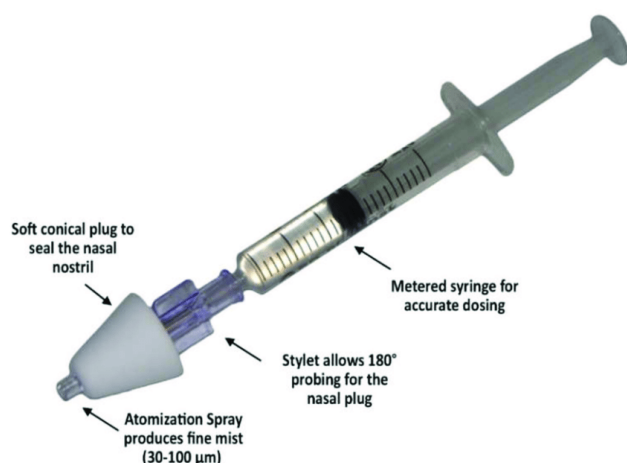
- Absorption may be incomplete or hindered

Equipment Needed

- MAD device
- Appropriate syringe
- Needle/Blunt tip
- Medication

Procedure

1. Draw up the appropriate amount of medication
 - a. (add an extra 0.1ml of medication to account for the dead space in the device)
2. Attach the MAD device
3. Confirm the 6 Rights and perform partner cross-check
4. Administer a half dose in each nostril (not to exceed 1mL per nostril)



Nasal Cannula

BLS procedure

Indications:

- Spontaneously breathing patients without airway compromise
- Patient unable to tolerate mask
- Apneic oxygenation

Contraindications

- None

Precautions

- Epistaxis

Equipment Needed

- Nasal Cannula
- Capnography Nasal Prongs
- Oxygen cylinder with regulator

Procedure

1. Explain procedure
2. Attach cannula to oxygen
3. Adjust flow to 2 to 6 lpm based on oxygenation of patient (do not exceed 5lpm if measuring EtCO₂)
4. Apply nasal cannula to patient
5. Adjust tightness
6. *If using capnography, plug device into monitor*

Nasopharyngeal Airway (NPA)

BLS procedure

Indications:

- Patient with gag reflex
- Any patient requiring airway assistance

Contraindications

- Basilar skull fracture

Precautions

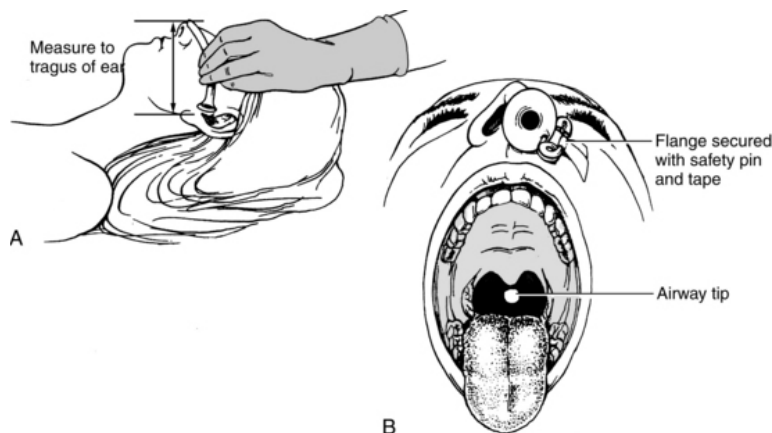
- Epistaxis
- Facial fracture

Equipment Needed

- Properly sized NPA
- Water soluble lubricant

Procedure

1. Explain procedure to patient if necessary
2. Select appropriate airway by measuring from the tip of the nose to the ear lobe
3. Lubricate airway with a water soluble lubricant
4. Lift the tip of the nose and insert the airway into the larger or more open nostril with the bevel facing towards the septum
5. If you meet resistance, gently rotate from side to side as you insert. If resistance continues remove and try the other nostril
6. Airway should rest against the flare of the nostril



Nebulizer Therapy

BLS procedure

Indications:

- Bronchospasm

Contraindications

- None

Precautions

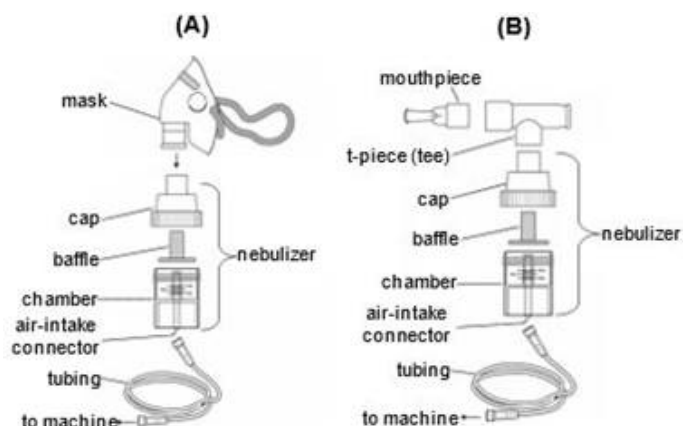
- If unable to ventilate, medication may not be absorbed/effective

Equipment Needed

- Proper medication
- Nebulizer device/kit
- Oxygen source

Procedure

1. Assemble nebulizer per manufacturer's instructions
2. Place medication in bowl of nebulizer
3. Attach to oxygen with tubing and place at 4-6 LPM
4. Have patient begin treatment when mist is visible
5. Repeat treatment as necessary per guideline



Picture 2 The nebulizer is used with either a mask (A) or a mouthpiece (B).

Neo-Tee® Resuscitator

BLS procedure

Indications:

- Infant/Neonate patient under 10 kg (22lb) requiring positive pressure ventilation
- Infant/Neonate patient under 10 kg (22lb) in respiratory arrest
- Infant/Neonate patient under 10 kg (22lb) in severe respiratory distress

Contraindications

- None

Precautions

- Assure good seal and proper mask size
- Do not exceed 15 lpm of O2 flow
- Uncover PEEP knob hole as soon as breath is delivered
- Remove red protective cap from the T-piece circuit prior to connecting the T-Piece

Equipment Needed

- Neo-Tee T-Piece Resuscitator
- Oxygen tubing
- Oxygen bottle with regulator
- Assorted masks.

Procedure

Pre-Use Check

1. Connect oxygen tubing to oxygen blender.
2. Adjust the flow on the flowmeter between 5 and 15 LPM (recommended 15 LPM).
3. Use Red protective cap for testing pressures prior to use.
4. Check the inspiratory pressure by occluding the PEEP knob hole with a finger. Use the adjustable PIP knob on the controller to set the PIP on the manometer (recommended 20 cm H2O).
5. Release the finger from PEEP knob hole and adjust variable PEEP knob to desired value as indicated on the manometer (recommended 5 cm H2O).

Directions for Use

1. Connect patient circuit to mask and place over patient's mouth and/or nose or connect patient circuit to masked laryngeal airway or endotracheal tube.
2. Resuscitate by placing and removing thumb or index finger over the PEEP knob hole to allow inspiration and expiration at the desired breath rate.
3. Adjust the input flow rate on the device to increase or decrease the depth of each breath to provide adequate chest excursion of the patient.
4. Adjust oxygen blender to provide desired delivered oxygen concentration.

Neo-Tee T-Piece Resuscitator

Step 1: **15**

Set the oxygen flowmeter at 15 lpm and attach the Neo-Tee tubing.

Step 2: **10**

Set the PIP controller to the 10 o'clock position.



Step 3: **20**

Twist the PIP controller to achieve 20 cm H₂O Peak Inspiratory Pressure.



Step 4: **5**

Set the PEEP to 5 cm H₂O.



Step 5: **Confirm**

Test ventilate closed system; confirm settings of 20/5

Step 6: **Ventilate**

Ventilate the neonate/infant

Oropharyngeal Airway (OPA)

BLS procedure

Indications:

- Patient without gag reflex
- Any unresponsive patient requiring airway assistance

Contraindications

- Intact gag reflex
- Responsive patient

Precautions

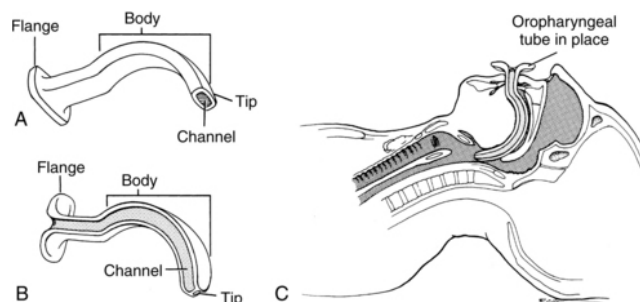
- None

Equipment Needed

- Properly sized OPA
- Suction

Procedure

1. Select appropriate size airway by measuring from the center of the mouth to the angle of the jaw or corner of the mouth to the ear lobe
2. Insert airway using the cross finger technique upside down with the tip pointing to the corner of the mouth
3. While inserting, gently rotate 90 degrees while continuing to advance the airway until the flat flange at the top of the airway rests on the patients front teeth
4. Alternate technique: utilize a tongue depressor to displace the tongue and follow the natural curvature of the airway
5. In pediatrics place directly in following the natural curvature of the airway



Pedi-Mate

BLS procedure

Indications:

- To transport of pediatric patients between **10–100 lbs (4.5–45.4 kg)** on the stretcher.

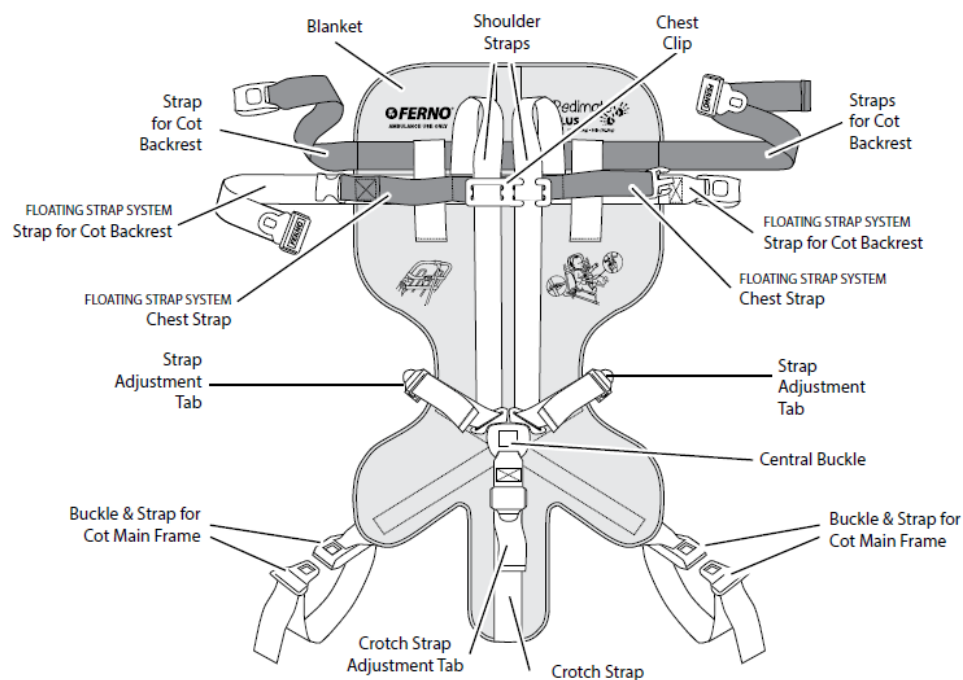
Contraindications/Precautions

- N/A

Equipment Needed

- Pedi-Mate device
- Stretcher

Procedure



1. Pre-Use Inspection

- Verify all straps, buckles, and stitching are intact
- Confirm device cleanliness and readiness for use
- If in doubt, remove from service and replace

2. Setup and Cot Positioning

1. **Remove existing cot restraints.**
 2. **Raise the cot backrest** to 15–45° to elevate the shoulders and maintain center of gravity.
 3. **Unroll and center** the Pedi-Mate® on the cot mattress.
 4. Align the **base of the blanket** with the cot joint; the black backrest strap should align with the patient's shoulders.
 5. Fasten and loosely secure **backrest and chest straps** around the cot backrest.
-

3. Securing the Pedi-Mate® to the Cot

1. Place the patient on the blanket; adjust the blanket if shoulder alignment is off.
 2. Tighten **backrest strap** until the mattress compresses.
 3. Attach both **main frame straps** by looping them around the cot frame at the head-end and fastening the buckles.
 4. Firmly tighten both main frame straps.
-

4. Securing the Patient

1. Pull **crotch strap** up between the legs and lay it on the abdomen.
 2. Wrap the **black chest strap** under the shoulder straps, and snug (but don't restrict breathing).
 3. Drape **shoulder straps** over each shoulder and buckle into the central buckle.
 4. Slide the **chest clip** on the left shoulder strap to secure strap placement.
 5. **Tighten all straps:**
 - Shoulder: Pull horizontally while stabilizing central buckle
 - Torso: Pull from side straps
 - Crotch: Pull upward for snug fit
-

5. Patient Monitoring

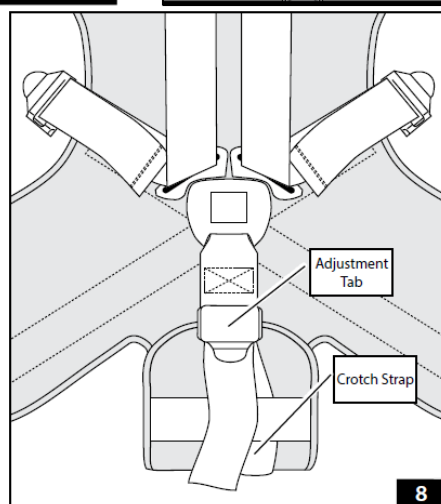
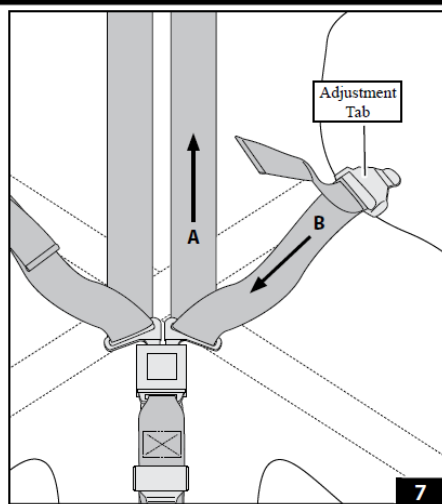
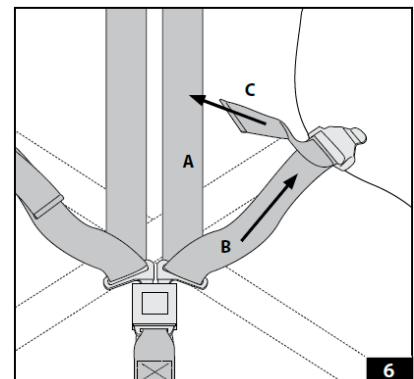
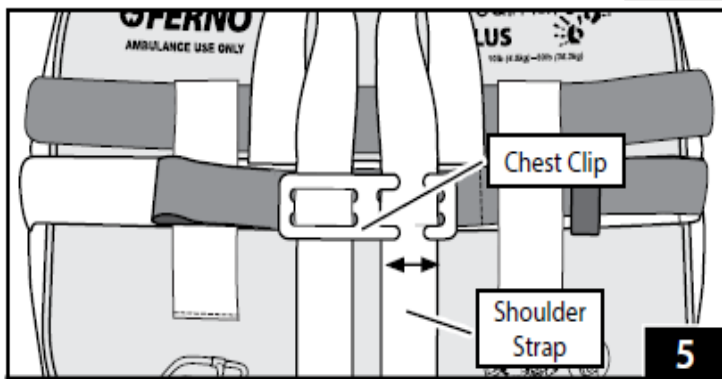
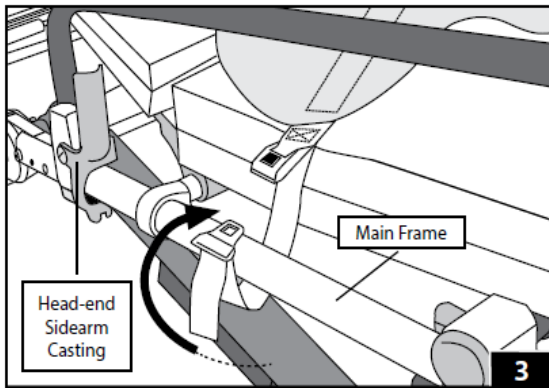
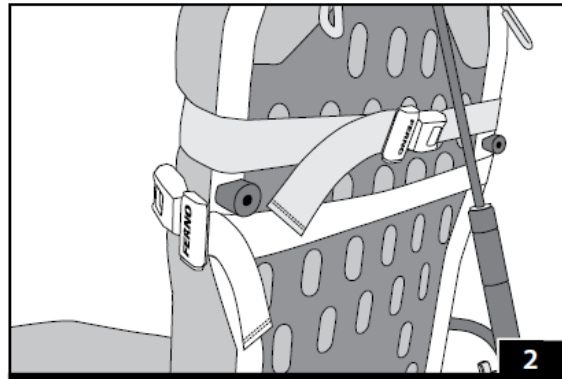
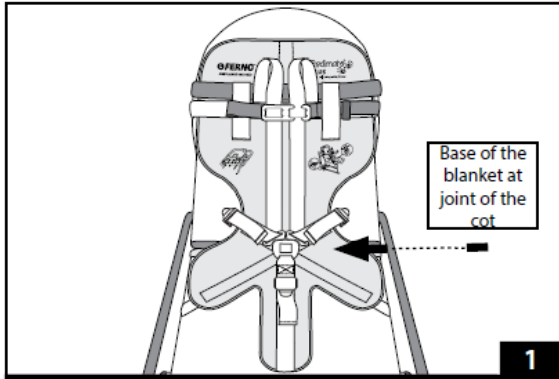
- Continuously monitor for:
 - Adequate ventilation (avoid chest strap over-tightening)
 - Strap security during transport
 - Re-tighten as needed if loosened
-

6. Release and Removal

1. Loosen crotch strap and chest clip.
 2. Press central buckle release and remove shoulder straps.
 3. Loosen chest strap and remove patient following local lifting protocols.
-

7. Post-Use

- **Clean** with warm, soapy water and air dry
- **Disinfect** per agency infection control policy (avoid bleach, phenolics, iodines)
- **Inspect** prior to returning to service



SAM Chest Seal

BLS procedure

Indications:

- Penetrating Back/Chest Wound
- Sucking Chest Wound

Contraindications

- None

Precautions

- None

Equipment Needed

- Something to clean the area
- Sam Chest Seal

Procedure

1. Clean area with absorbent pad.
2. Grip tab and remove clear liner.
3. Place dressing, adhesive side down, centered over wound.
4. Press dressing firmly to ensure adhesion

SAM Pelvic Sling

BLS procedure

Indications:

- Suspected pelvic fracture
- Open book pelvic fracture

Contraindications

- Hip fracture (isolated trochanter fracture)

Precautions

- Proper placement is imperative

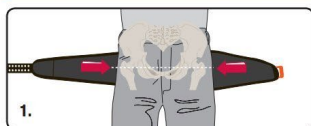
Equipment Needed

- SAM Pelvic Sling

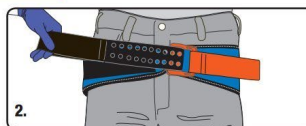
Procedure

1. Unfold sling with black surface up
2. Place black side of sling beneath patient at the level of the buttocks (greater trochanters / symphysis pubis)
3. Firmly close the sling by placing black Velcro side of flap down on black Velcro strip. Fold back material as needed. Try to place buckle close to midline
4. Grab orange free handle on outer surface of flap and release from flap by pulling upward
5. With or without assistance, firmly pull both handles in opposite directions to tighten sling
6. Keep pulling free handle until you feel or hear the buckle click
7. As soon as the buckle clicks, maintain tension and firmly press orange handle onto the black Velcro strip

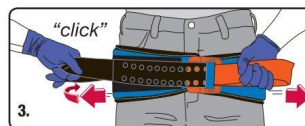
Applies in 3 Easy Steps no trimming, no cutting, no guessing



1. Remove objects from patient's pocket or pelvic area. Place SAM Pelvic Sling II black side up beneath patient at level of trochanters (hips).



2. Place **BLACK STRAP** through buckle and pull completely through.



3. Hold **ORANGE STRAP** and pull **BLACK STRAP** in opposite direction until you hear and feel the buckle click. Maintain tension and immediately press **BLACK STRAP** onto surface of SAM Pelvic Sling II to secure. You may hear a second click as the sling secures.

Slishman Traction Splint

BLS procedure

Indications:

- Simple midshaft femur fracture

Contraindications

- Open/complex fracture

Precautions

- None

Equipment Needed

- Slishman Traction Splint

Procedure

1. Attach ankle strap to effected extremity
2. Attach groin strap on same side
3. Apply coarse traction by extending pole and inserting distal end into ankle strap
4. Apply fine traction by releasing the thumb screw on red pole and pulling the cord to desired traction.
5. Tighten thumb screw on red pole clamp and release cord
6. Reassess and monitor

HOW TO APPLY



Prior to application assess CMS (circulation, motor and sensory) function and pain level per local protocol.

1. Attach Ankle Strap

- Remove ankle strap and end cap from pole
- Unroll ankle strap and apply with end cap lateral and facing up to receive splint pole
- Secure with Velcro wrap



NOTE: May apply ankle strap above calf in cases of lower leg injury



2. Attach Groin Strap

- Rest female buckle on anterior thigh
- Wrap male buckle and strap behind thigh
- Snap male to female buckle and tighten



3. Apply Coarse Traction

- Extend distal pole after releasing thumb screw on black pole clamp
- Insert distal pole into ankle strap end cap
- After achieving desired length, tighten thumb screw



4. Apply Fine Traction

- Release thumb screw on red pole clamp
- Pull cord to apply desired traction
- Tighten thumb screw on red pole clamp and release cord



5. Reassess and Monitor

- Reassess CMS and pain level
- Adjust traction as needed to minimize pain, while maintaining perfusion
- For rotational stability attach mid leg strap to splint and wrap (one or both legs) below knee



PEDIATRIC APPLICATION

For patients under 110 cm (approx. 43 inches) in height and/or 3 years or less in age, lengthen the groin strap allowing the splint to rest more proximal to the hip.



Click on QR code to see instructional video OR visit [YouTube.com](https://www.youtube.com) and search "Slushman Traction Splint"

Rev. 1.1 9/13

Splinting

BLS procedure

Indications:

- Suspected bone/joint injury

Contraindications

- None

Precautions

- Only attempt reduction for neurovascular compromise (absent distal pulse)

Equipment Needed

- SAM Splint
- Kling
- Triangular Bandage/Sling (for arm or shoulder)
- Vacuum Splint (for long bone)
- Pillow (for ankle)

Procedure

1. Check pulses, sensation, and motor function before splinting
2. Splint joints and bone ends above and below
3. Immobilize open and closed fractures in the same manner
4. Cover open fractures to minimize contamination
5. Check pulses, sensation, and motor function after splinting
6. Stabilize the extremity in the position of comfort
7. Apply cold to reduce swelling and pain

Suctioning

BLS procedure

Indications:

- Removal of small obstructions from the airway (blood, emesis, secretions, food particles, etc)

Contraindications

- None

Precautions

- Ensure oxygenation and adequate ventilation

Equipment Needed

- Suction device
- Suction catheter

Procedure

1. Ensure pre and post oxygenation
2. Turn on and prepare suction device
3. Assure presence of mechanical suction
4. Select proper suction catheter
5. Limit suction times: adult 10 seconds | infants and children 5 seconds

Tourniquet

BLS procedure

Indications:

- Severe arterial hemorrhage (spurting blood)

Contraindications

- Neck wound

Precautions

- Junctional injuries may be better managed with wound packing

Equipment Needed

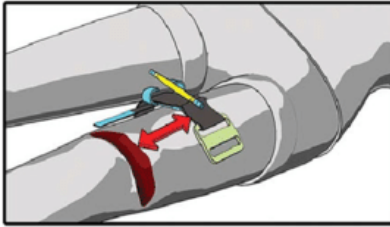
- Tourniquet

Procedure

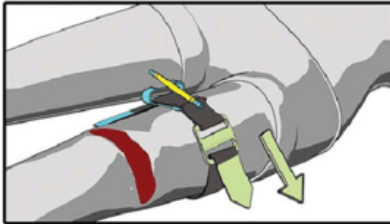
1. Remove any clothing on or near the extremity to avoid concealment of injury, tourniquet or to interfere with tourniquet placement and tightness
2. Place tourniquet proximal to wound
3. Tighten per manufacturer instructions until hemorrhage stops and distal pulse is eliminated
4. Secure tourniquet per manufacturers instruction
5. Note time of tourniquet application and communicate this to receiving care providers
6. Dress wound if possible
7. An additional tourniquet can be placed next to the initial tourniquet if bleeding control is inadequate following initial placement

Tourniquet Application Instructions

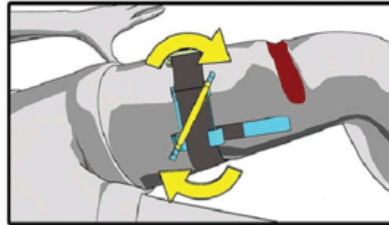
Place the tourniquet 2-3 inches above the wound. It should be between the torso and the wound.



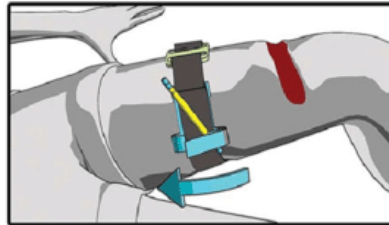
- 1. Pull** the free end of the Velcro strap. Thread it through the buckle. Securely fasten it back onto itself.



- 2. Twist** the rod. Keep twisting until bleeding stops. It is normal for this to cause some pain.



- 3. Clip** and secure the rod with the small Velcro strap so that it does not untwist. If bleeding hasn't stopped, apply a second tourniquet above the previous one, closer to the torso.



Wound Packing

BLS procedure

Indications:

- Severe hemorrhage

Contraindications

- Non-compressible wound (cavity of torso)

Precautions

- Central wound packing may be inappropriate

Equipment Needed

- Hemostatic Z-fold gauze

Procedure

1. Take appropriate PPE precautions
2. Expose the simulated injury by opening or cutting away clothing
3. If possible, remove excess blood from the wound while preserving any clots in the wound that may have formed.
4. Remove the hemostatic gauze from package.
 - a. Multiple gauze rolls may be required to control the hemorrhage.
5. Locate source of bleeding, hold direct pressure with glove hand.
6. Pack gauze into wound without releasing pressure over bleeding,
7. Apply Direct Pressure for three minutes.
8. Reassess the wound to ensure bleeding has stopped.
9. Apply a pressure dressing over the bandage to secure it in place.
10. Thermopreservation and immediate transport.